



What is Data Science

Exploring the World of Data

COMPREHENSIVE LEARNING NOTES



Understanding
Data Science



• Role of a
Data Scientist



• Career Paths in
Data Science



• Data & Data
Processing



• Machine Learning
& Deep Learning



• Data Mining &
Data Literacy



• Data-Driven
Decision Making



KASHIF MAQBOOL JOIYA
Data Scientist | AI Engineer



GitHub
<https://github.com/KashifMaqbool>



Table of Contents

1	SYLLABUS	1
1.1	DEFINING DATA SCIENCE AND WHAT DATA SCIENTISTS DO	1
1.1.1	TOPICS COVERED	1
1.2	DATA SCIENCE TOPICS	1
1.2.1	BIG DATA AND DATA MINING	1
1.2.2	MACHINE LEARNING, DEEP LEARNING, AND ARTIFICIAL INTELLIGENCE	2
1.3	APPLICATIONS AND CAREERS IN DATA SCIENCE	2
1.3.1	TOPICS COVERED	2
1.4	DATA LITERACY FOR DATA SCIENCE (OPTIONAL)	2
1.4.1	TOPICS COVERED	2
2	DEFINE DATA SCIENCE?	3
2.1	DEFINITION OF DATA SCIENCE	3
2.2	WHAT DOES DATA SCIENCE MEAN?	3
2.3	WHY IS DATA SCIENCE RELEVANT TODAY?	4
2.4	WHY THIS IS THE BEST TIME TO LEARN DATA SCIENCE	4
3	WHAT IS DATA SCIENCE?	4
3.1	THE DATA SCIENCE PROCESS	5
3.2	COMMUNICATING INSIGHTS	6
3.3	IMPACT OF DATA SCIENCE	6
4	DATA SCIENCE: THE SEXIEST JOB IN THE 21ST CENTURY	6
4.1	COURSE INFORMATION	6
4.2	INTRODUCTION	6
4.3	GROWTH OF THE DIGITAL UNIVERSE	7
4.4	THE BIG DATA TALENT SHORTAGE	7
4.5	WHY BUSINESSES NEED ANALYTICS EXPERTISE	7
4.6	ORGANIZATIONAL CHALLENGES IN USING DATA	8
4.7	INDUSTRY SURVEY FINDINGS	8
4.7.1	SAP SURVEY	8
4.7.2	ACCENTURE REPORT	8
4.7.3	KPMG SURVEY (2014)	8
4.8	LACK OF ANALYTICS TALENT	9
4.9	WALMART'S CROWDSOURCING APPROACH	9
4.10	HIGH DEMAND, HIGH SALARIES	9

4.11	KEY TAKEAWAYS	10
4.12	FURTHER READING	10
5	ADVICE FOR ASPIRING DATA SCIENTISTS	10
5.1	DEVELOP THE RIGHT MINDSET	10
5.2	GAIN COMFORT WITH ANALYTICS PLATFORMS	11
5.3	LEARN TO TELL A STORY WITH DATA	11
5.4	IDENTIFY OUR COMPETITIVE ADVANTAGE	11
5.5	BUILD EXPERTISE IN A SPECIFIC DOMAIN	12
5.6	LEARN THE RIGHT TOOLS	12
5.7	APPLY SKILLS TO REAL PROBLEMS	12
5.8	SHARE OUR WORK	12
6	DEFINING DATA SCIENCE – LESSON SUMMARY	13
6.1	OVERVIEW	13
6.2	WHAT IS DATA SCIENCE?	13
6.3	THE DATA SCIENCE PROCESS	13
6.4	ESSENTIAL QUALITIES OF A DATA SCIENTIST	14
6.5	SKILLS REQUIRED FOR A DATA SCIENTIST	14
6.6	THE FUTURE OF DATA SCIENCE	14
6.7	KEY TAKEAWAYS	15
7	DEFINING DATA SCIENCE LESSON GLOSSARY	15
8	REAL-WORLD APPLICATIONS OF ARTIFICIAL INTELLIGENCE AND DATA ANALYTICS	16
8.1	RECOMMENDATION ENGINE	16
8.2	PREDICTING ALGAE BLOOMS USING ARTIFICIAL NEURAL NETWORKS	16
8.3	CASE STUDY: ANALYZING PUBLIC TRANSIT COMPLAINTS	16
8.3.1	PROBLEM STATEMENT	16
8.3.2	UNDERSTANDING THE DATASET	16
8.3.3	INITIAL ANALYSIS	17
8.3.4	THE KEY OBSERVATION	17
8.3.5	INTEGRATING EXTERNAL DATA	17
8.3.6	FINDINGS	18
8.3.7	FINAL CONCLUSION	18
8.4	KEY TAKEAWAYS	18
9	NORMAN WHITE: TEACHING DATA SCIENCE WITH PYTHON AND JUPYTER NOTEBOOKS	19

9.1	INTRODUCTION	19
9.2	EDUCATIONAL BACKGROUND AND CAREER JOURNEY	19
9.3	TEACHING SCHEDULE AND COURSE STRUCTURE	19
9.4	COURSE ASSIGNMENTS	20
9.5	TEACHING WITH JUPYTER NOTEBOOKS	20
9.6	KEY TECHNOLOGIES AND CONCEPTS COVERED	21
10	UNDERSTANDING COMMON DATA FILE FORMATS	21
10.1	DELIMITED TEXT FILE FORMATS	21
10.1.1	COMMON DELIMITERS	21
10.1.2	STRUCTURE OF DELIMITED TEXT FILES	22
10.1.3	ADVANTAGES OF DELIMITED TEXT FILES	22
10.2	MICROSOFT EXCEL OPEN XML SPREADSHEET (XLSX)	22
10.2.1	STRUCTURE OF AN XLSX FILE	22
10.2.2	ADVANTAGES OF XLSX	23
10.3	EXTENSIBLE MARKUP LANGUAGE (XML)	23
10.3.1	XML CHARACTERISTICS	23
10.4	PORTABLE DOCUMENT FORMAT (PDF)	23
10.4.1	COMMON USES OF PDF	23
10.5	JAVASCRIPT OBJECT NOTATION (JSON)	23
10.5.1	ADVANTAGES OF JSON	23
10.6	SUMMARY	24
11	REGRESSION, DATA VISUALIZATION, NEURAL NETWORKS, AND TYPES OF DATA	24
11.1	KEY PERSPECTIVES FROM DATA SCIENCE PROFESSIONALS	24
11.2	STRUCTURED DATA VS. UNSTRUCTURED DATA	24
11.2.1	STRUCTURED DATA	25
11.2.2	UNSTRUCTURED DATA	25
11.3	UNDERSTANDING REGRESSION USING A TAXI FARE EXAMPLE	26
12	WHY DEFINING DATA SCIENCE IS DIFFICULT	26
12.1	INTRODUCTION	27
12.2	WHAT MAKES SOMEONE A DATA SCIENTIST?	27
12.3	DATA SCIENCE AS PRACTICE, NOT A NARROW DISCIPLINE	28
12.3.1	EXAMPLE: PROFESSOR RAGHURAM RAJAN	29
12.4	THE LIMITS OF TOOL- AND SIZE-BASED DEFINITIONS	29
12.4.1	DR. VINCENT GRANVILLE'S PERSPECTIVE	29
12.4.2	PROBLEMS WITH SIZE-BASED DEFINITIONS	30
12.5	CURIOSITY, COMMUNICATION, AND THE DATA SCIENTIST MINDSET	31
12.5.1	CURIOSITY: A KEY CHARACTERISTIC	31
12.5.2	RACHEL SCHUTT'S DEFINITION	31

12.6	KEY TAKEAWAYS	32
12.7	REFERENCES	32
13	A DAY IN THE LIFE OF A DATA SCIENTIST	33
13.1	WHAT DATA SCIENTISTS DO	33
13.2	EDUCATIONAL BACKGROUND FOR DATA SCIENTISTS	33
13.3	STATISTICAL MODELS AND MACHINE LEARNING	34
13.4	DIFFERENT PERSPECTIVES ON DATA SCIENCE	34
13.5	TYPES OF DATA USED BY DATA SCIENTISTS	35
13.6	COMMON DATA FORMATS	35
13.7	CHARACTERISTICS OF A SUCCESSFUL DATA SCIENTIST	35
13.8	KEY TAKEAWAYS	35
14	SUMMARY: WHAT DO DATA SCIENTISTS DO?	36
14.1	KEY DATA SCIENCE TERMS	36
15	GLOSSARY: WHAT DO DATA SCIENTISTS DO?	36
15.1	DATA SCIENCE GLOSSARY	36
16	DIGITAL TRANSFORMATION AND THE ROLE OF BIG DATA	38
16.1	HOW BIG DATA TRIGGERS DIGITAL TRANSFORMATION	39
16.2	CASE STUDY: HOUSTON ROCKETS AND BIG DATA	39
16.3	ORGANIZATIONAL IMPACT OF DIGITAL TRANSFORMATION	39
16.4	KEY TAKEAWAYS	40
17	INTRODUCTION TO CLOUD COMPUTING AND CLOUD DEPLOYMENT AND SERVICE MODELS	40
17.1	LEARNING OBJECTIVES	40
17.2	INTRODUCTION TO CLOUD COMPUTING	41
17.2.1	EXAMPLES OF CLOUD COMPUTING	41
17.3	BENEFITS OF CLOUD COMPUTING	41
17.4	ESSENTIAL CHARACTERISTICS OF CLOUD COMPUTING	41
17.4.1	ON-DEMAND SELF-SERVICE	41
17.4.2	BROAD NETWORK ACCESS	42
17.4.3	RESOURCE POOLING	42
17.4.4	RAPID ELASTICITY	42
17.4.5	MEASURED SERVICE	42
17.5	KEY CONCEPT OF CLOUD COMPUTING	42

17.6	CLOUD DEPLOYMENT MODELS	43
17.6.1	PUBLIC CLOUD	43
17.6.2	PRIVATE CLOUD	43
17.6.3	HYBRID CLOUD	43
17.7	CLOUD SERVICE MODELS	43
17.7.1	INFRASTRUCTURE AS A SERVICE (IAAS)	44
17.7.2	PLATFORM AS A SERVICE (PAAS)	44
17.7.3	SOFTWARE AS A SERVICE (SAAS)	44
17.8	SUMMARY	44
18	CLOUD COMPUTING FOR DATA SCIENCE	45
18.1	WHAT IS CLOUD COMPUTING?	45
18.2	BENEFITS OF CLOUD COMPUTING FOR DATA SCIENTISTS	45
18.2.1	SCALABILITY AND HIGH-PERFORMANCE COMPUTING	45
18.2.2	CENTRALIZED DATA STORAGE	45
18.2.3	COLLABORATION ACROSS TEAMS	45
18.2.4	ACCESS TO OPEN-SOURCE TECHNOLOGIES	46
18.2.5	ACCESS TO THE LATEST TOOLS AND LIBRARIES	46
18.2.6	ACCESSIBILITY	46
18.2.7	IMPROVED TEAM COLLABORATION	46
18.3	POPULAR CLOUD PLATFORMS	47
18.4	IBM SKILLS NETWORK LABS (SN LABS)	47
18.5	KEY TAKEAWAY	47
19	UNDERSTANDING BIG DATA	47
19.1	THE 5 Vs OF BIG DATA	48
19.1.1	VELOCITY	48
19.1.2	VOLUME	48
19.1.3	VARIETY	49
19.1.4	VERACITY	49
19.1.5	VALUE	50
19.2	BIG DATA ANALYTICS	50
19.3	REAL-WORLD PERSPECTIVE	51
20	DATA SCIENCE, BUSINESS ANALYTICS, AND BIG DATA	51
20.1	INTRODUCTION	51
20.2	PROGRAMMING BACKGROUND FOR DATA SCIENCE	51
20.3	GROWTH OF DATA SCIENCE AND BUSINESS ANALYTICS	52
20.4	INDUSTRY ADOPTION OF BIG DATA	52
20.5	INCREASING DEMAND FOR DATA SCIENCE EDUCATION	52
20.6	DEFINITION OF BIG DATA	53

20.7	ORIGINS OF BIG DATA	53
20.8	BIG DATA ANALYTICS	53
20.9	DEEP LEARNING	54
21	BIG DATA ARCHITECTURE AND THE EVOLUTION OF DATA SCIENCE	54
21.1	TRADITIONAL DATA PROCESSING VS. BIG DATA PROCESSING	54
21.2	ADVANTAGES OF BIG DATA CLUSTERS	54
21.3	HADOOP AND THE BIG DATA ECOSYSTEM	55
21.4	FOUNDATIONS OF DATA SCIENCE	55
21.5	MACHINE LEARNING AND LARGE-SCALE DATA ANALYSIS	55
21.6	DECISION SCIENCES	56
21.7	THE EMERGENCE OF DATA SCIENCE	56
21.8	WHAT IS DATA SCIENCE?	57
21.9	THE RISE OF DEEP LEARNING	57
21.10	KEY TAKEAWAYS	57
21.11	REFERENCES	58
22	BIG DATA PROCESSING TECHNOLOGIES: APACHE HADOOP, APACHE HIVE, AND APACHE SPARK	58
22.1	APACHE HADOOP	58
22.1.1	BENEFITS OF USING HADOOP	58
22.2	HADOOP DISTRIBUTED FILE SYSTEM (HDFS)	59
22.2.1	HOW HDFS WORKS	59
22.2.2	EXAMPLE OF HDFS	59
22.2.3	ADVANTAGES OF DATA REPLICATION	59
22.2.4	DATA LOCALITY	59
22.2.5	ADDITIONAL BENEFITS OF HDFS	60
22.3	APACHE HIVE	60
22.3.1	CHARACTERISTICS OF HIVE	60
22.3.2	LIMITATIONS OF HIVE	60
22.3.3	BEST USE CASES FOR HIVE	60
22.4	APACHE SPARK	61
22.4.1	APPLICATIONS OF SPARK	61
22.4.2	KEY FEATURES OF SPARK	61
22.5	KEY USE CASE OF APACHE SPARK	61
22.6	SUMMARY	61
23	DATA MINING	62
23.1	ESTABLISHING DATA MINING GOALS	62
23.2	SELECTING DATA	62
23.3	PREPROCESSING DATA	63

23.3.1	HANDLING MISSING DATA	63
23.4	TRANSFORMING DATA	63
23.5	STORING DATA	64
23.6	MINING DATA	64
23.7	EVALUATING MINING RESULTS	65
24	BIG DATA AND DATA MINING LESSON SUMMARY	65
24.1	OVERVIEW	65
24.2	CHARACTERISTICS OF BIG DATA	66
24.3	CLOUD COMPUTING AND BIG DATA	66
24.4	POPULAR OPEN-SOURCE BIG DATA TOOLS	67
24.4.1	APACHE HADOOP	67
24.4.2	APACHE HIVE	67
24.4.3	APACHE SPARK	67
24.5	THE DATA MINING PROCESS	67
24.5.1	GOAL SETTING	67
24.5.2	DATA SELECTION	67
24.5.3	DATA PREPROCESSING	67
24.5.4	DATA TRANSFORMATION	68
24.5.5	DATA MINING	68
24.5.6	EVALUATION	68
24.6	KEY TAKEAWAYS	68
25	GLOSSARY: BIG DATA AND DATA MINING	69
26	KEY DATA SCIENCE AND ARTIFICIAL INTELLIGENCE TERMINOLOGY	72
26.1	BIG DATA	72
26.2	DATA MINING	73
26.3	MACHINE LEARNING	73
26.4	DEEP LEARNING	73
26.5	ARTIFICIAL NEURAL NETWORKS	73
26.6	DATA SCIENCE VS. ARTIFICIAL INTELLIGENCE	74
26.6.1	DATA SCIENCE	74
26.6.2	ARTIFICIAL INTELLIGENCE	74
26.6.3	RELATIONSHIP BETWEEN DATA SCIENCE AND AI	75
26.7	BIG DATA IN DATA SCIENCE AND AI	75
26.8	SUMMARY	75
27	GENERATIVE AI AND DATA SCIENCE	76
27.1	LEARNING OBJECTIVES	76

27.2	INTRODUCTION TO GENERATIVE AI	76
27.3	HOW GENERATIVE AI OPERATES	76
27.4	APPLICATIONS OF GENERATIVE AI	76
27.5	HOW DATA SCIENTISTS USE GENERATIVE AI	77
27.5.1	SYNTHETIC DATA GENERATION	77
27.5.2	CODE GENERATION AND ANALYTICS AUTOMATION	77
27.5.3	BUSINESS INSIGHTS AND REPORTING	77
27.6	KEY TAKEAWAYS	78
28	NEURAL NETWORKS AND DEEP LEARNING	78
28.1	INTRODUCTION	78
28.2	HOW NEURAL NETWORKS WORK	78
28.3	LIMITATIONS OF EARLY NEURAL NETWORKS	79
28.4	EMERGENCE OF DEEP LEARNING	79
28.5	WHY DEEP LEARNING REQUIRES POWERFUL HARDWARE	79
28.5.1	GRAPHICS PROCESSING UNIT (GPU)	79
28.6	REAL-WORLD APPLICATIONS OF DEEP LEARNING	80
28.7	EXAMPLE FROM NYU AND FACEBOOK	80
28.8	LEARNING WITHOUT EXPLICIT PROGRAMMING	80
28.9	IMPORTANCE OF LINEAR ALGEBRA	81
28.10	COMPUTATIONAL REQUIREMENTS	81
29	APPLICATIONS OF MACHINE LEARNING	81
29.1	COMMON APPLICATIONS OF MACHINE LEARNING	81
29.1.1	MARKET BASKET ANALYSIS	82
29.1.2	PREDICTIVE ANALYTICS	82
29.2	APPLICATIONS OF MACHINE LEARNING IN FINTECH	82
29.2.1	RECOMMENDATION SYSTEMS IN FINTECH	83
29.2.2	FRAUD DETECTION	83
30	REGRESSION	83
30.1	WHY TALL PARENTS DON'T HAVE EVEN TALLER CHILDREN	84
30.2	THE DEPARTMENT OF OBVIOUS CONCLUSIONS	84
30.3	WHAT THE RESEARCH ACTUALLY DISCOVERED	85
30.4	IMPACT OF LOCATION ON HOUSING PRICES	85
30.4.1	TRANSPORT INFRASTRUCTURE	85
30.4.2	SHOPPING CENTERS	85
30.4.3	DISTANCE FROM DOWNTOWN	86
30.5	QUANTIFYING RELATIONSHIPS	86
30.6	WHY REGRESSION?	86

31	IBM CLOUD GALLERY	86
31.1	OVERVIEW	86
31.2	LEARNING OBJECTIVES	87
31.3	EXERCISE: EXAMINE A NUMERIC DATASET	87
32	DEEP LEARNING AND MACHINE LEARNING LESSON SUMMARY	87
32.1	OVERVIEW	87
32.2	ARTIFICIAL INTELLIGENCE AND ITS RELATED CONCEPTS	87
32.3	DEEP LEARNING	88
32.4	GENERATIVE AI	88
32.5	HOW DATA SCIENTISTS USE MACHINE LEARNING	88
32.6	REGRESSION	88
32.7	SUMMARY	89
33	DEEP LEARNING AND MACHINE LEARNING LESSON GLOSSARY	89
34	SUMMARY: DEEP LEARNING AND MACHINE LEARNING	91
35	BUILDING A STRONG DATA FOUNDATION FOR DATA SCIENCE	92
35.1	START BY CAPTURING DATA	92
35.2	ARCHIVE DATA INSTEAD OF OVERWRITING IT	92
35.3	IMPLEMENT DATA ARCHIVING BEST PRACTICES	93
35.4	THE IMPORTANCE OF DATA QUALITY	93
35.5	CLEAN EXISTING DATA OR START COLLECTING NEW DATA	93
35.6	BUILD A DATA SCIENCE TEAM	94
36	DATA SCIENCE APPLICATIONS FOR SOLVING REAL-WORLD PROBLEMS	94
36.1	DATA SCIENCE IN TRANSPORTATION	94
36.1.1	UBER'S USE OF REAL-TIME DATA	94
36.1.2	TORONTO TRANSPORTATION COMMISSION	94
36.2	DATA SCIENCE FOR ENVIRONMENTAL CHALLENGES	95
36.2.1	CYANOBACTERIA RESEARCH IN THE UNITED STATES	95
36.3	DEVELOPING EFFICIENT DATA-DRIVEN SOLUTIONS	96
37	DATA SCIENCE APPLICATIONS IN BUSINESS	96
37.1	HOW CONSUMERS GENERATE DATA	96
37.2	HOW DATA SCIENCE IMPACTS BUSINESS	97
37.2.1	INDUSTRY PERSPECTIVES	97

37.3	HOW NETFLIX GAINS A COMPETITIVE ADVANTAGE	97
37.3.1	HOUSE OF CARDS CASE STUDY	97
38	APPLICATIONS OF DATA SCIENCE IN HEALTHCARE AND DISASTER PREPAREDNESS	98
38.1	DATA SCIENCE IN HEALTHCARE	98
38.2	REAL-WORLD EXAMPLE: BOSTON CONSULTING GROUP AND ADVAMEDDX STUDY	99
38.3	ELECTRONIC MEDICAL RECORDS (EMR) AND DATA MINING	99
38.4	DATA SCIENCE IN DISASTER PREPAREDNESS	99
38.5	RESEARCH EXAMPLE: UNIVERSITY OF WARWICK	100
38.6	DATA SCIENCE EDUCATION	100
38.7	KEY TAKEAWAYS	100
39	THE FINAL DELIVERABLE	100
39.1	COURSE INFORMATION	100
39.2	ANALYTICS MUST LEAD TO COMMUNICATION	101
39.3	WHAT FINAL DELIVERABLES LOOK LIKE IN PRACTICE	101
39.4	A DELOITTE REPORT AS AN EXAMPLE OF STORYTELLING WITH DATA	101
39.5	USING DATA VISUALIZATION TO SUPPORT THE NARRATIVE	102
39.6	WHY THE DELOITTE REPORT IS EFFECTIVE	102
39.7	WHY NARRATIVE MATTERS	103
39.8	PLANNING THE DELIVERABLE BEFORE THE ANALYSIS	103
40	DATA SCIENCE APPLICATIONS – LESSON SUMMARY	103
40.1	PURPOSE OF DATA SCIENCE	103
40.2	HOW ORGANIZATIONS LEVERAGE DATA SCIENCE	104
40.2.1	RECOMMENDATION ENGINES	104
40.2.2	ROUTE OPTIMIZATION	104
40.2.3	DYNAMIC DRIVER ALLOCATION	105
40.3	DATA SCIENCE FOR COMPETITIVE ADVANTAGE	105
40.4	DATA SCIENCE IN HEALTHCARE	105
40.5	PREDICTING NATURAL DISASTERS	106
40.6	PRESENTING FINDINGS	106
40.6.1	ACADEMIA	106
40.6.2	CONSULTING AND BUSINESS	106
41	GLOSSARY: DATA SCIENCE APPLICATION DOMAINS	107
42	SKILLS, LEARNING PATH, AND CAREER OPPORTUNITIES IN DATA SCIENCE	107
42.1	SKILLS REQUIRED TO BECOME A DATA SCIENTIST	107

42.2	SELF-LEARNING AND HANDS-ON PRACTICE	108
42.3	LEARNING PLATFORMS AND PRACTICAL TOOLS	109
42.4	MOTIVATION AND CONTINUOUS LEARNING	109
42.5	WHERE DATA SCIENCE FITS IN AN ORGANIZATION	109
42.6	DEMAND FOR PHD-LEVEL DATA SCIENTISTS	110
43	HIRING THE RIGHT PEOPLE FOR A DATA SCIENCE TEAM	110
43.1	IMPORTANT QUALITIES WHEN HIRING DATA SCIENTISTS	111
43.1.1	CURIOSITY	111
43.1.2	SENSE OF HUMOR	111
43.1.3	SOCIAL SKILLS AND STORYTELLING	111
43.2	TECHNICAL AND PROFESSIONAL SKILLS REQUIRED	112
43.2.1	DATA SKILLS	112
43.2.2	RELATIONSHIP BUILDING	112
43.2.3	ANALYTICAL BACKGROUND	112
43.3	BUILDING A DATA SCIENCE TEAM	113
43.4	TECHNICAL SKILLS BASED ON THE WORKING ENVIRONMENT	113
43.4.1	STRUCTURED DATA ENVIRONMENT	113
43.4.2	BIG DATA ENVIRONMENT	113
43.5	RECOMMENDED PLATFORMS AND TOOLS	114
43.5.1	TRADITIONAL PREDICTIVE ANALYTICS (WITHOUT BIG DATA)	114
43.5.2	UNSTRUCTURED DATA	114
43.5.3	BIG DATA	114
43.6	COMMUNICATION AND STORYTELLING SKILLS	114
43.7	CREATING IMPACT THROUGH DATA STORYTELLING	115
44	WHY DATA SCIENCE IS ONE OF THE MOST PROMISING CAREER FIELDS	115
44.1	GROWING DEMAND FOR DATA SCIENCE	115
44.2	INDUSTRY INSIGHTS AND TALENT SHORTAGE	116
44.3	WHY CHOOSE A CAREER IN DATA SCIENCE?	116
44.4	WOMEN IN DATA SCIENCE INITIATIVE	116
44.5	PREPARING FOR A CAREER IN DATA SCIENCE	116
44.6	KEY TAKEAWAY	117
45	IMPORTANCE OF MATHEMATICS AND STATISTICS FOR DATA SCIENCE	117
45.1	BUILD A STRONG FOUNDATION	117
45.2	PREPARING FOR DATA SCIENCE IN HIGH SCHOOL	117
45.3	DEVELOP CURIOSITY AND CREATIVITY	118
45.4	ENCOURAGE EXPERIMENTATION	118
45.5	LEARN FROM REAL-WORLD DATA	118
45.6	WHY DATA SCIENCE IS A GREAT CAREER	118

45.7	DON'T BE DISCOURAGED BY MATHEMATICS	119
46	THE REPORT STRUCTURE	119
46.1	COURSE INFORMATION	119
46.2	PLANNING THE REPORT STRUCTURE	119
46.3	ESSENTIAL FRONT MATTER	120
46.3.1	COVER PAGE	120
46.3.2	TABLE OF CONTENTS (TOC)	120
46.3.3	EXECUTIVE SUMMARY OR ABSTRACT	121
46.4	CORE SECTIONS OF AN ANALYTICAL REPORT	121
46.4.1	INTRODUCTION	121
46.4.2	LITERATURE REVIEW	121
46.4.3	METHODOLOGY	121
46.4.4	RESULTS	122
46.5	FROM RESULTS TO DISCUSSION AND CONCLUSION	122
46.5.1	DISCUSSION	122
46.5.2	CONCLUSION	123
46.6	FINAL ELEMENTS	123
46.7	HAVE WE DONE OUR JOB AS WRITERS?	123
47	CAREERS AND RECRUITING IN DATA SCIENCE – LESSON SUMMARY	124
47.1	OVERVIEW	124
47.2	WHAT COMPANIES LOOK FOR IN A DATA SCIENTIST	124
47.3	PASSION AND CURIOSITY	124
47.4	ESSENTIAL TECHNICAL SKILLS	125
47.5	COMMUNICATION AND STORYTELLING	125
47.6	DR. HADER'S STORYTELLING ANALOGY	126
47.7	KEY TAKEAWAYS	126
48	GLOSSARY: CAREERS AND RECRUITING IN DATA SCIENCE	126
49	CAREERS AND RECRUITING IN DATA SCIENCE	127
49.1	CHARACTERISTICS OF AN EFFECTIVE DATA SCIENCE REPORT	128
50	CASE STUDY: FINAL ASSIGNMENT	128
50.1	EDUCATION AND SKILL ACQUISITION	128
50.2	BUILDING A STRONG FOUNDATION	129
50.3	VISUALIZATION FOR STORYTELLING	129
50.4	HANDS-ON EXPERIENCE	129

50.5	DATA WRANGLING AND PREPROCESSING	129
50.6	COMMUNICATION AND STORYTELLING	130
50.7	NETWORKING AND COLLABORATION	130
50.8	DOMAIN EXPERTISE	130
50.9	LANDING THE FIRST JOB	131
50.10	LILA'S APPROACH TO WORKING ON HER FIRST TASK AS A DATA SCIENTIST	131
50.11	DATASET SELECTION AND SOURCING	131
50.12	DATA UNDERSTANDING AND CLEANING	131
50.13	EXPLORATORY DATA ANALYSIS (EDA)	132
50.14	FEATURE ENGINEERING	132
50.15	STATISTICAL ANALYSIS AND MACHINE LEARNING	132
50.16	PRESENTATION AND REPORTING	133
50.17	CONTINUOUS LEARNING	133
50.18	MACHINE LEARNING SKILLS	133

Module 1: Defining Data Science and What Data Scientists Do

1 Syllabus

This topic provides an introduction to the field of **Data Science**, including its fundamental concepts, various career paths, and essential skills. We will explore what Data Science is, what Data Scientists do, and gain guidance for pursuing a successful career in this exciting field.

1.1 Defining Data Science and What Data Scientists Do

This section introduces the fundamental concepts of Data Science, the responsibilities of Data Scientists, and the skills required to succeed in the field.

1.1.1 Topics Covered

- Defining Data Science
- Video: **What is Data Science?**
- Fundamentals of Data Science
- The Many Paths to Data Science
- **Data Science: The Sexiest Job in the 21st Century**
- Defining Data Science
- Advice for New Data Scientists
- What Do Data Scientists Do?
- A Day in the Life of a Data Scientist
- Data Science Skills & Big Data
- Working on Different File Formats
- Data Science Topics and Algorithms
- Discussion Prompt: Introduce Yourself
- Reading: **What Makes Someone a Data Scientist?**

1.2 Data Science Topics

This section explores the foundational technologies, concepts, and tools used in modern Data Science.

1.2.1 Big Data and Data Mining

- Big Data and Data Mining
- How Big Data is Driving Digital Transformation
- Introduction to Cloud
- Cloud for Data Science
- Foundations of Big Data
- Data Scientists at New York University
- What is Hadoop?
- Big Data Processing Tools:
 - Hadoop

- HDFS (Hadoop Distributed File System)
 - Hive
 - Spark
- Reading: **Data Mining**

1.2.2 Machine Learning, Deep Learning, and Artificial Intelligence

- Deep Learning and Machine Learning
- Artificial Intelligence and Data Science
- Generative AI and Data Science
- Neural Networks and Deep Learning
- Applications of Machine Learning
- Reading: **Regression**
- Lab: **Exploring Data using IBM Cloud Gallery**

1.3 Applications and Careers in Data Science

This section focuses on the real-world applications of Data Science, industry use cases, and career opportunities.

1.3.1 Topics Covered

- Data Science Application Domains
- How Should Companies Get Started in Data Science?
- Old Problems with New Data Science Solutions
- Applications of Data Science
- How Data Science is Saving Lives
- Reading: **The Final Deliverable**
- Careers and Recruiting in Data Science
- How Can Someone Become a Data Scientist?
- Recruiting for Data Science
- Careers in Data Science
- Importance of Mathematics and Statistics for Data Science (*only name change*)
- The Report Structure
- Reading: **Infographic on Roadmap**

1.4 Data Literacy for Data Science (Optional)

This optional section introduces the concepts of Data Literacy, different types of data sources, data storage technologies, and modern data management platforms.

1.4.1 Topics Covered

- Understanding Data
- Understanding Data
- Data Sources
- Working on Varied Data Sources and Types
- Reading: **Metadata**
- Data Literacy

- Data Collection and Organization
- Relational Database Management System (RDBMS)
- NoSQL
- Data Marts, Data Lakes, ETL (Extract, Transform, Load), and Data Pipelines
- Considerations for Choice of Data Repository
- Data Integration Platforms

2 Define Data Science?

Data Science is a **process, not an event**. It is the process of using data to understand different things and to better understand the world.

Data Science is the art of uncovering the insights and trends that are hidden within data. It involves translating data into a meaningful story by using **storytelling** (presenting data clearly and engagingly to communicate insights effectively) to generate valuable insights. These insights help organizations make strategic decisions and informed choices.

2.1 Definition of Data Science

Data Science is a field that focuses on the processes and systems used to extract knowledge from data, whether the data is **structured** (organized data stored in predefined formats such as tables and databases) or **unstructured** (data without a predefined format, such as text, images, videos, and audio).

Data Science is the study of data, just as:

- Biological Sciences is the study of biology.
- Physical Sciences is the study of physical reactions.

Data is real and has real properties. Therefore, we need to study data if we want to understand it and work with it effectively.

Data Science involves two key components:

- Data
- Science

The term **Data Science** originated during the 1980s and 1990s when professors reviewing the statistics curriculum believed that the field had evolved beyond traditional statistics and should be called **Data Science**.

2.2 What Does Data Science Mean?

Data Science can be viewed as the process of working with data to find answers to questions we are exploring.

In a nutshell, Data Science is more about **data** than it is about **science**. If we have data and curiosity, and we are:

- Working with data
- Manipulating data
- Exploring data
- Analyzing data
- Searching for answers and insights

then the very process of performing these activities is Data Science.

2.3 Why is Data Science Relevant Today?

Data Science has become more relevant than ever because we now have access to enormous amounts of data.

Previously:

- We worried about the lack of data.
- Algorithms were limited or unavailable.
- Software was expensive.
- Storing large amounts of data was costly.

Today:

- We have a **data deluge** (an overwhelming amount of data generated from various sources).
- Powerful algorithms are widely available.
- Most data science software is open source and free.
- Massive amounts of data can be stored at a fraction of the previous cost.
- Organizations and individuals can access and analyze billions of datasets economically.

2.4 Why This Is the Best Time to Learn Data Science

Today, the essential resources for Data Science are more accessible than ever:

- Data is widely available.
- Data storage is inexpensive.
- Powerful analytical tools are easily accessible.
- Open-source software has reduced the cost of performing data analysis.
- Computing resources have become affordable and readily available.

3 What Is Data Science?

Everyone we ask may give us a slightly different description of what **Data Science** is, but most people agree that it has a significant **data analysis** component.

Data analysis itself is not new. What is new is the vast quantity of data available from a wide variety of sources, including:

- Log files
- Emails

- Social media
- Sales data
- Patient information files
- Sports performance data
- Sensor data
- Security camera data
- Many other sources

At the same time that more data is available than ever before, we also have the computing power needed to perform meaningful analysis and uncover valuable insights.

Data Science helps organizations:

- Understand their environments
- Analyze existing issues
- Reveal previously hidden opportunities

Data scientists use data analysis to expand an organization's knowledge by investigating data and exploring the best ways to use it to provide value to the business.

3.1 The Data Science Process

Many organizations use Data Science to solve a specific business problem. Therefore, the first and most important step is to clearly define the question the organization wants answered. This initial step determines how the entire Data Science project progresses.

Good data scientists are naturally curious and ask questions to clarify the business need before beginning the analysis.

After defining the problem, the next questions become:

- What data do we need to solve the problem?
- Where will that data come from?

Data scientists work with both **structured data** (organized data stored in predefined formats such as tables and databases) and **unstructured data** (data without a predefined structure, such as emails, images, videos, social media posts, and documents). Depending on the nature of the problem, they choose the most appropriate methods for analyzing the data.

Using multiple analytical models to explore the data helps reveal:

- Patterns (recurring relationships or trends within the data)
- Archetypes (typical or representative examples within a dataset)
- Outliers (data points that differ significantly from the rest of the data)

Sometimes these findings confirm what the organization already suspects. In other cases, they reveal completely new knowledge that leads the organization toward a new approach.

3.2 Communicating Insights

Once the data has revealed its insights, the role of the data scientist shifts to that of a **storyteller**. Data scientists communicate the results to project stakeholders in a clear and meaningful way.

To make the findings easier to understand, they use powerful **data visualization tools** that help stakeholders:

- Understand the nature of the results
- Interpret the insights correctly
- Make informed decisions based on the recommended actions

3.3 Impact of Data Science

Data Science is transforming the way we work.

It is changing:

- The way we use data
- The way organizations understand their environments
- The way organizations discover opportunities and make data-driven decisions
- The way businesses gain knowledge and create value from data

4 Data Science: The Sexiest Job in the 21st Century

4.1 Course Information

Course Textbook: *Getting Started with Data Science*

Publisher: IBM Press, 1st Edition (December 13, 2015)

Author: Murtaza Haider

Prescribed Reading: Chapter 1, Page 4

4.2 Introduction

In today's data-driven world, **data scientists have emerged as one of the most sought-after professionals**. Organizations across industries are actively searching for skilled data science talent, and experts estimate that **millions of data science positions may remain unfilled due to a shortage of qualified professionals**.

The demand is not limited to statisticians or computer scientists. Instead, organizations seek **well-rounded professionals** who possess:

- Subject matter expertise
- Software programming experience
- Analytical skills
- Strong communication abilities

4.3 Growth of the Digital Universe

Our digital footprint has expanded dramatically over the past decade.

- In **1995**, the digital universe was approximately **130 billion gigabytes**.
- By **2020**, it was projected to grow to **40 trillion gigabytes**.
- Organizations were expected to compete for **hundreds of thousands or even millions of workers** capable of managing and analyzing this enormous amount of data.

Because of this growing demand, the **Harvard Business Review** described **Data Science** as the **"Sexiest Job of the 21st Century."**

4.4 The Big Data Talent Shortage

According to a report by the **McKinsey Global Institute**, there is a significant shortage of professionals with data and analytics expertise.

The report projected that by **2018**, the United States alone could face:

- A shortage of **140,000 to 190,000 professionals** with great analytical skills.
- A shortage of **1.5 million managers and analysts** capable of using big data analysis to make effective business decisions.

These projections highlight the increasing need for skilled data professionals across industries.

4.5 Why Businesses Need Analytics Expertise

The digital revolution has influenced almost every aspect of our lives. As a result, organizations now have unprecedented opportunities to understand customer behavior through data.

With the right data, marketers can:

- Study habit formation.
- Understand customer preferences.
- Improve marketing strategies.
- Make better business decisions.

Research in **neurology** (the study of the nervous system and brain) and **psychology** (the scientific study of human behavior and mental processes) is helping organizations understand how habits and preferences are formed.

Retail companies such as **Target** use these insights to improve their business strategies. However, they can only benefit if they employ skilled data scientists.

Former Amazon Chief Scientist **Andreas Weigend** compared the hiring of statisticians and data scientists to an **"arms race"** (an intense competition among organizations to acquire the best talent).

4.6 Organizational Challenges in Using Data

Although data analytics has become increasingly important, many organizations still face challenges in fully utilizing it.

One major challenge is convincing **C-suite executives** (senior executive leaders such as CEOs, CFOs, and COOs) of the value of data-driven decision-making.

It appears that:

- Middle management often has a better understanding of analytics than senior executives.
- Many executives reached leadership positions without extensive interaction with data.
- Organizations will experience significant transformation only when executives become well-versed in data and analytics.

Professor **Peter Fader**, who manages the **Customer Analytics Initiative at Wharton**, believes that meaningful organizational change depends on leadership developing strong data literacy.

4.7 Industry Survey Findings

Several industry surveys highlight the growing importance of data science and the existing skills gap.

4.7.1 SAP Survey

SAP reported that:

- **92%** of surveyed organizations experienced a significant increase in their data holdings.
- Approximately **75%** recognized the need for new data science skills within their organizations.

4.7.2 Accenture Report

Accenture estimated that the demand for data scientists could exceed the available supply by **250,000 professionals in 2015 alone**.

4.7.3 KPMG Survey (2014)

A survey involving **150 executives** found that:

- **85%** of respondents did not know how to analyze data effectively.

According to **Alwin Magimay**, Head of Digital and Analytics for KPMG UK, many organizations struggle to connect the dots because they do not fully understand how data and analytics can transform their businesses.

4.8 Lack of Analytics Talent

Writing for **Forbes**, **Bernard Marr** emphasized the shortage of analytics professionals.

He stated that there are simply not enough individuals with the skills required to:

- Analyze data.
- Interpret complex information.
- Transform raw numerical (or other) data into actionable insights (practical recommendations that support informed business decisions).

Bernard Marr also referenced a **Gartner** survey in which **more than 50% of business leaders** reported a lack of in-house expertise in data science.

4.9 Walmart's Crowdsourcing Approach

To address the shortage of analytics talent, **Walmart** adopted a crowdsourcing approach.

The company partnered with **Kaggle**, a platform for data science competitions, and:

- Shared sales data from selected stores.
- Asked participants to develop more accurate sales forecasting models.
- Focused on improving forecasts based on promotional campaigns.

This example demonstrates how organizations can leverage external data science communities to solve complex analytical problems.

4.10 High Demand, High Salaries

Because qualified data scientists are in short supply, organizations are willing to offer highly competitive salaries.

Michael Chui, a principal at McKinsey, stated:

"Data science has become relevant to every company. There's a war for this type of talent."

A notable example is **Paul Minton**:

- Previously earned **\$20,000** annually working in a restaurant.
- Held a college degree in mathematics.
- Completed a **three-month programming course**.
- Secured a data scientist position at a web startup in San Francisco.
- Earned **over \$100,000 in 2014**.

This example illustrates how acquiring the right technical skills can significantly improve career opportunities.

According to **The New York Times**:

- The average base salary of a **Software Engineer** was approximately **\$100,000**.
- The average base salary of a **Data Scientist** was approximately **\$112,000**.

These figures demonstrate the strong market demand and attractive compensation associated with careers in data science.

4.11 Key Takeaways

- Data Science is one of the fastest-growing and most in-demand professions.
- Organizations require professionals with a combination of technical, analytical, domain, and communication skills.
- The rapid growth of digital data has created an enormous demand for data scientists.
- There is a global shortage of professionals with data science expertise.
- Many organizations still face challenges in adopting data-driven decision-making due to limited analytics knowledge among senior leadership.
- Surveys from SAP, Accenture, KPMG, Gartner, and McKinsey consistently highlight the widening gap between the demand for and supply of data science talent.
- Companies such as Walmart have adopted innovative approaches like crowdsourcing through Kaggle to solve analytics problems.
- High demand has resulted in excellent career opportunities and competitive salaries for data scientists.

4.12 Further Reading

Is Data Science Still the Sexiest Job in the 21st Century?

Authors: Thomas H. Davenport and DJ Patil

Publisher: Harvard Business Review

Link: <https://hbr.org/2022/01/is-data-science-still-the-sexiest-job-of-the-21st-century>

5 Advice for Aspiring Data Scientists

This topic provides practical advice for aspiring data scientists, emphasizing that technical skills alone are not enough. Developing the right mindset, understanding a domain, and communicating insights effectively are equally important for success.

5.1 Develop the Right Mindset

The most important qualities of a successful data scientist are:

- **Curiosity** – Curiosity is an absolute must. If we are not curious, we would not know what to do with the data or what questions to ask.
- **Being Judgmental** (*forming initial assumptions or hypotheses before analyzing data*) – If we do not have preconceived notions about things, we would not know where to begin.
- **Being Argumentative** (*the ability to present and defend a logical point of view*) – If we can argue and plead a case, we can start somewhere, learn from the data, and then modify our assumptions and hypotheses based on the evidence.

We may start at the wrong point and think:

"We believed this, but now, with data, we know this."

This learning process allows us to continuously improve our understanding. Curiosity, the willingness to take a strong position, and the readiness to revise that position based on data are essential characteristics of a data scientist.

5.2 Gain Comfort with Analytics Platforms

A data scientist should also have some comfort and flexibility with analytics platforms, including:

- Analytics software
- Computing platforms
- Data analysis tools

However, these technical skills are secondary. The most important qualities remain curiosity and the ability to develop and refine ideas through data.

5.3 Learn to Tell a Story with Data

Once we have analyzed the data and obtained results, the next essential skill is **data storytelling** (*communicating analytical findings in a clear, meaningful, and engaging way*).

After creating analytics and tabulations, we should be able to tell a compelling story from the results.

If we fail to communicate our findings effectively:

- Our discoveries remain hidden.
- Valuable insights remain buried.
- Others may never recognize the value of our work.

Our professional growth and recognition depend greatly on our ability to transform data into meaningful stories.

5.4 Identify Our Competitive Advantage

A good starting point is identifying our competitive advantage.

We should ask ourselves:

- Do we want to become a data scientist in any field?
- Or do we want to specialize in a specific industry?

Different industries require different skill sets. For example:

- Working for an IT, web-based, or internet company requires one set of technical skills.

- Working in the healthcare industry requires another set of specialized skills.

Before choosing analytical tools, we should first determine:

- What interests us most.
- Where our competitive advantage lies.

5.5 Build Expertise in a Specific Domain

Our competitive advantage is not necessarily our analytical ability.

Instead, it comes from our deep understanding of a particular area where we know more than others.

Examples include:

- Film
- Retail
- Healthcare
- Computers
- Any other domain in which we possess strong knowledge

Once we identify our area of expertise, we can begin acquiring analytical skills that are relevant to that specific industry.

5.6 Learn the Right Tools

The platforms and tools we learn should match the industry we want to work in.

Different industries use different technologies, so selecting the appropriate analytical platforms and software should be based on our chosen domain.

5.7 Apply Skills to Real Problems

After gaining proficiency with analytical tools, the next step is to apply those skills to real-world problems.

Practical experience helps us:

- Strengthen our analytical abilities.
- Solve meaningful business or industry problems.
- Build confidence through hands-on projects.

5.8 Share Our Work

After solving real problems, we should share our work with others.

Demonstrating what we can accomplish with data allows us to showcase our skills, communicate our expertise, and establish our professional reputation.

6 Defining Data Science – Lesson Summary

6.1 Overview

This topic reviews the key concepts of defining data science and brings together the main ideas covered throughout the lesson. We will recap:

- What data science is
- The role of a data scientist in an organization
- The skills required to become a successful data scientist
- Expert advice on developing these skills
- The future of the data science profession

6.2 What Is Data Science?

Simply put, **data science is the study of data**. It uses data to help us understand the world around us. Some consider data science to be the **art of uncovering insights and trends hidden within data**.

Although data analysis is not new, the availability of vast amounts of data and enhanced computing power has enabled us to generate new insights and knowledge through analysis. With modern computing power, we can process large volumes of information and reveal valuable insights.

Like a detective uncovering secrets, **data scientists translate data into stories that generate meaningful insights**. These insights support strategic decision-making for companies and institutions.

Similar to biological and physical sciences, **data science works with both structured and unstructured data**.

6.3 The Data Science Process

The process of deriving insights from data includes:

- Clarifying the problem
- Collecting data
- Analyzing data
- Recognizing patterns
- Storytelling
- Data visualization

Each step contributes to transforming raw data into valuable information that supports informed decision-making.

6.4 Essential Qualities of a Data Scientist

According to **Professor Murtaza Haider** from the **Ted Rogers School of Management**, three qualities are essential for becoming a successful data scientist:

- **Curiosity** helps us explore data and ask meaningful questions.
- **Argumentation** (the ability to build logical and evidence-based explanations) helps us clearly communicate findings and encourage others to make informed decisions based on new information.
- **Judgment** helps us choose the right direction and make sound decisions throughout the data science process.

6.5 Skills Required for a Data Scientist

Skilled data scientists are more than statisticians or computer experts. Organizations seek versatile professionals who:

- Possess strong knowledge in a particular domain.
- Have experience in programming.
- Can analyze data effectively.
- Communicate findings clearly.
- Are comfortable with mathematics.
- Are naturally curious.
- Can tell compelling stories using data.

Data scientists come from diverse backgrounds, including:

- Economics
- Engineering
- Medicine
- Other academic and professional fields

Once we understand our strengths and interests, we should focus on mastering data analysis within our chosen domain and select tools that best fit our industry. As our expertise grows, we can apply data science techniques to solve real-world problems by interpreting data and uncovering meaningful patterns.

6.6 The Future of Data Science

The role of data scientists will continue to evolve as technology advances and data-related careers develop.

To ensure employees remain qualified, organizations are increasingly expected to require professional certifications.

Regardless of technological changes, successful data scientists will always need to:

- Think logically.
- Apply algorithms effectively.

- Follow a methodical approach.
- Gather data accurately.
- Carefully analyze the models being used.
- Focus on achieving successful business outcomes.

6.7 Key Takeaways

- Data science is the study of data used to understand the world and uncover hidden insights.
- Advances in computing power and data availability have significantly expanded the capabilities of data science.
- Data scientists transform raw data into meaningful stories that support strategic decision-making.
- The data science workflow includes problem clarification, data collection, analysis, pattern recognition, storytelling, and visualization.
- **Curiosity, argumentation, and judgment are fundamental qualities of successful data scientists.**
- Organizations value professionals who combine domain expertise, programming skills, analytical thinking, communication abilities, and storytelling.
- Data scientists come from a wide range of educational and professional backgrounds.
- Building expertise in a chosen domain and using appropriate tools enables us to solve real-world problems effectively.
- **As the field continues to evolve, certifications, logical thinking, algorithmic problem-solving, accurate data collection, and careful model analysis will remain essential for delivering successful business results.**

7 Defining Data Science Lesson Glossary

This alphabetically organized glossary contains many of the important terms covered in these topics. We can use these terms as a quick reference while learning Data Science, working in the industry, participating in user groups, or pursuing additional certificate programs.

Term	Definition	Topic Where the Term is Introduced
Algorithms	A set of step-by-step instructions used to solve a problem or complete a task.	What is Data Science?
Model	A representation of the relationships and patterns found in data to make predictions or analyze complex systems while retaining the essential elements needed for analysis.	What is Data Science?
Outliers	Data point(s) that occur significantly outside the range of most other data in a dataset, potentially indicating anomalies (unusual observations), errors, or unique phenomena that could impact statistical analysis or modeling.	What is Data Science?
Quantitative Analysis	A systematic approach that uses mathematical and statistical analysis to interpret numerical data.	Many Paths to Data Science

Structured Data	Data that is organized and formatted into a predictable schema (a predefined structure), typically in related tables consisting of rows and columns.	What is Data Science?
Unstructured Data	Data that lacks a predefined data model or organization, making it more difficult to analyze using traditional methods. This type of data commonly includes text, images, videos, audio, and other content that does not fit neatly into rows and columns like structured data.	What is Data Science?

8 Real-World Applications of Artificial Intelligence and Data Analytics

8.1 Recommendation Engine

We can build highly effective recommendation engines without making them unnecessarily complex. One successful recommendation engine was developed as part of a large organization by working with different engineering teams and considering various aspects of the problem.

The final solution was intentionally kept simple so that it was easy to understand at every level, from executives to engineers and developers. Despite its simplicity, it was just as efficient as a much more complex solution that would have required significantly more time and effort to develop.

8.2 Predicting Algae Blooms Using Artificial Neural Networks

At a university, we faced the problem of predicting algae blooms. These algae blooms could increase the toxicity of water, creating challenges for water treatment companies.

Since the problem could not be solved effectively using only a chemical engineering background, artificial neural networks (ANNs) were used to predict when these algae blooms would reoccur. This allowed water treatment companies to prepare in advance and better manage the problem.

8.3 Case Study: Analyzing Public Transit Complaints

8.3.1 Problem Statement

The public transit system in Toronto is operated by the Toronto Transit Commission (TTC), one of the largest transit authorities in North America.

The TTC requested assistance in analyzing its complaints data to identify patterns and gain useful insights.

The organization had approximately **half a million complaints**, making the analysis a large-scale data analytics problem.

8.3.2 Understanding the Dataset

The dataset consisted of both structured and unstructured data.

Structured (Tabular) Data

- Date and time when the complaint was received
- Person who received the complaint
- Type of complaint
- Complaint resolution status
- Person or department responsible

Unstructured Data

- E-mail conversations
- Fax communications

Analyzing more than half a million e-mail exchanges made the problem significantly more challenging.

8.3.3 Initial Analysis

The primary objective was to answer the following questions:

- Why do people complain?
- Is there a recurring pattern?
- Are there specific days when complaints increase?

The data was analyzed from multiple perspectives and in different formats. However, no obvious explanation (impetus — *the driving reason or cause behind an event*) could be found for why complaint volumes were higher on some days and lower on others.

This investigation continued for approximately one month without producing a meaningful pattern.

8.3.4 The Key Observation

One day, while getting off a bus in Toronto, an unexpected event provided an important insight.

Without noticing the ground, a puddle of water was stepped into, leaving one foot completely wet. This happened because it had rained unexpectedly, and there had been no preparation for the weather.

This experience led to an important question:

Could there be a relationship between extreme weather conditions and the number of complaints received by the TTC?

8.3.5 Integrating External Data

To test this hypothesis, weather data was collected from the **Environment Canada** website, including:

- Rainfall and precipitation
- Wind conditions
- Temperature changes
- Other weather information

The complaints data was then compared with the weather data.

8.3.6 Findings

A strong relationship was discovered between severe weather conditions and complaint volumes.

The **10 days with the highest number of complaints** all coincided with unfavorable weather conditions, including:

- Unexpected rainfall
- Extreme drops in temperature
- Heavy snowfall
- Very windy conditions

This demonstrated that external environmental factors had a significant influence on customer complaints.

8.3.7 Final Conclusion

The findings were presented to the TTC executives.

The conclusion contained both good news and bad news:

Good News

We identified the reason why complaints increased dramatically on certain days.

Bad News

The primary cause was severe weather conditions, which the organization could not control.

8.4 Key Takeaways

- Simple AI solutions can perform as effectively as highly complex ones.
- Artificial neural networks can solve real-world prediction problems such as forecasting algae blooms.
- Combining structured and unstructured data enables deeper analysis.
- Valuable insights often emerge by integrating external data sources with existing datasets.
- Weather data can explain patterns that are impossible to identify by analyzing complaint data alone.
- Real-world observations can inspire hypotheses that lead to meaningful data-driven discoveries.

9 Norman White: Teaching Data Science with Python and Jupyter Notebooks

9.1 Introduction

Norman White is a Clinical Faculty Member in the **Information, Operations and Management Science (IOMS) Department** at the Stern School of Business. He has been part of Stern since graduating from college and has spent most of his career there.

He describes himself as a technology enthusiast who enjoys exploring and experimenting with technology in his spare time.

Currently, he serves as the **Faculty Director of the Stern Center for Research Computing**, where the center operates a **private cloud** (a dedicated computing environment used exclusively within an organization) that supports a wide variety of research systems.

The center provides specialized hardware and software for faculty members and PhD students. When researchers require customized computing environments, virtual machines are created, configured, and supported according to their needs. Nearly all data scientists and many PhD students at Stern use these research computing facilities.

9.2 Educational Background and Career Journey

Norman White earned his undergraduate degree in **Applied Physics**. During his undergraduate studies, he also completed several economics courses, which eventually led him to pursue business school.

This was during the early years of computing, and he developed a strong interest in computers. After joining Stern (then known as the NYU Business School downtown), he decided to focus on learning two important skills:

- Programming
- Touch typing

Although he mentions that he never truly mastered touch typing, he successfully became a **self-taught programmer**.

Later, he completed several courses at IBM while serving as the director of the computer center and simultaneously pursuing a **PhD in Economics and Statistics** at Stern.

In 1973, the school established the **Computer Applications and Information Systems Department**, and Norman White became one of its first faculty members, where he has continued teaching ever since.

9.3 Teaching Schedule and Course Structure

A typical Monday includes two classes:

- **2:00 PM:** Dealing with Data
- **6:00 PM:** Design and Development of Web-Based Systems

The "Dealing with Data" course is built around **Python notebooks** and gradually introduces students to modern data science tools and technologies.

The course covers:

- Unix and Linux fundamentals
- Python programming
- Regular expressions (patterns used for searching and manipulating text)
- Relational databases
- Python Pandas (a Python library for data analysis and manipulation, similar in purpose to R for statistical computing)
- Mathematical and statistical calculations using Python
- Big Data concepts

Big Data is one of the major topics emphasized throughout the course.

9.4 Course Assignments

Throughout the semester, students complete:

- Weekly homework assignments
- Team-based collaboration
- A comprehensive final project

The final projects encourage students to solve practical problems and often result in creative and impressive solutions.

9.5 Teaching with Jupyter Notebooks

The entire course is taught using **Jupyter Notebooks**.

To ensure every student has the same computing environment, each student is provided with a **virtual machine** on **Amazon Web Services (AWS)**.

Each virtual machine is preconfigured with:

- A standard system image
- Required software packages
- Course materials
- Jupyter Notebooks
- Commands to download updates or install the latest software

This standardized environment ensures that everyone works under identical conditions regardless of:

- Whether we use macOS or Windows

- The age or specifications of our computer
- Differences in local software installations

As a result, everyone in the class can complete the same exercises and projects without compatibility issues.

9.6 Key Technologies and Concepts Covered

- Unix
- Linux
- Python
- Regular Expressions
- Relational Databases
- Python Pandas
- Mathematical and Statistical Computing
- Big Data
- Jupyter Notebooks
- Virtual Machines
- Amazon Web Services (AWS)
- Cloud Computing

10 Understanding Common Data File Formats

As data professionals, we work with a variety of data file types and formats. It is important to understand the underlying structure of these file formats, along with their benefits and limitations. This understanding helps us make the right decisions when selecting the most appropriate file format based on our data and performance requirements.

In this topic, we will cover the following standard file formats:

- Delimited Text File Formats
- Microsoft Excel Open XML Spreadsheet (XLSX)
- Extensible Markup Language (XML)
- Portable Document Format (PDF)
- JavaScript Object Notation (JSON)

10.1 Delimited Text File Formats

Delimited text files are text files used to store data as plain text, where each line (or row) contains values separated by a **delimiter** (a sequence of one or more characters used to specify the boundary between independent values).

10.1.1 Common Delimiters

The following characters are commonly used as delimiters:

- Comma (,)
- Tab

- Colon (:)
- Vertical Bar (|)
- Space

The most commonly used delimited text file formats are:

- **CSV (Comma-Separated Values):** Uses a comma (,) as the delimiter.
- **TSV (Tab-Separated Values):** Uses a tab as the delimiter.

When literal commas are present within text data and therefore cannot be used as delimiters, **TSV** serves as an alternative to the CSV format because tab characters are infrequent in running text.

10.1.2 Structure of Delimited Text Files

- Each row (horizontal line) represents a **record**.
- The **first row** acts as the **column header**.
- Each column can contain a different data type, such as:
 - Date
 - String (text)
 - Integer (whole number)

10.1.3 Advantages of Delimited Text Files

- Allow field values of any length.
- Considered a standard format for representing straightforward information.
- Can be processed by almost all existing applications.
- Delimiters also represent one of the various methods used to specify boundaries within a data stream.

10.2 Microsoft Excel Open XML Spreadsheet (XLSX)

Microsoft Excel Open XML Spreadsheet (**XLSX**) is a Microsoft Excel file format that belongs to the spreadsheet file format category.

It is an **XML-based** file format created by Microsoft.

10.2.1 Structure of an XLSX File

- An **.XLSX** file is also known as a **workbook**.
- A workbook can contain multiple **worksheets**.
- Each worksheet is organized into:
 - Rows
 - Columns
- The intersection of a row and a column is called a **cell**.
- Each cell contains data.

10.2.2 Advantages of XLSX

- Uses an **open file format**, making it accessible to most applications.
- Supports all functions available in Microsoft Excel.
- Considered one of the more secure file formats because it **cannot save malicious code**.

10.3 Extensible Markup Language (XML)

Extensible Markup Language (XML) is a markup language with a set of rules for encoding data.

The XML file format is readable by both humans and machines.

It is a self-descriptive language designed for transmitting information over the internet.

10.3.1 XML Characteristics

- Similar to HTML in some respects but has important differences.
- Unlike HTML, **XML does not use predefined tags**.
- Platform-independent.
- Programming language-independent.
- Simplifies data sharing between different systems.

10.4 Portable Document Format (PDF)

Portable Document Format (PDF) is a file format developed by Adobe to present documents independently of application software, hardware, and operating systems.

This ensures that a PDF document appears the same on any device.

10.4.1 Common Uses of PDF

- Legal documents
- Financial documents
- Forms that allow users to fill in data

10.5 JavaScript Object Notation (JSON)

JavaScript Object Notation (JSON) is a text-based open standard designed for transmitting structured data over the web.

It is a language-independent data format that can be read using virtually any programming language.

10.5.1 Advantages of JSON

- Easy to use.
- Compatible with a wide range of web browsers.

- Suitable for sharing data of any size and type.
- Can represent not only text but also multimedia data, including audio and video.
- Widely used by APIs and Web Services for returning data.

10.6 Summary

In this topic, we explored several widely used file and data formats, including:

- Delimited Text Files (CSV and TSV)
- Microsoft Excel Open XML Spreadsheet (XLSX)
- Extensible Markup Language (XML)
- Portable Document Format (PDF)
- JavaScript Object Notation (JSON)

Understanding the structure, advantages, and common use cases of these file formats enables us to choose the most appropriate format for storing, exchanging, and processing data efficiently.

11 Regression, Data Visualization, Neural Networks, and Types of Data

11.1 Key Perspectives from Data Science Professionals

Different data science professionals have different areas of interest, each highlighting the importance of specific concepts within the field.

- **Regression** is often considered one of the first concepts that helps in understanding data more effectively.
- **Data Visualization** is regarded as a key element for communicating insights to people who may not have a strong understanding of data science. Effective visualizations make complex information easier to understand.
- **Artificial Neural Networks (ANNs)** are inspired by the biological behavior of the human brain. By attempting to mimic how the brain works, we can develop algorithms capable of solving complex real-world problems.
- **Data Visualization with R** is a popular approach for creating meaningful and visually appealing data representations.
- **Nearest Neighbor Algorithm** is appreciated for its simplicity and effectiveness. Despite being one of the simplest machine learning algorithms, it often produces excellent results and can outperform more complex algorithms that are prone to overfitting.

11.2 Structured Data vs. Unstructured Data

Understanding the difference between structured and unstructured data is fundamental in data science.

11.2.1 Structured Data

Structured data refers to data that is organized into a tabular format.

Characteristics include:

- Stored in **rows and columns**.
- Similar to data commonly found in **Microsoft Excel spreadsheets**.
- Easy to organize, search, and analyze using traditional data analysis techniques.
- Suitable for statistical analysis and machine learning models without extensive preprocessing.

11.2.2 Unstructured Data

Unstructured data refers to data that does not follow a predefined tabular structure.

Examples include:

- Text documents
- Web pages
- Blog posts
- Images
- Audio recordings
- Videos

Characteristics include:

- Does not exist in rows and columns.
- Mostly originates from websites and other digital sources.
- Requires more sophisticated algorithms to extract meaningful information.
- Often needs significant preprocessing to convert it into structured data before analysis.

In many real-world data science projects, a substantial amount of time and effort is spent transforming unstructured data into a structured format for effective analysis.

Summary:

Structured Data	Unstructured Data
Organized into rows and columns	No predefined structure
Similar to Microsoft Excel tables	Includes text, images, audio, video, and web pages
Easy to analyze	Requires preprocessing before analysis
Ready for traditional analytics	Often requires advanced algorithms for information extraction

11.3 Understanding Regression Using a Taxi Fare Example

Regression is often explained using mathematical models and statistical distributions, making it difficult for beginners to understand. A much simpler way to understand regression is by using the example of a taxi fare.

Imagine we take a taxi ride.

When we first enter the taxi, the meter already shows a fixed starting fare (for example, **\$2.50**). This amount must be paid regardless of whether the taxi moves.

This starting amount is known as the **constant** (also called the **intercept**).

After the journey begins:

- As the **distance** traveled increases, the fare increases.
- If the taxi is stuck in **traffic**, the fare increases over **time**.
- Both **time** and **distance** contribute to the total fare.
- In addition to these changing costs, there is always the initial fixed base fare.

Regression helps us discover these relationships.

Specifically, regression estimates:

- The **base fare (constant/intercept)** that exists before any distance is traveled.
- The relationship between **distance** and **fare**.
- The relationship between **time** and **fare**.

Suppose we only know:

- How far people traveled.
- How long their trips lasted.
- The total amount they paid.

Regression can analyze this historical data to determine:

- The fixed starting fare (constant/intercept).
- How much the fare increases with additional distance.
- How much the fare increases with additional time.

In other words, regression identifies the mathematical relationship between the independent variables (**time** and **distance**) and the dependent variable (**fare**), allowing us to predict future fares and understand how each factor contributes to the total cost.

12 Why Defining Data Science Is Difficult

Course Textbook: *Getting Started with Data Science*

Publisher: IBM Press, 1st Edition (December 13, 2015), Print

Author: Murtaza Haider

Prescribed Reading: Chapter 1 (Pages 12–15)

12.1 Introduction

Now that we know what is covered in the book, it is time to establish some definitions. Despite their widespread use, there is still no universal agreement on the meanings of **Big Data** and **Data Science**. Likewise, the question "**Who is a Data Scientist?**" remains an active topic of discussion and debate. Some individuals defend narrow definitions to protect their own academic disciplines or professional domains.

In this topic, we explore why restricting the definitions of **Big Data** or **Data Science** based on specific tools, disciplines, or data sizes can exclude many professionals who actively contribute to the field.

Simon Rogers (2012), writing for **The Guardian**, stated:

"Everybody loves a data scientist."

He also traced the growing popularity of data science to a statement by Google's **Hal Varian**, who predicted that **statisticians would be the "sexy job" of the next decade**. Although Hal Varian referred to statisticians, many believe he was actually describing what we now call **data scientists**.

This raises several important questions:

- What is Data Science?
- How does it differ from Statistics?
- What makes someone a Data Scientist?

In the era of Big Data, even a seemingly simple question such as "**What is Data Science?**" can produce many different answers. In some cases, these differing opinions become highly controversial.

12.2 What Makes Someone a Data Scientist?

The author defines a **Data Scientist** as:

Someone who finds solutions to problems by analyzing **Big or Small Data** using appropriate tools and then communicates those findings through compelling stories to relevant stakeholders.

According to this definition:

- The size of the data does **not** determine whether someone is a Data Scientist.
- There is **no minimum data threshold** required.
- Data Science is **not limited to Machine Learning** or any specific analytical tool.
- A person can be considered a Data Scientist if they possess:
 - A curious mind.
 - Strong analytical skills.

- The ability to communicate findings effectively.

This definition emphasizes **problem-solving, analytics, and storytelling**, rather than specific technologies or datasets.

12.3 Data Science as Practice, Not a Narrow Discipline

The author argues that:

Data Science is simply what Data Scientists do.

To explain this perspective, the author shares a personal experience from studying Engineering at the **University of Toronto**.

While pursuing graduate research, the author's work focused on:

- Forecasting housing prices.
- Predicting what homebuilders build.
- Predicting when they build.
- Predicting where new housing developments occur.

Meanwhile, other researchers in Civil Engineering were designing:

- Buildings
- Bridges
- Tunnels
- Stable slopes

Because the author's research differed from traditional engineering work, many questioned whether it was truly engineering.

When these concerns were shared with doctoral supervisor **Professor Eric Miller**, he replied:

"Engineering is what engineers do."

Over the following 17 years, the author realized the wisdom behind this statement.

An engineer can perform many different roles, including:

- Designing tunnels.
- Writing software.
- Designing smartphone components.
- Developing aircraft systems.
- Leading economic policy.

The profession is not defined by a single task but by the ability to solve problems.

12.3.1 Example: Professor Raghuram Rajan

The author highlights **Professor Raghuram Rajan** as an example.

His career includes:

- Bachelor's degree in Electrical Engineering from the **Indian Institute of Technology (IIT)**.
- Graduate studies in Economics.
- University Professor.
- Chief Economist of the **International Monetary Fund (IMF)**.
- 23rd Governor of the **Reserve Bank of India**.

Although Professor Rajan became a renowned economist, the author argues that his engineering education contributed significantly to his analytical and problem-solving abilities.

The author extends this idea to other world leaders with engineering backgrounds, including:

- **Xi Jinping**, President of the People's Republic of China.
- **Alexis Tsipras**, former Prime Minister of Greece.

These individuals may not design physical infrastructure anymore, but they continue applying engineering principles to improve economies and societies.

The author's conclusion is straightforward:

Data Science is what Data Scientists do.

12.4 The Limits of Tool- and Size-Based Definitions

Many experts define Data Science differently.

For example, during a panel discussion organized by **BigDataUniversity.com** in Toronto (September 2015), one co-panelist argued that:

Data Science is Machine Learning.

Under this definition, anyone not using Machine Learning would not qualify as a Data Scientist, regardless of the impact of their work.

12.4.1 Dr. Vincent Granville's Perspective

Dr. Vincent Granville (2014), author of *Developing Analytic Talent*, proposes several requirements for becoming a Data Scientist.

According to his description, a modern Data Science professor:

- Is a non-tenured instructor.
- Works at a non-traditional university.

- Publishes research through online blogs.
- Avoids spending time writing research grants.
- Works from home.
- Earns more than many traditional tenured professors.

The author notes that many academic Data Scientists would disagree with this characterization.

Dr. Granville also defines a Data Scientist as someone who:

- Can process a **50-million-row dataset** within a few hours.
- Distrusts traditional statistical models.

However, he simultaneously lists the following as essential foundations for Data Science:

- Algebra
- Calculus
- Probability
- Statistics

This creates inconsistencies because Data Science is separated from Statistics while still relying heavily on statistical knowledge.

12.4.2 Problems with Size-Based Definitions

Some people believe that Big Data is determined solely by:

- Data size.
- Number of observations.
- Use of technologies such as **Hadoop**.

The author argues that these definitions are problematic because technology continues to evolve.

For example:

- Standard desktop computers can now process extremely large datasets.
- **Stata** (statistical software) announced support for processing approximately **2 billion to 24.4 billion rows** under certain limitations.

If processing billions of rows qualifies as Big Data, then modern desktop software challenges the idea that only Hadoop belongs in the "Big Data club."

The key takeaway is that arbitrary thresholds based on:

- Data size,
- Software,
- Platforms, or
- Analytical methods

are likely to become outdated as technology advances.

Instead, Data Science should be viewed as a discipline that focuses on:

- Data-centric problem solving.
- Analytical thinking.
- Effective storytelling.
- Platform independence.
- Discipline independence.
- Freedom from arbitrary data-size requirements.

12.5 Curiosity, Communication, and the Data Scientist Mindset

Given the ongoing debate, the author refers to **Dr. DJ Patil**, the first Chief Data Scientist of the United States.

According to Dr. Patil (quoted in **The Guardian**, 2012):

A Data Scientist possesses a unique blend of skills that enables them to unlock insights from data and communicate those insights through compelling stories.

This definition is valuable because it:

- Welcomes people from diverse educational backgrounds.
- Does not restrict Data Science to specific software or tools.
- Does not impose minimum data-size requirements.

12.5.1 Curiosity: A Key Characteristic

Beyond technical skills, the author identifies **curiosity** as one of the most important qualities of a successful Data Scientist.

A Data Scientist should:

- Be naturally curious.
- Explore ideas and hypotheses thoroughly.
- Invest significant effort in discovering hidden insights.

The author compares this quality to journalism.

Editors often describe outstanding journalists as having:

"A nose for news." (The natural ability to recognize important stories before others.)

Similarly, successful Data Scientists possess a natural instinct for identifying valuable patterns, interesting questions, and meaningful insights within data.

12.5.2 Rachel Schutt's Definition

Rachel Schutt, Chief Data Scientist at **News Corp** and instructor at **Columbia University**, is also the author of *Doing Data Science*.

In an interview with **The New York Times** (Miller, 2013), she described a Data Scientist as someone who is:

- Part Computer Scientist.
- Part Software Engineer.
- Part Statistician.

However, she emphasized that the **best** Data Scientists possess an additional quality:

They are deeply curious individuals who ask thoughtful questions, feel comfortable working with unstructured situations, and excel at discovering structure within complexity.

12.6 Key Takeaways

- There is no universally accepted definition of **Data Science** or **Data Scientist**.
- Data Science should not be limited by:
 - Data size.
 - Specific software.
 - Machine Learning.
 - Particular disciplines.
- A Data Scientist solves problems using data and effectively communicates insights to stakeholders.
- Storytelling is as important as analytical ability.
- Curiosity is a defining characteristic of outstanding Data Scientists.
- Strong foundations in mathematics, statistics, and analytics remain valuable regardless of the tools being used.
- Data Science is best understood as a **practice focused on solving real-world problems through data**, rather than a field defined by specific technologies or arbitrary thresholds.

12.7 References

Book

- Haider, M. *Getting Started with Data Science*. IBM Press, 1st Edition, December 13, 2015.

Articles and Sources Mentioned

- Simon Rogers (2012), **The Guardian**.
- Hal Varian (Google).
- Dr. Vincent Granville (2014), *Developing Analytic Talent*.
- Dr. DJ Patil, Chief Data Scientist of the United States.
- Rachel Schutt, *Doing Data Science*.
- Miller (2013), **The New York Times**.

Links Mentioned

- BigDataUniversity.com

13 A Day in the Life of a Data Scientist

Welcome to a realm where data is not just numbers. It is the gateway to innovation, discovery, and the endless possibilities that lie ahead. In this topic, we explore what data scientists do and how they use data to solve real-world problems, based on the concepts discussed throughout this lesson.

13.1 What Data Scientists Do

Data scientists investigate complex problems and uncover meaningful explanations through data analysis.

Some real-world examples include:

- **Public Transit Analysis:** Dr. Murtaza Haider investigated why nearly half a million customers complained about public transit in Toronto. After extensive analysis, he discovered a relationship between unexpected bad weather events and the number of complaints received on a particular day.
- **Environmental Protection:** Data scientists can predict algae blooms to help prevent water toxicity. By harnessing the power of artificial neural networks (computer systems inspired by the human brain that learn patterns from data), they help water treatment companies protect ecosystems.
- **Recommendation Systems:** Norman White, a Clinical Faculty Member at the Stern School of Business, shared his journey of developing a recommendation engine (a system that suggests relevant items or decisions based on data), simplifying complex problems across multiple departments.

These examples demonstrate that the essence of data science is solving real-world problems with innovative, data-driven solutions.

13.2 Educational Background for Data Scientists

Education serves as a cornerstone for building the skills required to become a successful data scientist.

Dr. White's classroom emphasizes practical learning through:

- Python notebooks
- Unix
- Linux
- Relational databases
- Pandas

According to Dr. Vincent Granville, a strong educational background in the following subjects is essential:

- Algebra
- Calculus
- Probability

- Statistics

Dr. Granville also distinguishes between a statistician and a data scientist.

A data scientist uses statistics but is not limited to statistical analysis alone. Instead, data scientists combine statistics with programming, machine learning, and domain expertise to solve practical problems.

13.3 Statistical Models and Machine Learning

Data scientists frequently use statistical models to analyze relationships within data.

One common example is:

- **Statistical Regression** (a technique used to estimate the relationship between two variables), such as understanding the relationship between:
 - Distance driven
 - Amount of fuel consumed

In addition to statistical methods, data scientists also use machine learning algorithms such as:

- Nearest Neighbor (a machine learning algorithm that predicts outcomes based on the most similar data points)

These algorithms are commonly used to process what is often referred to as **Big Data**.

However, the term **Big Data** should be used with caution because technological innovations continually redefine what is considered "big."

Technologies such as Hadoop and other software advancements have significantly expanded our ability to process massive datasets.

13.4 Different Perspectives on Data Science

Dr. Patel provides a broader definition of a data scientist.

According to his perspective:

- Data scientists are not limited to working with datasets of a particular size.
- They are not restricted to using specific software tools.
- Individuals from diverse academic backgrounds and training can become successful data scientists.

A data scientist not only discovers valuable insights from data but also communicates those insights effectively to stakeholders.

It is this combination of technical expertise and communication skills that distinguishes data scientists from many other professionals.

13.5 Types of Data Used by Data Scientists

Data scientists work with information collected from many different sources, including:

- Video
- Audio
- Unformatted text

Text-based data may be:

- **Structured** (organized into rows and columns, such as database tables)
- **Unstructured** (not organized in a predefined format, such as emails or system logs)

13.6 Common Data Formats

Data scientists regularly work with data stored in various formats, including:

- Delimited text files
- Spreadsheets
- XML (Extensible Markup Language)
- PDF documents
- JSON (JavaScript Object Notation)

13.7 Characteristics of a Successful Data Scientist

According to Rachel Schutt, Chief Data Scientist at News Corp, the defining characteristic of an exceptional data scientist is **curiosity**.

A successful data scientist combines the strengths of a:

- Computer scientist
- Software engineer
- Statistician

Their ability to transform unstructured information into structured insights is what makes them highly effective.

13.8 Key Takeaways

Data science is much more than a profession—it is a journey of exploration, innovation, and storytelling through data.

A successful data scientist:

- Solves real-world problems using data.
- Combines mathematics, statistics, programming, and machine learning.
- Works with structured and unstructured data from multiple sources.
- Uses a wide range of tools and technologies.

- Communicates insights clearly to stakeholders.
- Remains curious, analytical, and committed to continuous learning.

As we reflect on the role of a data scientist, we recognize that the world of data is vast. Armed with technical skills, curiosity, and determination, we can navigate this world to uncover meaningful insights and drive impactful decisions.

14 Summary: What Do Data Scientists Do?

At this point in the course, we know:

- **Data science** is the study of large quantities of data, which can reveal insights that help organizations make strategic choices.
- There are many paths to a career in data science; most, but not all, involve mathematics, programming, and curiosity about data.
- New data scientists need to be **curious**, **judgmental** (able to evaluate information critically), and **argumentative** (able to support conclusions with logical reasoning and evidence).
- Knowledgeable data scientists are in high demand. Jobs in data science pay high salaries for skilled professionals.
- The typical workday for a Data Scientist varies depending on the type of project being worked on.
- Many algorithms are used to uncover insights from data.

14.1 Key Data Science Terms

Some key data science-related terms learned in this lesson include:

- Outliers
- Model
- Algorithms
- JSON (JavaScript Object Notation)
- XML (eXtensible Markup Language)
- CSV (Comma-Separated Values)
- Regression

15 Glossary: What Do Data Scientists Do?

This glossary contains many of the important terms covered in these topics.

15.1 Data Science Glossary

Term	Definition	Topic Where the Term is Introduced
Comma-separated values (CSV) / Tab-separated values (TSV)	Commonly used formats for storing tabular data as plain text, where either a comma or a tab separates each value.	Working on Different File Formats

Data file types	A computer file configuration designed to store data in a specific way.	Working on Different File Formats
Data format	The way data is encoded so it can be stored within a specific data file type.	Working on Different File Formats
Data visualization	A visual representation of data, such as graphs or charts, that makes it easier to understand patterns, trends, and insights.	Data Science Topics and Algorithms
Delimited text file	A plain text file in which a specific character (such as a comma or tab) separates the data values.	Working on Different File Formats
Extensible Markup Language (XML)	A language designed to structure, store, and enable data exchange between different technologies.	Working on Different File Formats
Hadoop	An open-source framework designed to store and process large datasets across clusters of computers.	What Makes Someone a Data Scientist
JavaScript Object Notation (JSON)	A data format compatible with various programming languages that enables two applications to exchange structured data.	Working on Different File Formats
Jupyter notebooks	A computational environment that allows us to create and share documents containing code, equations, visualizations, and explanatory text. Also referred to as Python notebooks .	Data Science Skills & Big Data
Nearest neighbor	A machine learning algorithm that predicts a target variable based on its similarity to other values in the dataset.	Working on Different File Formats
Neural networks	A computational model used in deep learning that mimics the structure and functioning of the human brain's neural pathways. It accepts input, processes it using previous learning, and produces an output.	A Day in the Life of a Data Scientist
Pandas	An open-source Python library that provides tools for working with structured data and is widely used for data manipulation and analysis.	Data Science Skills & Big Data
Python notebooks	Also known as Jupyter notebooks , this computational environment allows us to create and share documents containing code, equations, visualizations, and explanatory text.	Data Science Skills & Big Data

R	An open-source programming language used for statistical computing, data analysis, and data visualization.	Data Science Skills & Big Data
Recommendation engine	A computer program that analyzes user input, such as behaviors or preferences, and provides personalized recommendations based on that analysis.	A Day in the Life of a Data Scientist
Regression	A statistical model that shows the relationship between one or more predictor variables and a response variable.	Data Science Topics and Algorithms
Tabular data	Data that is organized into rows and columns.	A Day in the Life of a Data Scientist
XLSX	The Microsoft Excel spreadsheet file format.	Working on Different File Formats

Module 2: Data Science Topics

16 Digital Transformation and the Role of Big Data

Digital Transformation affects business operations by updating existing processes and operations and creating new ones to harness the benefits of new technologies.

This digital change integrates digital technology into all areas of an organization, resulting in fundamental changes to how it operates and delivers value to customers.

It is an organizational and cultural change driven by **Data Science**, and especially **Big Data**.

The availability of vast amounts of data, and the competitive advantage that analyzing it brings, has triggered digital transformations throughout many industries.

Examples include:

- **Netflix** moved from being a postal DVD lending system to one of the world's foremost video streaming providers.
- The **Houston Rockets** NBA team used data gathered by overhead cameras to analyze the most productive plays.
- **Lufthansa** analyzed customer data to improve its service.

Organizations all around us are changing to their very core.

16.1 How Big Data Triggers Digital Transformation

Digital Transformation is not simply duplicating existing processes in digital form. The in-depth analysis of how a business operates helps organizations discover how to improve their processes and operations and harness the benefits of integrating Data Science into their workflows.

The availability of Big Data enables organizations to:

- Improve existing business processes.
- Create new and more efficient operational strategies.
- Make better data-driven decisions.
- Deliver greater value to customers.
- Gain a competitive advantage through data analysis.

16.2 Case Study: Houston Rockets and Big Data

In **2018**, the **Houston Rockets**, a **National Basketball Association (NBA)** team, raised their game using Big Data.

The Rockets were one of four NBA teams to install a **video tracking system** that mined raw data from games.

They analyzed video tracking data to investigate which plays provided the best opportunities for high scores and discovered something surprising.

Data analysis revealed that the shots providing the best opportunities for high scores were:

- **Two-point dunks** from inside the two-point zone.
- **Three-point shots** from outside the three-point line.

The analysis showed that these shots were more effective than **long-range two-point shots** from inside the three-point line.

This discovery entirely changed the way the team approached each game, increasing the number of three-point shots attempted.

During the **2017–18 NBA season**, the Rockets made more three-point shots than any other team in NBA history, which became a major reason they won more games than any of their rivals.

This example demonstrates how Big Data transformed not only one team's strategy but also influenced the overall approach to winning basketball games.

16.3 Organizational Impact of Digital Transformation

Most organizations realize that Digital Transformation requires fundamental changes to their approach toward:

- Data
- Employees
- Customers
- Organizational culture

Digital Transformation impacts every aspect of an organization, so it is handled by decision-makers at the highest levels to ensure success.

Successful Digital Transformation requires support from:

- **Chief Executive Officer (CEO)**
- **Chief Information Officer (CIO)**
- **Chief Data Officer (CDO)** (an emerging executive role responsible for managing organizational data)

In addition, support is required from executives responsible for:

- Budget decisions
- Personnel decisions
- Day-to-day operational priorities

Digital Transformation is a whole-organization process. We all must support it for it to succeed.

16.4 Key Takeaways

- Digital Transformation integrates digital technologies into every area of an organization.
- It is driven primarily by **Data Science** and **Big Data**.
- Big Data enables organizations to improve processes, operations, and customer value.
- The Houston Rockets demonstrated how data-driven insights can fundamentally transform decision-making and performance.
- Digital Transformation requires organizational, cultural, and operational changes.
- Success depends on leadership commitment, executive support, and organization-wide participation.
- Although Digital Transformation requires a new mindset, it is the path to success both now and in the future.

17 Introduction to Cloud Computing and Cloud Deployment and Service Models

17.1 Learning Objectives

After completing this topic, we will be able to:

- Describe cloud computing concepts.
- Define cloud deployment models.
- Define cloud service models.
- Identify the essential characteristics of cloud computing.

17.2 Introduction to Cloud Computing

Cloud computing, also referred to as **the cloud**, is the delivery of on-demand computing resources such as networks, servers, storage, applications, services, and data centers over the Internet on a **pay-for-use** basis.

The term **cloud computing** can also be used to describe applications and data that we access over the Internet rather than on a local computer.

17.2.1 Examples of Cloud Computing

Examples of cloud computing include:

- Using online web applications.
- Employees using secure online business applications to perform their work.
- Storing personal files on cloud-based storage platforms such as:
 - Google Drive
 - OneDrive
 - Dropbox

17.3 Benefits of Cloud Computing

Cloud computing provides several advantages:

- Instead of purchasing applications and installing them locally, we can use online versions of applications by paying a monthly subscription.
- This approach is generally more cost-effective initially.
- We can access the latest version of an application without purchasing an entirely new retail copy.
- Since applications are hosted online, we save a significant amount of local storage space.
- Cloud-based applications support real-time collaboration, allowing multiple users to work on the same files simultaneously while viewing each other's edits and updates.

17.4 Essential Characteristics of Cloud Computing

Cloud computing is composed of **five essential characteristics**, **three deployment models**, and **three service models**.

17.4.1 On-Demand Self-Service

On-demand self-service means that we can access cloud resources such as processing power, storage, and networking through a simple interface without requiring human interaction with the service provider.

17.4.2 Broad Network Access

Broad network access means that cloud resources are accessible over the network through standard devices and platforms, including:

- Mobile phones
- Tablets
- Laptops
- Workstations

17.4.3 Resource Pooling

Resource pooling enables cloud providers to achieve economies of scale (cost advantages gained by serving many customers), making cloud services more cost-efficient.

Using a **multitenant model** (a model where multiple customers securely share the same computing resources), computing resources are pooled to serve multiple consumers. Resources are dynamically assigned and reassigned according to demand without requiring customers to know their physical location.

17.4.4 Rapid Elasticity

Rapid elasticity means that we can quickly obtain additional resources when demand increases and release them when demand decreases because resources are provisioned and released dynamically.

17.4.5 Measured Service

Measured service means that we pay only for the resources we use or reserve.

- If resources are not being used, we do not pay for them.
- Resource usage is monitored, measured, and reported transparently based on consumer utilization.

17.5 Key Concept of Cloud Computing

Cloud computing is fundamentally about using technology "**as a service**", leveraging remote systems over the Internet on demand, scaling resources up or down as needed, and paying only for actual usage.

This approach has transformed how computing services are consumed by:

- Reducing costs.
- Improving operational efficiency.
- Increasing organizational agility in responding to changing market demands.

17.6 Cloud Deployment Models

Cloud deployment models define:

- Where the infrastructure resides.
- Who owns and manages the infrastructure.
- How cloud resources and services are made available to users.

There are **three cloud deployment models**.

17.6.1 Public Cloud

In the public cloud model:

- Cloud services are delivered over the public Internet.
- The infrastructure is owned and managed by the cloud provider.
- Resources are shared among multiple organizations.

17.6.2 Private Cloud

In the private cloud model:

- The cloud infrastructure is dedicated to a single organization.
- It may operate on-premises or be owned, managed, and operated by a cloud service provider.

17.6.3 Hybrid Cloud

A hybrid cloud combines both public and private cloud environments, allowing them to work together seamlessly.

17.7 Cloud Service Models

Cloud service models are based on the three layers of the computing stack:

- Infrastructure
- Platform
- Application

The three cloud service models are:

- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)

17.7.1 Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) allows us to access infrastructure and physical computing resources such as:

- Servers
- Networking
- Storage
- Data center space

With IaaS, we do not need to manage or operate the underlying infrastructure.

17.7.2 Platform as a Service (PaaS)

Platform as a Service (PaaS) provides access to a platform consisting of hardware and software tools that are typically required to develop and deploy applications over the Internet.

17.7.3 Software as a Service (SaaS)

Software as a Service (SaaS) is a software licensing and delivery model in which software and applications are centrally hosted and provided on a subscription basis.

SaaS is also commonly referred to as **on-demand software**.

17.8 Summary

In this topic, we learned that:

- Cloud computing is the delivery of on-demand computing resources over the Internet on a pay-for-use basis.
- Cloud computing consists of:
 - Five essential characteristics
 - Three deployment models
 - Three service models
- The five essential characteristics of cloud computing are:
 - On-demand self-service
 - Broad network access
 - Resource pooling
 - Rapid elasticity
 - Measured service
- The three cloud deployment models are:
 - Public Cloud
 - Private Cloud
 - Hybrid Cloud
- The three cloud service models are:
 - Infrastructure as a Service (IaaS)
 - Platform as a Service (PaaS)
 - Software as a Service (SaaS)

18 Cloud Computing for Data Science

Cloud computing is a valuable technology for data scientists because it enables us to store, process, and analyze large amounts of data without being limited by the physical capabilities of our local computers or organizational infrastructure.

18.1 What is Cloud Computing?

Cloud computing allows us to:

- Store our data and information in a centralized storage system.
- Overcome the physical limitations of our computers and local systems.
- Utilize the storage and computing power of advanced remote machines instead of relying solely on our own devices or our organization's hardware.
- Deploy powerful analytics and high-performance computing resources to process large datasets efficiently.

Instead of storing data locally, we can send it to cloud storage. Similarly, if we do not have the required computing algorithms or processing power on our own systems, we can access and run them directly in the cloud. This allows us to perform complex computations on very large datasets, even when our local computing environment is insufficient.

18.2 Benefits of Cloud Computing for Data Scientists

Cloud computing offers several important advantages.

18.2.1 Scalability and High-Performance Computing

The cloud enables us to:

- Store massive amounts of data on remote servers.
- Run advanced computing algorithms.
- Perform high-performance computing (HPC) (using powerful computing systems to process complex tasks much faster than standard computers).
- Analyze large datasets without requiring expensive local hardware.

18.2.2 Centralized Data Storage

By storing data in a centralized location, we can:

- Access our data whenever needed.
- Manage information more efficiently.
- Eliminate many of the storage limitations of personal computers.

18.2.3 Collaboration Across Teams

One of the greatest strengths of cloud computing is collaboration.

Multiple teams located in different countries—for example, Germany, India, and Ghana—can simultaneously access:

- The same datasets.
- Shared algorithms.
- Analytical tools.
- Results and outputs.

Since everything is stored in one centralized location, team members can work together efficiently regardless of their geographical location.

18.2.4 Access to Open-Source Technologies

Cloud platforms provide instant access to open-source technologies such as:

- Apache Spark

We can begin using these technologies immediately without installing or configuring them on our local machines.

18.2.5 Access to the Latest Tools and Libraries

Cloud environments provide access to:

- The latest software tools.
- Updated libraries.
- Modern development environments.

This eliminates the need to manually install, maintain, or update software.

18.2.6 Accessibility

Cloud services are available:

- From anywhere in the world.
- Across all time zones.
- On multiple devices, including:
 - Laptops
 - Tablets
 - Smartphones

This flexibility allows us to continue working regardless of our location or device.

18.2.7 Improved Team Collaboration

Multiple collaborators can access the same data simultaneously and work together on developing solutions. Since the data, tools, and computational resources are centrally available, collaboration becomes much easier and more efficient.

18.3 Popular Cloud Platforms

Several major technology companies provide cloud computing platforms.

- **IBM Cloud**
- **Amazon Web Services (AWS)**
- **Google Cloud Platform (GCP)**

These platforms allow us to become familiar with cloud technologies through pre-built development environments.

18.4 IBM Skills Network Labs (SN Labs)

IBM provides **Skills Network Labs (SN Labs)** to learners enrolled through IBM Developer Skills Network learning portals.

Within SN Labs, we have access to tools and services such as:

- Jupyter Notebooks
- Apache Spark clusters (groups of interconnected computers that work together to process large-scale data efficiently)

These resources enable us to:

- Create our own data science projects.
- Develop analytical solutions.
- Practice using real cloud-based data science tools.

18.5 Key Takeaway

Cloud computing significantly enhances the productivity of data scientists by providing:

- Centralized data storage.
- Virtually unlimited computing resources.
- High-performance computing capabilities.
- Easy collaboration among distributed teams.
- Instant access to modern tools and open-source technologies.
- Accessibility from anywhere using multiple devices.

With continued practice and familiarity, we can leverage cloud computing to build, analyze, and deploy data science solutions more efficiently than ever before.

19 Understanding Big Data

In today's digital world, everyone leaves a digital trace. From our travel habits to our workouts and entertainment, the increasing number of internet-connected devices that we interact with daily records vast amounts of data about us. This enormous amount of information is known as **Big Data**.

Ernst & Young defines Big Data as:

"Big Data refers to the dynamic, large and disparate volumes of data being created by people, tools, and machines. It requires new, innovative, and scalable technology to collect, host, and analytically process the vast amount of data gathered in order to derive real-time business insights that relate to consumers, risk, profit, performance, productivity management, and enhanced shareholder value."

Although there is no single universal definition of Big Data, most definitions share five common characteristics known as the **5 Vs of Big Data**:

- Velocity
- Volume
- Variety
- Veracity
- Value

19.1 The 5 Vs of Big Data

19.1.1 Velocity

Velocity is the **speed at which data accumulates**.

Data is generated continuously and at an extremely rapid pace. Near real-time or real-time streaming technologies, along with local and cloud-based systems, enable us to process information very quickly.

Example

Every 60 seconds, hours of video footage are uploaded to YouTube, continuously generating new data. We can imagine how quickly this data grows over hours, days, and years.

19.1.2 Volume

Volume refers to the **scale of data** or the increasing amount of stored data.

The growth in data volume is driven by:

- Increasing number of data sources
- Higher-resolution sensors
- Scalable infrastructure

Example

The world's population is approximately **seven billion people**, and the vast majority use digital devices such as:

- Mobile phones
- Desktop computers
- Laptop computers

- Wearable devices

These devices generate, capture, and store approximately **2.5 quintillion bytes of data every day**, which is equivalent to about **10 million Blu-ray DVDs**.

19.1.3 Variety

Variety refers to the **diversity of data**.

Data exists in different formats:

- **Structured Data** – Organized into rows and columns within relational databases.
- **Unstructured Data** – Data that is not organized in a predefined format, such as:
 - Tweets
 - Blog posts
 - Pictures
 - Numbers
 - Videos

Variety also means that data originates from numerous sources, including:

- Machines
- People
- Business processes
- Internal organizational systems
- External organizational systems

Major drivers of data variety include:

- Mobile technologies
- Social media
- Wearable technologies
- Geo technologies (location-based technologies)
- Video platforms
- Internet of Things (IoT) devices

Example

Different forms of data include:

- Text
- Images
- Videos
- Audio
- Health data collected from wearable devices
- Data generated by Internet of Things (IoT) devices

19.1.4 Veracity

Veracity refers to the **quality, origin, accuracy, and reliability of data**.

Important attributes of data veracity include:

- Consistency
- Completeness
- Integrity
- Ambiguity (uncertainty or the possibility of multiple interpretations)

Key drivers include:

- Cost
- Need for traceability (the ability to track where data originated and how it has changed)

With the enormous amount of data available today, an important question arises:

Is the information real, or is it false?

Approximately **80% of available data is unstructured**, making it essential for us to categorize, analyze, and visualize data in order to produce reliable and accurate insights.

19.1.5 Value

Value refers to our **ability and need to transform data into meaningful outcomes**.

Value is not limited to financial profit. It can also provide:

- Medical benefits
- Social benefits
- Customer satisfaction
- Employee satisfaction
- Personal satisfaction

The primary reason we invest time and resources in understanding Big Data is to derive valuable insights from it.

19.2 Big Data Analytics

Data Scientists derive valuable insights from Big Data while addressing the challenges presented by massive datasets.

The enormous scale of modern data makes it impractical to rely solely on conventional data analysis tools. Instead, distributed computing technologies enable efficient storage, processing, and analysis of large datasets.

Common Big Data technologies include:

- Apache Spark
- Hadoop
- Hadoop Ecosystem

These technologies help us:

- Extract data
- Load data
- Analyze data
- Process data across distributed computing resources

As a result, organizations gain deeper insights and discover new knowledge, allowing them to:

- Better understand their customers
- Improve decision-making
- Enhance the services they offer

19.3 Real-World Perspective

Every time we:

- Wear a smartwatch
- Unlock a smartphone
- Track a workout
- Use internet-connected devices

our data begins a journey. That data may travel through Big Data platforms around the world, where it is collected, processed, analyzed, and eventually transformed into valuable insights that can improve products, services, and our overall digital experience.

20 Data Science, Business Analytics, and Big Data

20.1 Introduction

Data science and business analytics have become some of the most in-demand fields over the past few years. The rapid growth of data, combined with new tools and analytical approaches, has transformed how organizations solve problems and make decisions. In this topic, we explore the importance of programming, computational thinking, the rise of data science, and the evolution of big data technologies.

20.2 Programming Background for Data Science

Everyone involved in data science typically has at least some programming experience. While the level of expertise varies, the ability to think computationally (solving problems using logical and algorithmic thinking) is considered the most important skill.

Examples of different backgrounds include:

- Some individuals hold a Master of Science in Computer Science and have extensive programming experience.
- Some MBA students come from technical fields and program regularly.
- Others may have taken a programming course several years ago but still possess computational thinking skills.

20.3 Growth of Data Science and Business Analytics

Over the last four or five years, data science and business analytics have become increasingly important because:

- We now have new tools and technologies.
- We have developed new analytical approaches.
- Organizations generate massive amounts of data.
- Traditional techniques can no longer efficiently store or process this growing volume of data.

As awareness has increased, companies and employers have recognized the need for professionals with data science and analytics skills across many industries.

20.4 Industry Adoption of Big Data

A major bank serves as an example of how organizations have embraced big data.

Initially:

- Only a small group within the bank experimented with a small computing cluster (a group of interconnected computers working together).

Later:

- The bank expanded to five or six major big data clusters.
- Credit card data was consolidated into these clusters.
- Various data science techniques were applied to analyze the data from multiple perspectives to generate valuable insights.

This demonstrates how organizations have significantly increased investments in data infrastructure and analytics.

20.5 Increasing Demand for Data Science Education

The popularity of data science is also reflected in education.

An undergraduate course focused on dealing with data experienced significant enrollment growth:

- One year: 28 students
- Following year: 140 students

This increase indicates growing awareness among students and parents regarding career opportunities in data science and analytics.

Parents who previously encouraged careers such as:

- Accounting

- Financial Services
- Marketing

are now also encouraging students to:

- Take more STEM (Science, Technology, Engineering, and Mathematics) classes in high school.
- Prepare for careers in data science.
- Enter fields where analytics has become increasingly important.

20.6 Definition of Big Data

The definition of big data can vary depending on who is asked.

One practical definition is:

Big data is data that has enough volume (amount of data) and velocity (speed at which data is generated and processed) that it cannot be handled using traditional database systems.

Some statisticians humorously define big data as:

- Data that cannot fit on a thumb drive.

20.7 Origins of Big Data

According to this perspective, big data began with Google.

When Larry Page and Sergey Brin developed Google's PageRank algorithm (the algorithm used to rank web pages in search results), they faced a major challenge:

- They needed to store and process the entire web.
- Existing technologies were unable to handle such enormous amounts of data.
- No suitable infrastructure existed for this task.

As a result:

- Google developed its own distributed storage and processing approach (data distributed across multiple computers working together).
- Hadoop later adopted similar concepts, making large-scale data storage and processing widely accessible.
- These technologies became the foundation for today's large big data clusters.

20.8 Big Data Analytics

Big data is no longer limited to storing large datasets.

Modern big data also focuses on analyzing enormous volumes of information using:

- Advanced analytical techniques.
- New statistical methods.
- Scalable computing technologies capable of processing extremely large datasets.

These methods enable organizations to extract meaningful insights from data that traditional systems cannot efficiently process.

20.9 Deep Learning

As data science continues to evolve, advanced topics such as **Deep Learning** (a branch of machine learning that uses multi-layer neural networks to learn complex patterns from large datasets) become an important part of big data analytics and modern artificial intelligence.

21 Big Data Architecture and the Evolution of Data Science

21.1 Traditional Data Processing vs. Big Data Processing

Traditionally, in computation and data processing, we would bring the data to the computer. We would write a program and bring the data into that program for processing.

In a **big data cluster**, Larry Page and Sergey Brin introduced a simple but powerful approach:

- The data is sliced into smaller pieces.
- Each piece is distributed across many computers.
- Each piece is replicated or triplicated (multiple copies are stored for reliability).
- Initially, the data was distributed across hundreds of computers, but today it is distributed across thousands or even tens of thousands of computers.

Instead of moving all the data to one computer, we send the same program to every computer in the cluster. Each computer processes only its assigned portion of the data and sends the results back.

The processing consists of two main stages:

- **Map (Mapper) Process:** Each computer processes its own piece of the data independently.
- **Reduce Process:** The results are sorted, combined, and redistributed to produce the final output.

Although these concepts are fairly simple, they enable us to solve many different kinds of problems involving extremely large datasets.

21.2 Advantages of Big Data Clusters

One of the major advantages of big data clusters is **linear scalability** (performance increases proportionally as more servers are added).

For example:

- If we double the number of servers, we get approximately double the processing performance.
- We can also handle approximately twice the amount of data.

This approach removed a major bottleneck for large social media companies by making it possible to process massive amounts of information efficiently.

21.3 Hadoop and the Big Data Ecosystem

Yahoo later adopted this approach and hired **Doug Cutting**, who had been working on an open-source implementation (a clone/copy) of Google's big data architecture.

This implementation became known as **Hadoop**.

If we search for **Hadoop**, we will find that it has become one of the most popular technologies in the big data ecosystem. Today, there are hundreds of thousands of companies with some form of presence in the big data industry.

21.4 Foundations of Data Science

Most components of data science have existed for many decades. The key difference today is that they have come together with modern computational capabilities.

The fundamental components of data science include:

- Probability and Statistics
- Algebra and Linear Algebra
- Programming
- Databases

These disciplines have always existed, but modern computing power now allows us to apply advanced techniques such as **Machine Learning**.

21.5 Machine Learning and Large-Scale Data Analysis

With modern computational capabilities, we can process extremely large datasets.

Instead of:

- Taking a small sample of data
- Creating a hypothesis
- Testing that hypothesis

We can now:

- Analyze very large datasets.
- Search directly for patterns.
- Generate new hypotheses based on the discovered patterns.

This represents a shift from traditional hypothesis testing toward pattern discovery.

Some traditional statisticians find this approach controversial because classical statistics expects us to develop a hypothesis independently of the data before testing it.

However, many machine learning techniques became the only practical way to analyze extremely large social media datasets.

21.6 Decision Sciences

Modern data science combines several traditional disciplines into what is often called **Decision Sciences**.

These disciplines include:

- Computer Science
- Probability
- Statistics
- Mathematics
- Operations Research
- Information Systems

Organizations with expertise across these areas were well positioned to contribute to the rapid growth of data science.

One example mentioned is the Stern School, which combines:

- A statistics department with PhD-level statisticians
- An Operations Management department
- An Information Systems department

This combination of expertise enabled the school to actively contribute to the field of Decision Sciences.

The topic also mentions that **Foster Provost** became the first director of the **NYU Center for Data Science**.

21.7 The Emergence of Data Science

Around five years before this discussion, the term **Data Science** was rarely used.

If we search Google Trends for **Data Science**, we would observe:

- Almost no usage initially.
- A sudden and dramatic increase in popularity.

A similar trend occurred earlier with the term **Big Data**, which experienced rapid growth approximately seven or eight years before.

21.8 What Is Data Science?

One of the first questions people ask is:

What exactly is Data Science?

Even faculty members and researchers continue to refine their understanding of the relationship between:

- Business Analytics
- Data Science

Although we understand the major components of data science, the field continues to evolve, grow, and change.

21.9 The Rise of Deep Learning

In recent years, **Deep Learning** has become an important part of data science.

Neural networks have existed for approximately 20 to 30 years. However, around 20 years ago, they had limited practical applications because computing power was insufficient.

Researchers, particularly at the **University of Toronto**, developed **multi-layer neural networks** (deep neural networks consisting of many hidden layers that can learn highly complex patterns from large datasets).

This technology has expanded rapidly and is now widely used by companies such as:

- Google
- Facebook
- Many other technology companies

21.10 Key Takeaways

- Traditional computing brings data to a single computer, whereas big data processing distributes data across thousands of computers.
- Big data processing relies on the **Map** and **Reduce** processes.
- Big data clusters scale linearly, allowing both processing power and storage capacity to increase proportionally with additional servers.
- Hadoop is an open-source implementation of Google's big data architecture developed with the contributions of Doug Cutting.
- Data science integrates probability, statistics, mathematics, programming, databases, and computer science.
- Machine learning enables us to analyze very large datasets and discover patterns instead of relying solely on traditional hypothesis testing.
- Decision Sciences combines multiple quantitative and computational disciplines to support data-driven decision-making.
- Data Science has experienced rapid growth over the past decade and continues to evolve.

- Deep learning and multi-layer neural networks have significantly expanded the capabilities of artificial intelligence and are now widely adopted across industry.

21.11 References

- Hadoop (mentioned in the topic): <https://hadoop.apache.org/>

22 Big Data Processing Technologies: Apache Hadoop, Apache Hive, and Apache Spark

Big Data processing technologies provide ways to work with large sets of **structured, semi-structured, and unstructured data** so that we can derive valuable insights from big data.

Some of the key Big Data technologies include **NoSQL databases, Data Lakes, Apache Hadoop, Apache Hive, and Apache Spark**. This topic focuses on the role of **Apache Hadoop, Apache Hive, and Apache Spark** in Big Data Analytics.

22.1 Apache Hadoop

Apache Hadoop is a **Java-based open-source framework** that provides **distributed storage and processing of large datasets** across clusters of computers.

- Hadoop is a collection of tools that provides distributed storage and processing of big data.
- A **node** is a single computer in a Hadoop distributed system.
- A **cluster** is a collection of multiple nodes working together.
- Hadoop can scale from a single node to any number of nodes, with each node providing local storage and computation.
- Hadoop offers a reliable, scalable, and cost-effective solution for storing data without requiring a specific data format.

22.1.1 Benefits of Using Hadoop

Using Hadoop, we can:

- Incorporate emerging data formats such as:
 - Streaming audio
 - Video
 - Social media sentiment
 - Clickstream data
 - Structured, semi-structured, and unstructured data that is not traditionally stored in a data warehouse
- Provide real-time, self-service access for all stakeholders.
- Optimize and streamline enterprise data warehouse costs by:
 - Consolidating data across the organization.
 - Moving **cold data** (data that is not accessed frequently) to a Hadoop-based system.

22.2 Hadoop Distributed File System (HDFS)

One of the four main components of Hadoop is the **Hadoop Distributed File System (HDFS)**, which is a storage system for big data that runs on multiple commodity hardware (standard, low-cost computers) connected through a network.

HDFS provides **scalable and reliable storage** by partitioning files across multiple nodes.

22.2.1 How HDFS Works

- HDFS splits large files into smaller blocks and distributes them across multiple computers.
- Multiple nodes can access different parts of the same file simultaneously, enabling parallel processing.
- Computations run on the nodes where the data is stored.
- HDFS automatically replicates file blocks across different nodes to prevent data loss, making the system fault-tolerant.

22.2.2 Example of HDFS

Consider a phonebook containing phone numbers for everyone in the United States.

- Entries with last names starting with **A** may be stored on **Server 1**.
- Entries starting with **B** may be stored on **Server 2**.
- The remaining entries are distributed similarly across the cluster.

To reconstruct the complete phonebook, the program gathers data blocks from every server in the cluster.

By default, HDFS stores **two additional replicas** of every data block on different servers. If one server fails, the replicated copies ensure that the data remains available.

22.2.3 Advantages of Data Replication

Data replication provides several benefits:

- Higher availability when hardware failures occur.
- Better scalability by dividing work into smaller tasks that run simultaneously across the cluster.
- Improved reliability through fault tolerance.

22.2.4 Data Locality

Another important concept in Hadoop is **Data Locality** (moving computation closer to where the data resides).

Instead of transferring massive datasets across the network, Hadoop sends the computation to the node where the data is already stored.

Benefits of Data Locality include:

- Reduced network congestion.
- Increased throughput (the amount of data processed over a given period).
- Faster processing of large datasets.

22.2.5 Additional Benefits of HDFS

HDFS also provides:

- **Fast recovery from hardware failures**, as it automatically detects faults and recovers from failures.
- **Support for streaming data**, enabling high data throughput rates.
- **Accommodation of very large datasets**, as HDFS can scale to hundreds of nodes within a single cluster.
- **Portability**, since HDFS works across multiple hardware platforms and supports various operating systems.

22.3 Apache Hive

Apache Hive is an **open-source data warehouse software** used for reading, writing, and managing large datasets stored directly in **HDFS** or other storage systems such as **Apache HBase**.

22.3.1 Characteristics of Hive

- Hive is built on top of Hadoop.
- It is designed primarily for querying and analyzing large datasets.
- Hive provides SQL-based access to data, making it easier to perform analytical queries.

22.3.2 Limitations of Hive

Because Hive relies on Hadoop:

- Queries have **high latency** (longer response times), making Hive unsuitable for applications that require immediate responses.
- Hive is **read-oriented**, making it unsuitable for transaction processing systems that involve frequent write operations.

22.3.3 Best Use Cases for Hive

Hive is well suited for:

- ETL (Extract, Transform, Load)
- Reporting
- Data Analysis
- Data Warehousing
- SQL-based querying of large datasets

22.4 Apache Spark

Apache Spark is a **general-purpose distributed data processing engine** designed to extract and process large volumes of data for a wide range of applications.

22.4.1 Applications of Spark

Spark supports:

- Interactive Analytics
- Stream Processing
- Machine Learning
- Data Integration
- ETL (Extract, Transform, Load)

22.4.2 Key Features of Spark

- Uses **in-memory processing** to significantly speed up computations.
- Writes data to disk only when memory becomes constrained.
- Provides interfaces for major programming languages, including:
 - Java
 - Scala
 - Python
 - R
 - SQL
- Can run using its standalone clustering technology or on top of other infrastructures such as Hadoop.
- Can access data from a wide variety of data sources, including:
 - HDFS
 - Hive

These capabilities make Spark highly versatile for modern data processing workloads.

22.5 Key Use Case of Apache Spark

The primary use case of **Apache Spark** is its ability to:

- Process streaming data at high speed.
- Perform complex analytics in real time.
- Support large-scale distributed data processing for advanced analytics and machine learning workloads.

22.6 Summary

[Apache Hadoop](#), [Apache Hive](#), and [Apache Spark](#) each play an important role in the Big Data ecosystem.

- **Apache Hadoop** provides scalable, reliable, and distributed storage and processing of large datasets.
- **HDFS** enables fault-tolerant, scalable storage through data partitioning, replication, and data locality.
- **Apache Hive** serves as a SQL-based data warehouse solution optimized for ETL, reporting, and analytical workloads.
- **Apache Spark** delivers high-performance, in-memory distributed computing for real-time analytics, stream processing, machine learning, and large-scale data integration.

23 Data Mining

Course Textbook: *Getting Started with Data Science*

Publisher: IBM Press, 1st Edition (December 13, 2015)

Author: Murtaza Haider

Prescribed Reading: Chapter 12, Pages 529–531

23.1 Establishing Data Mining Goals

The first step in data mining is to establish clear goals for the exercise. We must identify the key questions that need to be answered while also considering the costs and benefits of the data mining process.

Additionally, we should determine in advance the expected level of accuracy and usefulness of the results. If cost were not a concern, unlimited resources could be invested to obtain the desired answers. However, in practice, the **cost-benefit trade-off** plays a crucial role in defining the goals and scope of a data mining exercise.

The desired level of accuracy directly influences the overall cost:

- Higher accuracy generally requires greater investment.
- Lower accuracy typically reduces costs.
- Beyond a certain point, increasing accuracy provides only **diminishing returns** (a situation where additional investment results in only small improvements).

Therefore, balancing cost, expected accuracy, and business value is essential when establishing data mining goals.

23.2 Selecting Data

The success of a data mining exercise largely depends on the quality of the data being used.

In some cases, the required data is readily available. For example, retailers often maintain extensive databases containing customer purchases and demographic information.

However, when suitable data is unavailable, we must:

- Identify alternative data sources.
- Plan new data collection initiatives, such as surveys.

The following factors directly affect the cost of a data mining project:

- Type of data
- Size of the dataset
- Frequency of data collection

Selecting the appropriate data that can answer the required questions at a reasonable cost is therefore a critical step.

23.3 Preprocessing Data

Preprocessing is one of the most important stages of data mining because raw data is often incomplete, inconsistent, or contains errors.

During preprocessing, we should:

- Identify and remove irrelevant attributes.
- Detect erroneous data and flag it appropriately.
- Verify data integrity.
- Develop a formal strategy for handling missing data.

For example, human errors may accidentally merge data incorrectly or place information into the wrong columns. Such issues should be identified and corrected before analysis.

23.3.1 Handling Missing Data

We must determine whether missing data occurs:

- **Randomly** (missing without any specific pattern)
- **Systematically** (missing due to a consistent reason or pattern)

If data is missing randomly, relatively simple solutions may be sufficient.

However, systematic missing data can introduce **systematic bias** (consistent errors that distort analysis results).

For example, if a group of individuals refuses to report their income, any analysis involving income will exclude those individuals, potentially producing biased conclusions.

Therefore, before beginning analysis, we should decide whether observations or variables containing missing values should be excluded entirely or only from specific parts of the analysis.

23.4 Transforming Data

Once the relevant data attributes have been retained, the next step is to determine the most appropriate format for storing and analyzing the data.

An important objective of data transformation is to reduce the number of attributes required to explain the phenomenon being studied.

One common technique is **Principal Component Analysis (PCA)**, which reduces the number of attributes while preserving most of the important information.

Variables may also require transformation to better represent the underlying phenomenon.

For example, an individual's total income may come from multiple sources, including:

- Wage income
- Rental property income
- Government support payments
- Other income sources

Combining these sources creates a more representative measure of total individual income.

Variables may also be converted from one type to another.

For example, a continuous income variable can be transformed into categorical groups such as:

- Low income
- Medium income
- High income

This transformation can help capture **non-linear relationships** (relationships that do not follow a straight-line pattern) in the data.

23.5 Storing Data

After transformation, data should be stored in a format that supports efficient data mining.

The storage system should:

- Provide unrestricted read and write access for data scientists.
- Allow newly created variables to be written back into the database efficiently.
- Support fast reading and writing operations.
- Store data in a centralized manner rather than scattering it across multiple servers or storage devices.
- Ensure data security and privacy.

Proper storage improves both the efficiency and reliability of the data mining process.

23.6 Mining Data

Once the data has been properly processed, transformed, and stored, it is ready for data mining.

This stage involves applying various analytical techniques, including:

- Parametric methods
- Non-parametric methods
- Machine learning algorithms

A good starting point is **data visualization**, which provides multidimensional views of the data using advanced graphing capabilities.

Visualization helps us develop a preliminary understanding of:

- Hidden trends
- Patterns
- Relationships within the dataset

Later sections of the chapter discuss various data mining algorithms and analytical methods in greater detail.

23.7 Evaluating Mining Results

After the data mining process is complete, we perform a formal evaluation of the results.

Evaluation may include testing the predictive performance of models using observed data to determine how effectively the algorithms reproduce known outcomes. This process is known as an **in-sample forecast** (evaluating a model using the same dataset on which it was trained).

The results are then shared with key stakeholders for feedback.

Based on this feedback:

- Improvements are identified.
- Models are refined.
- Better algorithms are selected.
- The data mining process is repeated.

As a result, data mining becomes an **iterative process** (a repeating cycle of improvement), where each iteration produces more accurate and useful results by incorporating stakeholder feedback and improved analytical methods.

24 Big Data and Data Mining Lesson Summary

24.1 Overview

In this topic, we recap the fundamentals of **Big Data**, how it drives **digital transformation**, how **cloud computing** enables us to analyze big data, the **data mining process**, and the most common **open-source tools** used to process big data.

The availability of vast amounts of data has transformed businesses, industries, and everyday life. Organizations now recognize the need for fundamental changes in their business approaches because data influences nearly every aspect of operations. The enormous amount

of data generated by people, tools, and machines requires innovative, scalable technologies to store, process, and analyze it effectively.

Big Data enables us to derive real-time business insights related to consumers, risk, profit, performance, productivity, and management, ultimately enhancing business value.

24.2 Characteristics of Big Data

Although there is no universally accepted definition of Big Data, data scientists generally agree that it is characterized by the following five attributes:

- **Value** – The expected benefit or meaningful insights gained by investing time and resources in analyzing big data.
- **Volume** – Refers to the scale or amount of data. Increasing data sources and scalable infrastructure are major drivers of data volume.
- **Velocity** – Represents the continuously increasing speed at which data is generated and processed from nonstop sources.
- **Variety** – Indicates that data comes from multiple sources and exists in different formats, including both structured and unstructured data.
- **Veracity** – Refers to the quality, reliability, and origin of the data, ensuring that it accurately conforms to facts.

24.3 Cloud Computing and Big Data

The development of cloud technologies enables us to work efficiently with Big Data.

Cloud computing refers to the delivery of on-demand computing resources on a pay-for-use basis.

Cloud computing provides the following five essential characteristics:

- **On-Demand Service** – We can access processing power, storage, and networking resources whenever needed.
- **Broad Network Access** – Computing resources are accessible through networks with internet connectivity.
- **Resource Pooling** – Cloud providers dynamically allocate shared resources among multiple users according to demand, making cloud computing cost-efficient.
- **Elasticity** – Resources automatically scale up when demand increases and scale back when demand decreases.
- **Measured Service** – We pay only for the computing resources that we use or reserve.

Cloud computing also addresses several important challenges by providing:

- Scalability
- Collaboration
- Accessibility
- Simplified software maintenance

Another major advantage is that the cloud gives us instant access to technologies without requiring installation or configuration, while automatically providing updated versions of these tools as they are released.

24.4 Popular Open-Source Big Data Tools

Several open-source technologies are commonly used to process and analyze Big Data.

24.4.1 Apache Hadoop

Apache Hadoop provides distributed storage and processing capabilities across clusters of computers, enabling efficient handling of massive datasets.

24.4.2 Apache Hive

Apache Hive is a data warehouse system built on top of Hadoop for data querying and analysis.

Hive allows us to:

- Read large datasets
- Write large datasets
- Manage datasets directly within the Hadoop Distributed File System (**HDFS**) or **Apache HBase**

24.4.3 Apache Spark

Apache Spark is a general-purpose data processing engine designed to extract and process large volumes of data for a wide range of applications with high performance.

24.5 The Data Mining Process

Big Data becomes valuable through a structured **six-step data mining process**.

24.5.1 Goal Setting

Identify the key questions that need to be answered. Cost and benefit considerations should guide this stage.

24.5.2 Data Selection

Identify appropriate data sources or plan new data collection initiatives based on the defined goals.

24.5.3 Data Preprocessing

Identify irrelevant attributes and erroneous data, then clean and flag the data where necessary.

24.5.4 Data Transformation

Convert the processed data into an appropriate format for storage and analysis.

24.5.5 Data Mining

Determine the analytical methods and machine learning algorithms that will be used to process and analyze the data.

24.5.6 Evaluation

Evaluate the mined results by:

- Testing the predictive capabilities of the models using observed data.
- Assessing the effectiveness and efficiency of the algorithms.
- Sharing the results with stakeholders.

The entire data mining process is **iterative** (repeated continuously to improve results), meaning the outcomes from one iteration guide future data mining efforts.

24.6 Key Takeaways

- Big Data is transforming businesses, industries, and everyday life.
- The five commonly accepted characteristics of Big Data are:
 - Value
 - Volume
 - Velocity
 - Variety
 - Veracity
- Cloud computing enables us to efficiently manage and analyze Big Data through:
 - On-demand service
 - Broad network access
 - Resource pooling
 - Elasticity
 - Measured service
- Cloud technologies simplify scalability, collaboration, accessibility, and software maintenance while providing instant access to the latest analytical tools.
- Popular open-source Big Data tools include:
 - Apache Hadoop
 - Apache Hive
 - Apache Spark
- The six-step data mining process consists of:
 1. Goal Setting
 2. Data Selection
 3. Data Preprocessing
 4. Data Transformation
 5. Data Mining
 6. Evaluation
- The data mining process is iterative, allowing each cycle to improve future analyses.

- Together, cloud computing and open-source technologies enable us to effectively and efficiently mine Big Data and generate valuable insights that drive digital transformation.

25 Glossary: Big Data and Data Mining

The following glossary contains the key terms introduced throughout this topic. These terms are important for us to recognize when working in the data science industry, participating in professional communities, and continuing our learning through certificate programs.

Term	Definition	Topic Where the Term is Introduced
Analytics	The process of examining data to draw conclusions and make informed decisions. It is a fundamental aspect of data science that involves statistical analysis and data-driven insights.	Data Scientists at New York University
Big Data	Vast amounts of structured, semi-structured, and unstructured data characterized by the five V's: Volume, Velocity, Variety, Veracity, and Value . When analyzed effectively, Big Data can provide competitive advantages and drive digital transformation.	How Big Data is Driving Digital Transformation
Big Data Cluster	A distributed computing environment consisting of thousands or tens of thousands of interconnected computers that collectively store and process large datasets.	What is Hadoop?
Broad Network Access	The ability to access cloud resources through standard network mechanisms using devices such as mobile phones, laptops, and workstations.	Introduction to Cloud
Chief Data Officer (CDO)	An executive responsible for overseeing data-related initiatives, governance, and strategies, ensuring that data plays a central role in digital transformation efforts.	How Big Data is Driving Digital Transformation
Chief Information Officer (CIO)	An executive responsible for managing an organization's information technology and computer systems, contributing to technology-related aspects of digital transformation.	How Big Data is Driving Digital Transformation
Cloud Computing	The delivery of on-demand computing resources—including networks, servers, storage, applications, services, and data centers—over the Internet on a pay-as-you-use basis.	Introduction to Cloud
Cloud Deployment Models	Categories that define where cloud infrastructure resides, who manages it, and how cloud resources are made available to users. Common deployment models include Public Cloud, Private Cloud, and Hybrid Cloud .	Introduction to Cloud

Cloud Service Models	Cloud computing service categories based on different layers of the computing stack, including Infrastructure as a Service (IaaS) , Platform as a Service (PaaS) , and Software as a Service (SaaS) .	Introduction to Cloud
Commodity Hardware	Standard, off-the-shelf hardware components used in Big Data clusters to provide cost-effective storage and processing without requiring specialized hardware.	What is Hadoop?
Data Algorithms	Computational procedures and mathematical models used to process and analyze data efficiently, especially large datasets made accessible through cloud computing.	Cloud for Data Science
Data Replication	A strategy that duplicates data across multiple nodes in a cluster to improve data durability and availability while reducing the risk of data loss caused by hardware failures.	What is Hadoop?
Data Science	An interdisciplinary field focused on extracting knowledge and insights from data using programming, statistics, machine learning, and analytical techniques.	Data Scientists at New York University
Deep Learning	A subset of machine learning that uses artificial neural networks inspired by the human brain to automatically learn patterns and make complex decisions from data.	Data Scientists at New York University
Digital Change	The integration of digital technology into business processes and operations, leading to improvements and innovation in organizational performance and customer value.	How Big Data is Driving Digital Transformation
Digital Transformation	A strategic and cultural organizational change driven by data science—especially Big Data—to integrate digital technologies across all areas of an organization, fundamentally transforming operations and value delivery.	How Big Data is Driving Digital Transformation
Distributed Data	The practice of dividing data into smaller chunks and distributing them across multiple computers within a cluster, enabling parallel processing.	What is Hadoop?
Hadoop	A distributed storage and processing framework designed for handling and analyzing massive datasets. Hadoop is widely used for Big Data analytics and data science applications.	Data Scientists at New York University
Hadoop Distributed File System (HDFS)	Hadoop's storage system that partitions and distributes files across multiple nodes, enabling parallel access, scalability, and fault tolerance.	What is Hadoop?
Infrastructure as a Service (IaaS)	A cloud service model that provides computing infrastructure such as servers, storage, and networking without requiring users to manage the underlying hardware.	Introduction to Cloud

Java-Based Framework	Hadoop is implemented in Java (an open-source, high-level programming language), providing the foundation for distributed storage and processing systems.	Big Data Processing Tools: Hadoop, HDFS, Hive, and Spark
Map Process	The first phase of Hadoop's MapReduce programming model, where data is processed in parallel across cluster nodes, typically performing data transformation tasks.	What is Hadoop?
Measured Service	A cloud computing characteristic where resource usage is automatically monitored, measured, and reported so users are billed based on actual consumption.	Introduction to Cloud
On-Demand Self-Service	The ability for users to provision computing resources such as processing power, storage, and networking automatically through self-service interfaces without requiring human interaction with the service provider.	Introduction to Cloud
Rapid Elasticity	The capability of cloud platforms to quickly scale computing resources up or down based on demand.	Introduction to Cloud
Reduce Process	The second phase of Hadoop's MapReduce model, where outputs from the Map phase are aggregated and further processed to generate the final analytical results.	What is Hadoop?
Replication	The process of creating multiple copies of data within a Big Data cluster to improve fault tolerance and ensure data availability if hardware or nodes fail.	What is Hadoop?
Resource Pooling	A cloud computing characteristic where computing resources are shared among multiple users and dynamically allocated based on demand, improving efficiency and reducing costs.	Introduction to Cloud
Skills Network Labs (SN Labs)	IBM-provided learning resources that include tools such as Jupyter Notebooks and Spark clusters for cloud-based data science projects and skill development.	Cloud for Data Science
Spilling to Disk	A technique used when memory becomes full, where data is temporarily written to disk storage to allow processing to continue without interruption.	Big Data Processing Tools: Hadoop, HDFS, Hive, and Spark
STEM Classes	Science, Technology, Engineering, and Mathematics courses commonly offered in schools to prepare students for technical careers, including data science.	Data Scientists at New York University
Variety	One of the five V's of Big Data. It refers to the diversity of data types, including structured, semi-structured, and unstructured data such as text, images, audio, and video.	Foundations of Big Data

Velocity	One of the five V's of Big Data. It describes the speed at which data is generated, collected, and processed, often in real time or near real time.	Foundations of Big Data
Veracity	One of the five V's of Big Data. It refers to the quality, reliability, accuracy, consistency, and completeness of data.	Foundations of Big Data
Video Tracking System	A system that captures and analyzes video data from games to study player movements and game dynamics, supporting data-driven decision-making in sports.	How Big Data is Driving Digital Transformation
Volume	One of the five V's of Big Data. It represents the enormous amount of data generated and stored due to increasing data sources, higher-resolution sensors, and scalable infrastructure.	Foundations of Big Data
V's of Big Data	The five key characteristics of Big Data: Volume, Velocity, Variety, Veracity, and Value . These characteristics describe the scale, speed, diversity, quality, and business value of Big Data.	Foundations of Big Data

26 Key Data Science and Artificial Intelligence Terminology

In data science, many terms are often used interchangeably. Understanding the differences between these concepts helps us better understand how modern data-driven systems work and how they relate to one another.

26.1 Big Data

Big Data refers to datasets that are:

- So massive in size
- Generated so quickly
- So diverse in format

that they cannot be efficiently analyzed using traditional methods such as relational databases (structured database systems that organize data into tables).

The development of enormous computing power through distributed networks (systems where processing is spread across multiple connected computers), along with advanced data analysis tools and techniques, enables organizations to analyze these vast datasets and uncover new knowledge and valuable insights.

Big Data is commonly described using the Five Vs:

- **Velocity** – The speed at which data is generated and processed.
- **Volume** – The enormous amount of data being produced.
- **Variety** – The different types and formats of data (structured, semi-structured, and unstructured).
- **Veracity** – The quality, accuracy, and reliability of data.
- **Value** – The useful insights and business benefits that can be extracted from data.

26.2 Data Mining

Data Mining is the process of automatically searching and analyzing data to discover previously unrevealed patterns.

The process involves:

- Preprocessing the data to prepare it for analysis.
- Transforming the data into an appropriate format.
- Mining and extracting insights and patterns using various tools and techniques.

These techniques range from:

- Simple data visualization tools
- Machine learning algorithms
- Statistical models

26.3 Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that uses computer algorithms to analyze data and make intelligent decisions based on what they have learned, without being explicitly programmed.

Machine learning algorithms:

- Are trained using large datasets.
- Learn from examples rather than following predefined rule-based algorithms.
- Enable machines to solve problems independently.
- Make accurate predictions using the provided data.

26.4 Deep Learning

Deep Learning is a specialized subset of Machine Learning that uses layered neural networks to simulate human decision-making.

Deep learning algorithms can:

- Label and categorize information.
- Identify complex patterns within data.
- Continuously learn from experience.
- Improve the quality and accuracy of results by determining whether previous decisions were correct.

26.5 Artificial Neural Networks

Artificial Neural Networks (ANNs), commonly referred to as Neural Networks, are inspired by biological neural networks (the interconnected structure of human brain cells), although they function quite differently.

A neural network in AI consists of many small computing units called neurons, which:

- Receive incoming data.
- Learn patterns over time.
- Make decisions based on learned information.

Neural networks typically contain multiple layers, which is why deep learning algorithms become more effective as the size of the dataset increases. In contrast, some traditional machine learning algorithms may reach a performance plateau (stop improving significantly) as more data becomes available.

26.6 Data Science vs. Artificial Intelligence

Although Data Science and Artificial Intelligence are closely related, they are not the same.

26.6.1 Data Science

Data Science is the process and methodology for extracting knowledge and insights from large volumes of disparate data (data collected from multiple different sources and formats).

It is an interdisciplinary (combining multiple branches of study) field involving:

- Mathematics
- Statistical analysis
- Data visualization
- Machine learning
- Other analytical techniques

Data Science enables us to:

- Organize and interpret information.
- Identify patterns within data.
- Extract meaningful insights.
- Make informed business decisions.

Data Science can also use many AI techniques to derive insights from data. For example, it may use:

- Machine learning algorithms
- Deep learning models

to extract meaning and draw inferences from data.

26.6.2 Artificial Intelligence

Artificial Intelligence (AI) includes everything that enables computers to learn how to solve problems and make intelligent decisions.

AI encompasses technologies such as:

- Machine Learning
- Deep Learning
- Neural Networks
- Other intelligent algorithms

26.6.3 Relationship Between Data Science and AI

There is interaction between Data Science and Artificial Intelligence, but neither is a subset of the other.

- Data Science is a broad discipline that encompasses the entire data processing methodology, from collecting and preparing data to analyzing and interpreting it.
- Artificial Intelligence focuses on developing systems that can learn, reason, make decisions, and solve problems intelligently.
- Data Science frequently uses AI techniques, such as Machine Learning and Deep Learning, to analyze data and generate insights.

26.7 Big Data in Data Science and AI

Both Data Science and Artificial Intelligence involve the use of Big Data, which consists of significantly large volumes of data.

The availability of Big Data, combined with modern computing power and advanced analytical techniques, allows organizations to discover valuable insights, improve decision-making, and develop intelligent systems.

26.8 Summary

The key concepts covered in this topic include:

- **Big Data** – Extremely large, fast-growing, and diverse datasets characterized by the Five Vs: Velocity, Volume, Variety, Veracity, and Value.
- **Data Mining** – The process of discovering hidden patterns and insights from prepared data.
- **Machine Learning** – A subset of AI that enables computers to learn from data without explicit programming.
- **Deep Learning** – A specialized branch of Machine Learning that uses layered neural networks for complex pattern recognition and decision-making.
- **Artificial Neural Networks** – Layered computing models inspired by biological neurons that power deep learning systems.
- **Data Science** – An interdisciplinary field that extracts meaningful knowledge and insights from data.
- **Artificial Intelligence** – The broader field focused on enabling machines to perform intelligent tasks and make decisions.
- **Relationship** – Data Science and AI complement one another, and both frequently leverage Big Data to solve real-world problems.

27 Generative AI and Data Science

27.1 Learning Objectives

After completing this topic, we will be able to:

- Describe Generative AI.
- Explain how data scientists use Generative AI in data science.

27.2 Introduction to Generative AI

Generative AI is a subset of artificial intelligence that focuses on producing new data rather than just analyzing existing data.

It allows machines to create content, including:

- Images
- Music
- Natural language (human-like text)
- Computer code
- And more

Generative AI creates content that mimics creations made by people.

27.3 How Generative AI Operates

Deep learning models such as **Generative Adversarial Networks (GANs)** and **Variational Autoencoders (VAEs)** are at the foundation of Generative AI.

These models create new instances that replicate the underlying distribution (the overall pattern and statistical characteristics) of the original data by learning patterns from enormous volumes of data.

27.4 Applications of Generative AI

Generative AI has found diverse applications across various industries.

- **Natural Language Processing (NLP):** Models such as OpenAI's GPT-3 can generate human-like text, revolutionizing content creation and chatbots.
- **Healthcare:** Generative AI can synthesize medical images, aiding in the training of medical professionals.
- **Art and Creativity:** It can create unique and visually stunning artworks, generating endless creative visual compositions.
- **Game Development:** Game developers use Generative AI to generate realistic environments, characters, and game levels.
- **Fashion:** Generative AI assists in designing new styles and creating personalized shopping recommendations.

27.5 How Data Scientists Use Generative AI

27.5.1 Synthetic Data Generation

Training and testing a model requires large amounts of data. Sometimes, the data we want to study does not contain enough observations to build an effective model.

Interest in **synthetic data** (artificially generated data that closely resembles real-world data) as a tool for analysis and model creation has increased due to recent developments in Generative AI.

Data scientists can augment (expand and enrich) their datasets using Generative AI to create synthetic data.

This synthetic data is generated with properties similar to the real dataset, including:

- Distribution (how data values are spread)
- Clustering (how similar data points are grouped)
- Many other characteristics learned from the original dataset

Data scientists can then combine synthetic data with real data for model training and testing.

27.5.2 Code Generation and Analytics Automation

Data scientists, researchers, and analysts frequently find themselves constrained by time when examining data patterns and insights.

Due to this limitation, they can only conceive, develop, and evaluate a small number of hypotheses, leaving many other possibilities untested.

Generative AI enables data scientists to:

- Generate software code for constructing analytical models.
- Test software code automatically.
- Automate coding tasks for analytics.

Coding automation has the potential to revolutionize analytics by allowing data scientists to focus on higher-level activities, such as:

- Identifying and clarifying the problem the models are intended to solve.
- Evaluating hypotheses from a wider range of data sources.

27.5.3 Business Insights and Reporting

Generative AI can generate accurate business insights and comprehensive reports, making it possible to update these insights as data evolves.

Furthermore, it can autonomously explore data to uncover hidden patterns and insights that might go unnoticed during manual analysis, ultimately enhancing decision-making.

27.5.3.1 Example: IBM Cognos Analytics

IBM's **Cognos Analytics** provides AI-powered automation that helps unlock the full potential of data through its natural language AI assistant.

With Cognos Analytics, we can:

- Describe the question we want to answer.
- Specify the hypothesis we want to test.
- Generate the insights needed using natural language.

27.6 Key Takeaways

- Generative AI is a subset of artificial intelligence that focuses on producing new data rather than analyzing existing data.
- It uses deep learning models such as **Generative Adversarial Networks (GANs)** and **Variational Autoencoders (VAEs)** to learn patterns from existing data and generate new data with similar characteristics.
- Generative AI has applications in natural language processing, healthcare, art, game development, fashion, and many other industries.
- Data scientists use Generative AI to create synthetic data, helping overcome data limitations during model training and testing.
- Generative AI automates code generation and analytical model development, allowing data scientists to focus on problem-solving and hypothesis evaluation.
- It generates business insights, creates comprehensive reports, and uncovers hidden patterns within data to improve decision-making.
- AI-powered platforms such as IBM Cognos Analytics enable users to generate insights using natural language queries and hypotheses.

28 Neural Networks and Deep Learning

28.1 Introduction

Neural networks are Computer Science's attempt to mimic how the neurons in our brain actually function. They are computer programs designed to imitate the way neurons and synapses process information by building complex networks that can be trained to perform specific tasks.

28.2 How Neural Networks Work

Approximately **20–23 years ago**, a neural network typically worked as follows:

- Inputs were fed into the network.
- The inputs passed through different processing nodes.
- Each node performed transformations and aggregated the information.
- The transformed data then moved through additional layers of nodes.
- Finally, the network produced an output.

One early application of neural networks was recognizing **handwritten digits**.

A neural network starts with some **inputs** and some **outputs**. We repeatedly feed the inputs into the network to determine what kinds of transformations are required to produce the desired outputs. This process is repeated over and over until the network **converges** (reaches a stable solution where the transformations consistently produce the correct outputs).

28.3 Limitations of Early Neural Networks

Although the theory behind neural networks was well established and they worked successfully on relatively small problems such as recognizing handwritten digits, they had significant limitations:

- They were **computationally very intensive**.
- The available computing power at the time was insufficient for solving large-scale problems efficiently.
- As a result, neural networks gradually went out of favor, and many instructors stopped teaching them **approximately 15 years ago**.

28.4 Emergence of Deep Learning

A few years later, the term **Deep Learning** became increasingly popular. Deep learning can be thought of as "**neural networks on steroids**" (an expression used to describe a much more powerful version of traditional neural networks).

Instead of using only a few processing layers, deep learning:

- Uses **multiple layers of neural networks**.
- Relies on **lots and lots of computing power**.
- Solves much larger and more complex problems than traditional neural networks.

28.5 Why Deep Learning Requires Powerful Hardware

A young faculty member in the **Marketing Department** conducted research partially based on deep learning.

To perform this research, she required a computer equipped with a **Graphics Processing Unit (GPU)** because deep learning involves enormous amounts of:

- Matrix calculations
- Linear algebra calculations

These mathematical computations are fundamental to training neural networks.

28.5.1 Graphics Processing Unit (GPU)

A **GPU (Graphics Processing Unit)** is specialized hardware designed to perform a massive number of calculations simultaneously.

At one university's **South Data Center**, there were piles of **Dell** servers, each equipped with GPUs.

The speaker noted:

- One application at the university required **200 servers**.
- Each server contained a **Graphics Processing Unit (GPU)**.
- Each GPU had the equivalent of approximately **600 processing cores**.
- Together, this provided **tens of thousands of processing cores** dedicated to deep learning.

28.6 Real-World Applications of Deep Learning

Modern neural networks and deep learning systems can perform many tasks that were previously impossible for traditional neural networks.

These include:

- Speech recognition
- Face recognition
- Recognizing people
- Image recognition
- Image classification

The speaker also stated that the **NSA (National Security Agency)** likely performs extensive work involving neural networks.

28.7 Example from NYU and Facebook

One instructor who teaches the **Deep Learning** class at **NYU (New York University)** and is also the **Head Data Scientist at Facebook** demonstrates deep learning during class.

The instructor brings a notebook computer that contains multiple **Graphics Processing Units (GPUs)**.

During the lecture:

- Students begin speaking to the computer.
- The notebook trains a neural network **during the class**.
- The neural network learns to recognize speech in real time.

28.8 Learning Without Explicit Programming

Traditional neural networks performed relatively small tasks.

Modern deep learning systems can perform much larger tasks and are capable of learning patterns automatically.

For example, a deep learning model can:

- Learn the difference between a **cat** and a **dog**.
- Recognize different kinds of objects.

- Learn without being explicitly programmed for every feature.

This ability to automatically learn patterns from data is one of the reasons it is called **Deep Learning**.

When listening to a deep learning system learn speech recognition and speech generation, the speaker describes it as sounding **like a baby learning to talk**. As training progresses, the machine gradually improves until it can speak intelligibly.

28.9 Importance of Linear Algebra

A significant portion of deep learning is based on:

- Matrix operations
- Linear algebra

Therefore, we should understand:

- Linear algebra concepts
- Matrix transformations
- How these mathematical operations transform data inside neural networks

Although many modern deep learning libraries perform these mathematical calculations automatically, we should still understand what is happening underneath the software.

28.10 Computational Requirements

Deep learning requires extremely high computational power.

Key points include:

- High-performance computing resources are essential for serious deep learning work.
- Standard notebook computers can be used for experimentation.
- Serious deep learning projects require specialized computational resources, particularly systems equipped with powerful **Graphics Processing Units (GPUs)**.

29 Applications of Machine Learning

Machine learning is now used across many domains, with recommender systems being one of its major applications. It is also widely used for classification, cluster analysis, predictive analytics, and solving complex business and financial problems.

29.1 Common Applications of Machine Learning

Machine learning is applied in many areas, including:

- **Recommender Systems**
- **Classification**

- **Cluster Analysis**
- **Market Basket Analysis**
- **Predictive Analytics**

29.1.1 Market Basket Analysis

Market basket analysis helps determine **what goods tend to be bought together**.

About **20 years ago**, solving market basket analysis problems was computationally very difficult. Today, we perform these analyses routinely using machine learning techniques.

29.1.2 Predictive Analytics

Predictive analytics is another important application of machine learning.

We use new machine learning techniques to predict outcomes using methods such as:

- Decision Trees
- Bayesian Analysis
- Naive Bayes

These techniques are widely available in software packages such as **R**.

Rather than knowing every mathematical detail of these techniques, it is important that we understand:

- How the techniques are used
- What their results mean
- The trade-offs involved when applying them

Some important concepts include:

- **Precision vs. Recall** (the balance between correctly identifying relevant results and avoiding false positives)
- **Oversampling** (creating additional samples for underrepresented classes to balance the dataset)
- **Overfitting** (when a model learns the training data too closely and performs poorly on new, unseen data)

Someone with basic knowledge of data science can apply these techniques effectively, provided they understand the trade-offs rather than only the implementation details.

29.2 Applications of Machine Learning in FinTech

Machine learning has numerous applications in the financial technology (FinTech) industry. Two major applications are recommendation systems and fraud detection.

29.2.1 Recommendation Systems in FinTech

Recommendation systems work in FinTech in much the same way as they do in online platforms.

For example:

- **Netflix** recommends another show based on the shows we have already watched.
- **Facebook** recommends people we may want to follow based on our existing connections.

A similar approach is used in FinTech.

For example, if we are investment professionals and have explored one investment idea, the system may recommend:

- A similar investment idea
- A similar asset
- A similar company
- A similar investment technique

Machine learning enables these recommendations throughout many different areas of FinTech.

29.2.2 Fraud Detection

Fraud detection is one of the most important machine learning applications in the retail side of banking and finance.

The objective is to determine **whether a credit card transaction is fraudulent or not in real time**.

To accomplish this, the machine learning system:

- Learns from all previous transactions.
- Builds a predictive model using historical transaction data.
- Evaluates each new transaction as it occurs.
- Determines whether the transaction appears legitimate or suspicious.
- Routes suspicious transactions to the fraud team for further investigation.

A typical decision made by such a system may be:

- **"We think that's OK."**
- **"That's not so good. Let's route it to our fraud people to check."**

30 Regression

Course Textbook: *Getting Started with Data Science*

Publisher: IBM Press; 1st Edition (December 13, 2015) – Print

Author: Murtaza Haider

Prescribed Reading: Chapter 7, Pages 235–236

30.1 Why Tall Parents Don't Have Even Taller Children

We might have noticed that taller parents often have tall children who are not necessarily taller than their parents—and that is a good thing. This does not suggest that children born to tall parents are not necessarily taller than the rest. That may be the case, but they are not necessarily taller than their own "tall" parents.

To understand why this is a good thing, we can perform a simple mental simulation. Imagine if every successive generation born to tall parents were taller than their parents. In a matter of a couple of millennia, human beings would become uncomfortably tall for their own good, requiring even bigger furniture, cars, and planes.

In 1886, Sir Frances Galton studied this same question and developed a statistical technique that we now know as **regression models**.

Regression models have become the **workhorse of statistical analysis**. In almost all empirical research, whether in academic or professional fields, regression models—or their variants—are used extensively.

Some applications of regression models include:

- Developing more effective medicines.
- Improving methods for medical operations.
- Optimizing resources for small and large hospitals.
- Analyzing consumer behavior.
- Measuring firm productivity.
- Evaluating the competitiveness of public and private sector organizations.

To explain the usefulness of regression models, the author narrates a story from his **Master's thesis**.

30.2 The Department of Obvious Conclusions

In 1999, the author completed his **Master's research** on developing **hedonic price models** (statistical models used to estimate the value of a product based on its characteristics) for **residential real estate properties**.

The research project:

- Took **3 years** to complete.
- Included **500,000 real estate transactions**.

As the author was preparing for his thesis defense, his wife offered to drive him to the university. During the drive, she asked:

"Tell me, what have you found in your research?"

The author proudly replied:

"Well, I have been studying the determinants of housing prices. I have found that larger homes sell for more than smaller homes."

As soon as she heard this, she pulled the car onto the shoulder of the highway, applied the brakes, and responded:

"I can't believe that they are giving you a Master's degree for finding just that. I could have told you that larger homes sell for more than smaller homes."

At that moment, the author felt like a professor teaching in the **Department of Obvious Conclusions**.

From an everyday perspective, the finding appeared obvious. However, the actual contribution of the research was much deeper than simply identifying that larger houses sell for more.

30.3 What the Research Actually Discovered

The research did not merely establish a **correlation** (a relationship between two variables) between housing prices and house size. It also quantified the **magnitude** (the size or strength) of those relationships.

For example, the author found that, **all else being equal** (assuming every other influencing factor remains the same):

- An additional **washroom** adds more to the housing price than an additional **bedroom**.
- The **marginal increase** (the extra increase resulting from one additional unit) in the price of a house is higher for an additional washroom than for an additional bedroom.

The author later discovered that **real estate brokers in Toronto** appreciated this finding because it provided measurable evidence rather than general observations.

30.4 Impact of Location on Housing Prices

The research also showed that location-related factors significantly affect housing prices.

30.4.1 Transport Infrastructure

- Houses located **closer to subways** sold for more than those situated farther away.
- Houses located **near freeways or highways** sold for less than other houses.

30.4.2 Shopping Centers

The study found a **nonlinear impact** (the relationship does not increase or decrease at a constant rate) of proximity to large shopping centers on housing prices.

Specifically:

- Houses located **less than 2.5 km** from shopping centers sold for **less** than the rest.
- Houses located **less than 5 km but more than 2.5 km** from shopping centers sold for **more** than those located farther away.

30.4.3 Distance from Downtown

The author also found that **housing values in Toronto declined with distance from downtown**.

30.5 Quantifying Relationships

As the author explained these findings, his wife became **mildly impressed**.

The likely reason for her lukewarm response was that many of the findings aligned with everyday experience. However, the true value of the research was not simply confirming what people already believed—it was **quantifying the magnitude of those relationships**.

Regression models allow us to move beyond common knowledge by measuring **how much** one factor influences another.

30.6 Why Regression?

Regression analysis can answer a wide variety of practical questions.

Some examples include:

- How much more can a house sell for with an additional bedroom?
- What is the impact of lot size on housing price?
- Do homes with brick exteriors sell for less than homes with stone exteriors?
- How much does a finished basement contribute to the price of a housing unit?
- Do houses located near high-voltage power lines sell for more or less than the rest?

31 IBM Cloud Gallery

31.1 Overview

IBM Cloud Resource Hub is a growing collection of datasets, notebooks, and project templates. In this topic, we use the IBM Cloud Resource Hub to explore different datasets. As we learned in the course, data can be more than just numbers. Data can be numeric, text, images, videos, audios, and more.

We will explore the following three samples:

- **Sample 1** contains data with only **numeric attributes**.
- **Sample 2** contains data with **numeric and text attributes**.
- **Sample 3** contains a **Jupyter Notebook**, a tool that data scientists use to create models.

Through these samples, we will examine how data scientists work with different types of datasets.

31.2 Learning Objectives

By the end of this topic, we will be able to:

- Explore the **IBM Cloud Resource Hub**.
- Examine a **numeric dataset**.
- Examine a dataset with **non-numeric attributes**.
- Examine a **Jupyter Notebook**.

31.3 Exercise: Examine a Numeric Dataset

Open the following link to access the **IBM Cloud Resource Hub**:

<https://dataplatform.cloud.ibm.com/gallery>

32 Deep Learning and Machine Learning Lesson Summary

32.1 Overview

In this topic, we review the following concepts:

- Commonly used terms in Artificial Intelligence (AI) and how to differentiate them
- How data scientists use Artificial Intelligence (AI)
- The relationship between Machine Learning and Data Science
- The regression model used to show relationships between things

Artificial Intelligence (AI) has boomed and become accessible to almost everyone. Data scientists use AI regularly when analyzing data.

32.2 Artificial Intelligence and Its Related Concepts

The following terms are fundamental to understanding AI:

- **Artificial Intelligence (AI):** The branch of computer science that includes the development of systems that can mimic many of the tasks associated with human intelligence.
- **Machine Learning:** A subset of AI that uses computer algorithms to learn about data and make predictions without needing to explicitly program the analysis methods into the system.
- **Deep Learning:** A subset of Machine Learning that uses layered neural networks to simulate human decision-making.
- **Neural Network:** A collection of small computing units, called **neurons**, that take incoming data and learn to make decisions over time, such as distinguishing between a dog and a cat.
- **Generative AI:** A subset of AI that focuses on producing new data rather than just analyzing existing data.

32.3 Deep Learning

Deep Learning uses layered neural networks to learn patterns from data.

A neural network is made up of small computing units called **neurons**, which process incoming data and improve their decision-making over time.

For example, a neural network can learn to distinguish the difference between a dog and a cat.

One important characteristic of Deep Learning is that its algorithms become more efficient as the amount of data increases in volume, unlike many other Machine Learning algorithms, which tend to plateau (reach a point where additional data provides little improvement).

32.4 Generative AI

Generative AI focuses on producing new data rather than only analyzing existing data.

It enables machines to create content such as:

- Images
- Music
- Languages
- Computer code

These outputs mimic human creations.

Generative AI can also create datasets with characteristics similar to a raw dataset.

Sometimes, when data scientists do not have enough data, they create **synthetic data** (artificially generated data that resembles real data) and use it to train and test their models.

32.5 How Data Scientists Use Machine Learning

As data scientists, we use Machine Learning algorithms to derive insights from data.

Machine Learning algorithms are frequently applied for:

- Predictive analytics
- Recommendation systems
- Fraud detection

For example, Machine Learning algorithms can identify fraudulent credit card purchases based on previous purchasing habits.

32.6 Regression

Machine Learning algorithms rely heavily on a statistical technique called **regression**.

Regression identifies:

- The strength of the correlation between one or more input variables and an output variable.
- The amount of the correlation between one or more input variables and an output variable.

For example, regression can help answer questions such as:

- How much does the price of a house increase based on its square footage and number of bedrooms?
- How confident can we be about this relationship?

32.7 Summary

Key takeaways from this topic include:

- Artificial Intelligence (AI) is the branch of computer science that develops systems capable of mimicking many tasks associated with human intelligence.
- Machine Learning is a subset of Artificial Intelligence.
- Deep Learning is a subset of Machine Learning.
- Deep Learning utilizes neural networks to teach itself patterns in inputs and outputs.
- Deep Learning algorithms become more efficient as the volume of data increases, whereas many other Machine Learning algorithms tend to plateau.
- Generative AI produces new data that is similar to a raw dataset.
- Data scientists can generate synthetic data when real-world data is insufficient for training and testing models.
- Machine Learning is widely used for predictive analytics, recommendation systems, and fraud detection.
- Regression is a statistical technique used to determine the strength and amount of the correlation between input variables and an output.
- Using big data, we use all of these areas of AI to make predictions.

33 Deep Learning and Machine Learning Lesson Glossary

This alphabetized glossary contains many of the terms covered in this course. These terms are important for us to recognize when working in the industry, participating in user groups, and participating in other certificate programs.

Term	Definition	Topic Where the Term Is Introduced
Artificial Neural Networks	Collections of small computing units (neurons) that process data and learn to make decisions over time.	Artificial Intelligence and Data Science

Bayesian Analysis	A statistical technique that uses Bayes' theorem to update probabilities based on new evidence.	Applications of Machine Learning
Business Insights	Accurate insights and reports generated by generative AI can be updated as data evolves, enhancing decision-making and uncovering hidden patterns.	Generative AI and Data Science
Cluster Analysis	The process of grouping similar data points together based on certain features or attributes.	Neural Networks and Deep Learning
Coding Automation	Using generative AI to automatically generate and test software code for constructing analytical models, freeing data scientists to focus on higher-level tasks.	Generative AI and Data Science
Data Mining	The process of automatically searching and analyzing data to discover patterns and insights that were previously unknown.	Artificial Intelligence and Data Science
Decision Trees	A type of machine learning algorithm used for decision-making by creating a tree-like structure of decisions.	Applications of Machine Learning
Deep Learning Models	Includes Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) that create new data instances by learning patterns from large datasets.	Generative AI and Data Science
Five V's of Big Data	Characteristics used to describe big data: Velocity, Volume, Variety, Veracity, and Value.	Neural Networks and Deep Learning
Generative AI	A subset of AI that focuses on creating new data, such as images, music, text, or code, rather than just analyzing existing data.	Generative AI and Data Science
Market Basket Analysis	Analyzing which goods tend to be bought together is often used for marketing insights.	Neural Networks and Deep Learning
Naive Bayes	A simple probabilistic classification algorithm based on Bayes' theorem.	Applications of Machine Learning

Natural Language Processing (NLP)	A field of AI that enables machines to understand, generate, and interact with human language, revolutionizing content creation and chatbots.	Generative AI and Data Science
Precision vs. Recall	Metrics used to evaluate the performance of classification models.	Applications of Machine Learning
Predictive Analytics	Using machine learning techniques to predict future outcomes or events.	Neural Networks and Deep Learning
Synthetic Data	Artificially generated data with properties similar to real data, used by data scientists to augment their datasets and improve model training.	Generative AI and Data Science

34 Summary: Deep Learning and Machine Learning

At this point in the course, we know the following:

- **Big Data** has five characteristics:
 - Velocity
 - Volume
 - Variety
 - Veracity
 - Value
- The **five cloud computing characteristics** are:
 - On-demand self-service
 - Broad network access
 - Resource pooling
 - Rapid elasticity
 - Measured service
- **Data mining** follows a six-step process:
 1. Goal setting
 2. Selecting data sources
 3. Preprocessing
 4. Transforming
 5. Mining
 6. Evaluation
- The availability of so many disparate (coming from many different sources and formats) amounts of data created by people, tools, and machines requires new, innovative, and scalable technology to drive transformation.
- **Deep learning** utilizes neural networks to teach itself patterns in inputs and outputs.
- **Machine learning** is a subset of Artificial Intelligence (AI) that uses computer algorithms to learn about data and make predictions without explicitly programming the analysis methods into the system.
- **Regression** identifies the strength and amount of the correlation between one or more inputs and an output.

- Skills involved in processing **Big Data** include the application of:
 - Statistics
 - Machine learning models
 - Computer programming
- **Generative AI**, a subset of Artificial Intelligence, focuses on producing new data rather than just analyzing existing data. It allows machines to create content, including:
 - Images
 - Music
 - Language
 - Computer code
 - More, mimicking creations by people

Module 3: Applications and Careers in Data Science

35 Building a Strong Data Foundation for Data Science

For businesses, one principle remains true: if we are unable to measure something, we are unable to improve it. If we are unable to measure our costs, we are unable to reduce them. If we are unable to measure our profits, we are unable to increase them.

35.1 Start by Capturing Data

The first thing a company has to do is start recording information and capturing data. This includes collecting data about:

- Costs
 - Labor costs
 - Material costs
 - The cost of selling one product
 - Total costs
- Revenue
 - Where revenue is coming from
 - Whether **80% of the revenue comes from 20% of the customers**, or if the situation is the other way around.

The first step is always to **start capturing data**. Once we have data, we can apply algorithms and analytics to it.

35.2 Archive Data Instead of Overwriting It

If we are not capturing data, we should start capturing it. If we are already capturing data, we should archive(save) it.

We should **not overwrite old data** thinking that it is no longer needed.

Data never gets old. Data is always relevant, even if it is **100 years old** or **200 years old**. It remains valuable for the business, the organization, and long-term success.

To preserve the value of data, we should:

- Keep all collected data.
- Archive data properly.
- Ensure that nothing goes to waste.
- Maintain consistency in the data.
- Create proper documentation so that someone **20 years later** can still understand and use the data effectively.

35.3 Implement Data Archiving Best Practices

We should put the best practices for data archiving in place as soon as a business starts.

If a business is already operating without proper data archiving practices, it should begin implementing them immediately.

The key actions include:

- Start measuring everything that matters.
- Archive historical data.
- Maintain consistent documentation.
- Preserve data for future analysis.

35.4 The Importance of Data Quality

Many companies fail to measure important business information for **a decade**, and only later decide that they want to implement data science.

However, data science inside a company is only as valuable as the data that has been collected.

Garbage in, garbage out (poor-quality input data produces poor-quality results) is a fundamental rule in data analysis.

If something is not measured, it becomes very difficult to improve or change it.

Therefore, the **very first step is measurement**.

35.5 Clean Existing Data or Start Collecting New Data

Depending on the company's current situation:

- If the company already has existing data, it should begin reviewing and cleaning the data.
- If the company does not have existing data, it should immediately start collecting it.

Only after reliable data has been collected and prepared can meaningful analytics and data science techniques be applied.

35.6 Build a Data Science Team

A successful data science initiative requires more than a single data scientist.

Companies should first have employees who are genuinely interested in data science because, without interest, there will be little engagement.

Organizations should remember that success depends on building **a team of data scientists**, where each team member has strengths in different areas of data science. Combining these complementary skills enables the team to solve a wider range of data science problems effectively.

36 Data Science Applications for Solving Real-World Problems

Organizations can leverage the almost unlimited amount of data now available to them in a growing number of ways. However, all organizations ultimately use data science for the same reason—to discover optimum solutions to existing problems.

This topic explores examples of how data science provides innovative solutions to long-standing challenges across different industries and environments.

36.1 Data Science in Transportation

36.1.1 Uber's Use of Real-Time Data

Uber collects real-time user data to discover:

- How many drivers are available.
- If more drivers are needed.
- Whether a surge charge should be allowed to attract more drivers.

Uber uses data to put the right number of drivers in the right place, at the right time, for a cost the rider is willing to pay.

36.1.2 Toronto Transportation Commission

The **Toronto Transportation Commission** has made great strides in solving an old problem with traffic flows by restructuring those flows in and around the city.

Using data science tools and analysis, they have:

- Gathered data to better understand streetcar operations and identify areas for interventions.
- Analyzed customer complaints data.

- Used probe data to better understand traffic performance on main routes.
- Created a team to better capitalize on big data for planning, operations, and evaluation.

By focusing on peak hour clearances and identifying the most congested routes, the monthly hours lost for commuters due to traffic congestion dropped from **4.75 hours in 2010** to **3 hours in mid-2014**.

36.2 Data Science for Environmental Challenges

Data science can also play a proactive role in addressing environmental issues.

Freshwater lakes supply a variety of human and ecological needs, such as:

- Providing drinking water.
- Producing food.

However, lakes across the world are threatened by increasing incidences of **harmful cyanobacterial blooms** (rapid growth of toxic blue-green algae that can harm people, animals, and aquatic ecosystems).

There are many projects and studies working to solve this long-existing dilemma.

36.2.1 Cyanobacteria Research in the United States

In the **United States**, a team of scientists from research centers stretching from **Maine** to **South Carolina** is developing and deploying high-tech tools to explore cyanobacteria in lakes across the east coast.

The team is using:

- Robotic boats.
- Buoys.
- Camera-equipped drones.

These technologies measure:

- Physical data.
- Chemical data.
- Biological data.

The measurements are collected in lakes where cyanobacteria are detected, resulting in large volumes of data related to the lakes and the development of the harmful blooms.

The project is also building new algorithmic models to assess the findings.

The information collected will lead to better predictions of when and where cyanobacterial blooms take place, enabling proactive approaches to protect public health in recreational lakes and in lakes that supply drinking water.

Such interdisciplinary training prepares the next generation of scientists to address societal issues with the proper modernized data science tools.

36.3 Developing Efficient Data-Driven Solutions

It takes gathering a lot of data, cleaning and preparing it, and then analyzing it to gain the insight needed to develop better solutions for today's enterprises.

To develop a better and more efficient solution, we must:

- Identify the problem and establish a clear understanding of it.
- Gather the data for analysis.
- Identify the right tools to use.
- Develop a data strategy.

Case studies are also helpful in customizing a potential solution.

Once these conditions exist and the available data is extracted, we can develop a machine learning model.

It takes time for an organization to refine best practices for data strategy using data science, but the benefits are worth it.

37 Data Science Applications in Business

Data science and big data are making an undeniable impact on businesses, changing day-to-day operations, financial analytics, and especially interactions with customers. Businesses can gain enormous value from the insights that data science provides, although it is sometimes difficult to see exactly how. The following examples illustrate how data science is applied in real-world scenarios.

37.1 How Consumers Generate Data

In the era of big data, almost everyone generates massive amounts of data every day, often without being aware of it. This digital trace reveals patterns of our online lives.

Some common ways in which we generate data include:

- When we search for or purchase a product on websites like **Amazon**, the platform begins making recommendations related to our searches. This type of system is known as a **recommendation engine** (a system that suggests products or content based on our preferences and previous behavior), which is a common application of data science.
- Companies such as **Amazon**, **Netflix**, and **Spotify** use algorithms to make specific recommendations derived from customer preferences and historical behavior.
- Personal assistants such as **Siri** on **Apple** devices use data science to devise answers to the infinite number of questions that end users may ask.
- **Google** tracks our online activities, shopping habits, and social media interactions. It then analyzes this data to recommend restaurants, bars, shops, and other attractions based on the information collected from our devices and our current location.

- Wearable devices such as **Fitbit**, **Apple Watch**, and **Android watches** contribute additional data, including:
 - Activity levels
 - Sleep patterns
 - Heart rate

37.2 How Data Science Impacts Business

Now that we understand how consumers generate data, we can see how data science is transforming businesses.

37.2.1 Industry Perspectives

- In **2011**, **McKinsey & Company** stated that data science was going to become the **key basis of competition**, supporting new waves of productivity, growth, and innovation.
- In **2013**, **UPS** announced that it was using data from customers, drivers, and vehicles in a new route guidance system aimed at saving time, money, and fuel.

Initiatives like these support the statement that data science will fundamentally change the way businesses compete and operate.

37.3 How Netflix Gains a Competitive Advantage

A well-known example of data science creating a competitive advantage is **Netflix**.

Netflix collects and analyzes massive amounts of data from millions of users, including:

- Which shows people are watching.
- What time of day people are watching.
- When people pause, rewind, and fast-forward.
- Which directors and actors people search for.

By analyzing users' preferences for certain directors and acting talent, and discovering which combinations people enjoy, Netflix can be confident that a show will be a hit before filming even begins. When this information is combined with the success of earlier versions of a show, Netflix can make informed investment decisions.

37.3.1 House of Cards Case Study

Netflix analyzed multiple data sources before investing in **House of Cards**.

The company knew that:

- Many of its users had streamed the work of **David Fincher**.
- Films featuring **Robin Wright** had always performed well.
- The British version of **House of Cards** was very successful.
- Significant numbers of people who liked **David Fincher** also liked **Robin Wright**.

All of this information combined suggested that purchasing the series would be a good investment for the company.

The prediction proved to be correct. The series became a huge success.

Thanks to data science, **Netflix knows what people want before they do.**

38 Applications of Data Science in Healthcare and Disaster Preparedness

Using Data Science techniques to understand and analyze the large data sets available today has a huge impact on human lives. It can provide targeted information to help healthcare professionals give the best treatment to patients, help predict natural disasters so that people can prepare early, and much more besides.

38.1 Data Science in Healthcare

In healthcare, data scientists use **predictive analytics** (using historical and current data to predict future outcomes) developed from:

- Data mining
- Data modeling
- Statistics
- Machine learning

These techniques help us find the best options for patients.

Predictive analytics examines all known factors for a disease, including:

- Gene markers (specific DNA sequences associated with particular diseases or traits)
- Associated conditions
- Environmental factors

It then recommends:

- Appropriate tests
- Suitable trials
- Suggested treatments

Every individual physician has their own store of knowledge gained from their studies, interests, and experiences. Data science systems that use predictive analytics ensure that all physicians can also access the latest information about diseases, tests, and treatment plans, tailored to their specific patient.

With this type of system, every physician has access to the same knowledge, allowing the best options to be consistently offered and improving patient outcomes.

38.2 Real-World Example: Boston Consulting Group and AdvaMedDx Study

A study by the **Boston Consulting Group** and **AdvaMedDx** (an industry association of medical diagnostics companies) examined the barriers to the adoption of potentially lifesaving diagnostic tests for patients with a specific cancer and a particular gene marker.

The study discovered that the biggest factor in a patient being offered a specific test was the patient's **oncologist** (a physician specializing in the diagnosis and treatment of cancer), who may or may not have known about the test and its relationship to the gene marker.

By providing extra information through data science tools, physicians can be made aware of the most helpful tests and treatments for a specific patient.

38.3 Electronic Medical Records (EMR) and Data Mining

There are many opportunities to explore other ways to mine data, such as from **Electronic Medical Records (EMR)** (digital versions of patients' medical histories) for different types of medical research.

NorthShore University HealthSystem in suburban Chicago, a leader in the implementation of Electronic Medical Records (EMR) systems, now offers guidance on data mining.

Notable achievements include:

- It is the first healthcare provider in America to be awarded the highest level of EMR deployment for both inpatient and outpatient care.
- This remarkable effort has generated much-anonymized data available for innovative analytics research.
- Developing more sophisticated big data analytics capabilities helps healthcare organizations move from basic descriptive analytics toward predictive insights through data science.

38.4 Data Science in Disaster Preparedness

In the field of **Disaster Preparedness**, the ability to save lives using Data Science tools has been under development for many years.

The use of predictive analytics tools is improving and providing new data analysis in a multitude of ways, alerting populations to danger faster than ever before.

Large, high-quality data sets can be used to predict the occurrence of numerous types of natural disasters, which can be the difference between life and death for thousands of people.

Natural disasters that can be predicted with the help of Data Science include:

- Earthquakes
- Hurricanes

- Tornados
- Floods
- Volcanic eruptions

38.5 Research Example: University of Warwick

Recent research at the **University of Warwick** in the UK used social media content such as photos and keywords to track the development of:

- Floods
- Hurricanes
- Other weather events

When added to the information recorded by scientists and weather stations, this type of data can be used to improve the predictions for localised weather events.

38.6 Data Science Education

Because the real benefit of this knowledge is so important, schools are starting to include this type of Data Science education in their curriculum.

For instance, the **University of Chicago Graham School** offers a **Master of Science** course in **Threat and Response Management**.

38.7 Key Takeaways

Data Science tools enable organizations to:

- Analyse vast quantities of data from widely different sources.
- Present information in a way that allows data scientists to gain new knowledge.
- Improve healthcare decision-making through predictive analytics.
- Support medical research using Electronic Medical Records (EMR).
- Predict natural disasters more accurately.
- Alert populations to danger faster than ever before.
- Improve localized weather predictions using scientific and social media data.
- Save hundreds of lives in some cases through data-driven insights.

39 The Final Deliverable

39.1 Course Information

Course Textbook: *Getting Started with Data Science*

Author: Murtaza Haider

Publisher: IBM Press, 1st Edition (December 13, 2015), Print

Prescribed Reading: Chapter 3, Pages 52–53

39.2 Analytics Must Lead to Communication

The ultimate purpose of analytics is to communicate findings to the concerned stakeholders who might use these insights to formulate policy or strategy.

Analytics summarize findings in **tables** and **plots**. We should then use these insights to build a **narrative** that effectively communicates the findings.

In academia, the final deliverable is typically in the form of:

- Essays
- Reports

These deliverables are usually **1,000 to 7,000 words** in length.

39.3 What Final Deliverables Look Like in Practice

In consulting and business, the final deliverable can take several forms, including:

- A small document of **fewer than 1,500 words**, illustrated with tables and plots.
- A comprehensive document comprising **several hundred pages**.

Large consulting firms, such as **McKinsey** and **Deloitte**, routinely generate analytics-driven reports to:

- Communicate their findings.
- Establish their expertise in specific knowledge domains.

39.4 A Deloitte Report as an Example of Storytelling with Data

One example of an effective analytics deliverable is the "**United States Economic Forecast**" published by **Deloitte University Press**.

This **24-page report** focuses on the state of the **U.S. economy** as observed in **December 2014**.

The report begins with a strong opening (grabber) by highlighting that, contrary to popular perception, **economic growth and job growth in the United States have been quite robust**.

Rather than being a simple collection of facts, the report is a carefully crafted narrative that:

- Cites **Voltaire**.
- Follows a distinct theme.
- Focuses on the positive developments in the U.S. economy.

Some of the positive developments discussed include:

- Increased investment in manufacturing equipment in the United States.
- The likelihood of higher consumer consumption resulting from lower oil prices.

39.5 Using Data Visualization to Support the Narrative

The Deloitte report uses **time series plots** (graphs showing changes over time) to illustrate trends in different markets.

Examples include:

- A **GDP growth chart** showing:
 - How the economy contracted during the **Great Recession**.
 - How the economy rebounded afterward.
 - Four likely future economic scenarios.
- A **consumer spending plot** showing changes in spending patterns.

The accompanying narrative discusses **income inequality in the United States** and refers to **Thomas Piketty's** book on the same topic.

The report explains that:

- Many consumers did not experience an increase in their real incomes over the years.
- Despite stagnant real incomes, they continued to maintain their level of spending.

Additional graphics focus on:

- Housing
- Business sector
- Government sector
- International trade
- Labor markets
- Financial markets
- Prices

The appendix includes **four tables** documenting the data for the **four scenarios** discussed in the report.

39.6 Why the Deloitte Report Is Effective

Deloitte's "**United States Economic Forecast**" fulfills the purpose intended by its authors.

The report:

- Uses data and analytics to generate likely economic scenarios.
- Builds a powerful narrative supporting the thesis that the **U.S. economy is performing much better than most people believe**.
- Demonstrates Deloitte's capability to analyze economic data.
- Shows Deloitte's ability to prescribe strategies for addressing economic challenges.

39.7 Why Narrative Matters

Consider what would happen if the narrative were removed from the Deloitte report and the findings were presented only as a **PowerPoint presentation** containing:

- **Eight graphics**
- **Four tables**

Without the narrative, the presentation would fail to communicate the carefully crafted message that references **Thomas Piketty** and **Voltaire**.

The Deloitte report is an excellent example of **storytelling with data**, where the narrative transforms data and visualizations into a compelling and meaningful message.

We are encouraged to read the report and determine for ourselves whether the deliverable would have been equally powerful without the narrative.

39.8 Planning the Deliverable Before the Analysis

Working backward from the Deloitte report demonstrates the importance of planning the final deliverable before beginning the analysis.

Before the authors started their analysis, they would have:

- Discussed the scope of the final deliverable.
- Deliberated on the key message of the report.
- Determined the data and analytics needed to support their case.

The initial planning and conceptualization of the final deliverable are therefore extremely important for producing a compelling document.

Embarking on analytics without giving due consideration to the final deliverable is likely to result in a poor-quality document where the analytics and narrative struggle to blend effectively.

40 Data Science Applications – Lesson Summary

Data science enables organizations to drive business goals, improve efficiency, make predictions, and even save lives. As data scientists, we follow a structured process to help organizations achieve these outcomes by transforming data into actionable insights.

40.1 Purpose of Data Science

All organizations ultimately use data science for the same reason—to discover **optimal solutions to existing problems**.

To discover these solutions, an organization should:

- Identify the problem.
- Establish a clear understanding of the problem.
- Measure the relevant data.
- Capture and gather the required data.

Measurement is the first step for an organization to solve its problems using data. If something is not measured, it is challenging to improve or change it. If an organization is not capturing the data, we should help determine how to capture it.

Important Principle:

- **Never overwrite old data—it is always relevant and never gets old.**

Once the data has been collected, we can begin examining and cleaning it.

As data scientists, our responsibilities include:

- Helping the organization identify appropriate analysis tools.
- Developing an effective analysis strategy.
- Considering case studies when customizing a potential solution.
- Identifying suitable analysis tools.
- Developing machine learning and statistical models.

It takes time for an organization to refine data strategy best practices, but the benefits are worth the effort.

40.2 How Organizations Leverage Data Science

Everyone who uses the Internet generates massive amounts of data every day, often without being aware of it. Organizations can leverage this data to reveal meaningful patterns and improve decision-making.

40.2.1 Recommendation Engines

Companies such as **Amazon** use recommendation engines to suggest products related to a user's searches.

These recommendation engines are built by data scientists using:

- Machine learning
- Statistical models

40.2.2 Route Optimization

UPS uses data science to support business decisions by analyzing data from:

- Customers
- Drivers
- Vehicles

This enables the company to create optimized routes for drivers, resulting in more efficient use of:

- Time
- Money
- Fuel

40.2.3 Dynamic Driver Allocation

Uber uses data to:

- Place the correct number of drivers in the right location.
- Match supply with rider demand.
- Offer rides at a price the rider is willing to pay.

Initiatives like these demonstrate how data science fundamentally changes the way businesses compete and operate.

40.3 Data Science for Competitive Advantage

Businesses also use data science to gain competitive advantages.

Streaming companies collect and analyze massive amounts of data from millions of users, including:

- Which shows users watch.
- What time of day they watch them.
- When they pause, rewind, or fast-forward.
- Which directors they search for.
- Which actors they search for.

By thoroughly analyzing user preferences and behaviors, a streaming company can confidently predict the success of a show **before filming begins**.

40.4 Data Science in Healthcare

Beyond improving business performance, data science also helps save lives.

In healthcare, data scientists use **predictive analytics** (analytics that predict future outcomes using historical and current data) developed from:

- Data mining
- Data modeling
- Statistics
- Machine learning

Predictive analytics examines factors related to diseases, including:

- Gene markers

- Associated conditions
- Environmental factors

Data scientists use predictive models to help physicians:

- Recommend appropriate tests.
- Identify suitable clinical trials.
- Suggest treatments.

Developing more extensive data analytics capabilities enables healthcare organizations to move from **descriptive analytics** (understanding what has happened) toward **predictive insights** (forecasting what is likely to happen) through data science.

40.5 Predicting Natural Disasters

Data science tools enable organizations to analyze vast amounts of data from multiple sources.

Large, high-quality datasets help predict the occurrence of natural disasters such as:

- Earthquakes
- Hurricanes
- Tornados
- Floods
- Volcanic eruptions

Data scientists present this information in a way that enables organizations to gain new understandings and, in some cases, save lives.

40.6 Presenting Findings

When presenting findings, we provide organizations with a **final deliverable** that explains and summarizes the conclusions.

The format of the final deliverable depends on the domain.

40.6.1 Academia

The final deliverable typically includes:

- Research papers
- Reports

40.6.2 Consulting and Business

The final deliverable is often illustrated using:

- Tables
- Plots

We generate analytics-driven reports to communicate findings effectively and establish powerful, convincing, and evidence-based narratives.

41 Glossary: Data Science Application Domains

This alphabetized glossary contains many of the terms in this course. These terms are important for us to recognize when working in the industry, participating in user groups, and participating in other certificate programs.

Term	Definition	Topic Where the Term is Introduced
Arithmetic Models	Data science often uses mathematical models to analyze data and predict outcomes.	<i>Old problems, new data science solutions</i>
Case Study	In-depth analysis of an instance of a chosen subject to draw insights that inform theory, practice, or decision-making.	<i>Old problems, new data science solutions</i>
Data Mining	Extracting information from raw data, such as making decisions, predicting trends, or understanding phenomena.	<i>How Data Science is Saving Lives</i>
Data Science	The field involves collecting, analyzing, and interpreting data to extract valuable insights and make informed decisions.	<i>Old problems, new data science solutions</i>
Data Strategy	A plan that outlines how an organization will collect, manage, and use data to achieve its goals.	<i>Old problems, new data science solutions</i>
Predictive Analytics	Using data, algorithms, models, and machine learning to make predictions.	<i>How Data Science is Saving Lives</i>

42 Skills, Learning Path, and Career Opportunities in Data Science

42.1 Skills Required to Become a Data Scientist

A real data scientist, especially a high-end data scientist, is mostly a **PhD**. They often come from backgrounds such as:

- Physics
- Statistics
- Computer Science
- Mathematics

They also need strong knowledge of:

- Databases
- Statistics
- Probability

However, if we are joining a data science team, the first skills we need include:

- Knowing how to program.

- Having computational thinking.
- Completing at least one programming course.
- Understanding algebra, at least up to analytic geometry.
- Having knowledge of calculus (preferably).
- Understanding basic probability.
- Understanding basic statistics.
- Knowing the difference between different statistical distributions.
- Understanding databases.

One of the easiest places to start is **relational databases**, which store lots and lots of data. We can first walk before we run by understanding:

- Computers
- Databases
- How data is stored

If we understand relational databases, we can still use **big data clusters** as if they were one large relational database. We do not have to understand the entire **MapReduce programming model** at the beginning.

As we progress further in the field, we need a much deeper understanding of:

- Computer Science theory
- Statistics
- Probability

High-end, PhD-level data scientists work at the intersection of these disciplines.

42.2 Self-Learning and Hands-On Practice

Self-learning is an important part of becoming a data scientist.

For example:

- Hadoop was learned through self-study by reading articles, watching videos, experimenting, and building projects.
- A **Beowulf cluster** (a group of computers connected through a **TCP/IP network** to work together as a high-performance computing system) was explored through experimentation.

After receiving a grant from **Citi Bank**, a **five-node HPC (High Performance Computing) cluster** was built at a university. Although it was small, researchers appreciated it because they could run their workloads **40 and 50 times faster**.

One of the best ways to learn is by actually doing the work. Building systems and experimenting with technology helps us understand concepts much better than only reading about them.

42.3 Learning Platforms and Practical Tools

Modern online learning platforms make it possible to learn by practicing immediately.

Some useful tools and platforms include:

- IPython
- Jupyter Notebooks
- Zeppelin

These tools allow us to:

- Take courses.
- Execute code immediately.
- Observe the results.
- Experiment with different ideas.
- Develop a better understanding of what is actually happening.

42.4 Motivation and Continuous Learning

Motivation is one of the biggest challenges in learning.

One approach to keeping learners motivated is the **badge system** used by **Big Data University**. Such systems encourage learners to continue progressing through courses.

Ultimately, continuous learning depends on the individual. We have to understand our goals and remain motivated to achieve them.

42.5 Where Data Science Fits in an Organization

Data science should probably not sit under the **CIO (Chief Information Officer)**.

In many companies, CIOs come from:

- Accounting backgrounds
- Finance backgrounds

Instead, data science is more effective when it comes from the **research side** of the organization.

Data scientists are commonly found in organizations with strong research agendas, including:

- Pharmaceutical companies
- Financial organizations
- Technology companies

42.6 Demand for PhD-Level Data Scientists

The demand for **PhD-level data scientists** is extremely high.

Many graduates move to companies such as:

- Facebook
- LinkedIn
- Uber
- Lyft

These professionals:

- Earn large amounts of money.
- Work on challenging and interesting real-world problems.

One example is determining **how to schedule Uber cars**, which requires working with **enormous amounts of data**.

43 Hiring the Right People for a Data Science Team

Companies hiring for a data science team—whether for a **Data Scientist**, **Data Analyst**, or **Chief Data Scientist**—often tend to look for someone who has every possible skill:

- Domain-specific knowledge
- Excellent ability to analyze structured and unstructured data
- Strong presentation skills
- Great storytelling skills

When we put all these requirements together, we realize we are looking for a **"unicorn"** (a person with every desired skill), and the chances of finding such a person are very rare.

Instead, we should evaluate the pool of applicants and identify the person who has the greatest resonance with the company's **DNA** (culture, values, and business needs).

Although analytics skills can be taught if someone dedicates time and effort, what matters most is whether the person is passionate about the type of business the company does.

For example:

- Someone could be an excellent data scientist in a retail environment but may not be excited about working in IT-related firms or with gigabytes of weblogs.
- Someone who is genuinely excited about weblogs or health-related data can contribute much more to productivity.

43.1 Important Qualities When Hiring Data Scientists

When building a data science team, the following qualities are considered more important than technical skills.

43.1.1 Curiosity

The first quality to look for is curiosity.

We should ask whether a person is curious not only about data science but about everything around them, such as:

- Why is this room painted a certain way?
- Why do the bookshelves have books?
- What kinds of books are on the shelves?

A data scientist should have a certain degree of curiosity about everything within their field of vision.

43.1.2 Sense of Humor

The second quality is having a sense of humor.

Being lighthearted is important because if someone is too serious, they may take everything too seriously and may not be able to appreciate the lighter aspects of situations.

43.1.3 Social Skills and Storytelling

If we had to establish a hierarchy, technical skills would come after:

- Social skills
- Curiosity
- Sense of humor
- The ability to tell a story
- The ability to recognize that there is a story within the data

Once these qualities are present, we can then evaluate whether the person has the required technical skills.

If there is some indication of technical ability, the remaining technical skills can be taught through training.

However, we cannot teach:

- Curiosity
- Storytelling
- Sense of humor

43.2 Technical and Professional Skills Required

There is no hard and fast rule for hiring data scientists. Every hiring decision depends on the specific situation.

However, a data scientist should possess certain technical capabilities.

43.2.1 Data Skills

A data scientist should be able to:

- Work with data
- Manipulate data
- Analyze data
- Communicate the findings effectively

Many people do not really care about measures such as **R-square** (a statistical measure that indicates how well a model explains the variation in the data) or the **confidence interval** (a range of values that is likely to contain the true value of a parameter).

Therefore, we should be able to introduce these concepts and explain them in a compelling and understandable way.

43.2.2 Relationship Building

Because data science is still a relatively new field, people working in these roles need to:

- Build relationships
- Work across different departments
- Be relatable to others

43.2.3 Analytical Background

A strong data scientist should have:

- A good mathematics background
- A good statistics background
- Strong problem-solving abilities
- Analytical thinking
- The ability to analyze problems effectively

They should also:

- Love playing with data
- Know how to create data visualizations

43.3 Building a Data Science Team

Before hiring anyone, a company should clearly understand what it expects from its data science team.

The required roles may change as the team grows.

Depending on business needs, the company may require:

- Data Engineers
- Architects
- Visualization Designers
- More specialists capable of multiplying large matrices

The hiring strategy should always align with the objectives of the data science team.

43.4 Technical Skills Based on the Working Environment

The required technical skills depend on the type of environment in which we want to work.

43.4.1 Structured Data Environment

If we plan to work in a structured data environment, such as traditional market research, we should develop:

- Statistical knowledge
- Knowledge of basic statistical algorithms
- Knowledge of machine learning algorithms

These become the primary technical tools in such environments.

43.4.2 Big Data Environment

Working with big data requires a different skill set.

The process consists of three major steps:

1. Store large amounts of data.
2. Manipulate large amounts of data.
3. Apply algorithms to large datasets.

This is a three-step process.

Most importantly, it begins with deciding:

- Where we want to work
- Which field interests us
- Which domain excites us

43.5 Recommended Platforms and Tools

The choice of tools depends on the type of work we want to perform.

43.5.1 Traditional Predictive Analytics (Without Big Data)

Recommended tools include:

- R
- Stata
- Python

43.5.2 Unstructured Data

If we are working mostly with unstructured data, **Python** is more suitable than **R**.

43.5.3 Big Data

For big data environments, the common platforms are:

- Hadoop
- Spark

Ultimately, the tools we choose should depend upon:

- Where we would like to work
- What kind of work excites us

43.6 Communication and Storytelling Skills

Technical skills alone are not enough.

An equally important aspect of data science is the ability to communicate findings effectively.

These include:

- Communication skills
- Presentation skills
- Storytelling skills

After completing the analysis, we should be able to tell a compelling story from the results.

For example:

- If we have a very large table, we should be able to synthesize it into a more appealing and understandable format.
- Whether the findings are presented on a screen, in a presentation, or in a document, the results should clearly communicate the insights.

The ability to present findings:

- Verbally
- Through presentations
- In written documents

is just as important as technical expertise.

43.7 Creating Impact Through Data Storytelling

When presenting results, we should create a sense of discovery for the audience.

A useful analogy is driving on a mountain road:

- We approach a sharp turn without seeing what lies ahead.
- As we make the turn, a tremendous valley suddenly appears.
- We experience a great sense of awe because we did not know what was beyond the turn.

Similarly, when we communicate a valuable data insight effectively:

- People experience something they were not expecting.
- They become aware of something they did not know before.
- They gain new knowledge.
- They feel empowered.
- They begin generating new ideas based on the newly discovered insight.

This creates a great sense of joy.

As data scientists, we are able to share this value with our clients because we enable them to discover meaningful insights through data.

44 Why Data Science Is One of the Most Promising Career Fields

The emergence of the **Internet of Things (IoT)** and advances in **distributed computing** have brought vast amounts of data and the technological capability to analyze it. Now that we can extract useful insights and new knowledge, we need to know how to shape that data to focus on what to do with it and what it can do for us.

This is where **Data Science** comes in.

44.1 Growing Demand for Data Science

Companies such as **LinkedIn**, **Glassdoor**, **Indeed**, and **Dice** track employment trends, which show that a career in Data Science moved up the list of the most promising jobs to become **number one since 2016**. It remained **one of the top three career choices for 2020**.

According to **Dice**, job postings come from companies across a wide variety of industries, not just the technology sector.

Global Industry Analysts Incorporated predicted that the **Data Science platform market** would grow by **US\$314.8 billion by 2025**, driven by a **compound growth rate of 38.2%**.

44.2 Industry Insights and Talent Shortage

The **McKinsey Global Institute** warned of huge talent shortages for **data and analytics** by **2018**.

Forrester Research Analyst Brandon Purcell stated in **January 2019** that the demand for Data Scientists would only continue to grow as organizations increasingly rely on **data-driven insights**.

We are now well into that period, and recruiters are finding it difficult to fill the growing need for talented Data Scientists.

44.3 Why Choose a Career in Data Science?

One of the biggest motivations for pursuing a career in Data Science is that it applies to **almost every discipline**.

If we have:

- An aptitude and desire to work with data
- An interest in coding
- No problem learning mathematics and statistics
- Good storytelling skills (the ability to communicate data insights clearly)

then we can certainly enter the field of Data Science and excel.

For most people, this means acquiring additional tools and skills and continuously learning new tools and techniques used in the field.

44.4 Women in Data Science Initiative

The **Women in Data Science (WiDS) Initiative**, spearheaded by the **Stanford Institute for Computational and Mathematical Computing**, is committed to inspiring and educating Data Scientists worldwide, regardless of gender, while supporting women in the field.

44.5 Preparing for a Career in Data Science

When seeking a career in Data Science, we need to make sure our skill set matches the role we are targeting.

We can tailor our skill set to the specific area we want to enter by adding any missing skills through one of the many excellent online training resources.

By doing so, we will be well prepared for a fascinating and rewarding career.

44.6 Key Takeaway

This is an excellent time to move into the field of Data Science because there are diverse career opportunities available along with abundant educational resources that make entering the field a reality.

45 Importance of Mathematics and Statistics for Data Science

45.1 Build a Strong Foundation

To prepare for a career in data science, we should:

- Learn how to program.
- Learn some math.
- Take a course in probability.
- Learn a little bit of statistics.
- Then, play.
- Build something and write something.

When we say *build*, programming and building systems is not just physical. We can build computer systems, statistical systems, or other types of systems.

Once we try to build something, we naturally discover the tools we need. We may ask questions such as:

- What does an **inner product** mean? (*a mathematical operation used to measure the relationship between vectors and widely used in machine learning and linear algebra*)
- How do we use it?

As we encounter these concepts while building projects, we can learn them as needed.

Starting early gives us a significant advantage. When we get to college, we will have a big jump on many other college students. By the time we graduate, that advantage becomes even greater. Besides creating better career opportunities, learning data science is also enjoyable and rewarding.

45.2 Preparing for Data Science in High School

If we are in high school and considering data science, we should:

- Get familiar with databases.
- Start learning SQL.
- Start thinking about computer science if it interests us.
- Take a computer science course if our school offers one.

These are all valuable parts of becoming a data scientist.

45.3 Develop Curiosity and Creativity

Beyond technical skills, we should foster our creativity and curiosity.

Activities that can help include:

- Playing detective games.
- Participating in treasure hunts.
- Engaging in activities that encourage problem-solving and exploration.

These experiences help us become naturally curious, which is an important characteristic of successful data scientists.

45.4 Encourage Experimentation

We should encourage curiosity and experimentation.

Science fairs are an excellent example because they encourage us to:

- Ask questions.
- Answer them using the scientific method.
- Learn through experimentation.

In data science, we follow the same process, but instead of creating vinegar volcanoes, we experiment with datasets to answer questions and discover insights.

45.5 Learn from Real-World Data

Real-world events provide excellent opportunities to understand data science.

For example, during an election season, news about polls and survey results can help us start conversations about questions such as:

- How do the people who run polls know the likely outcome?
- How can they predict what will happen in the election?

These discussions demonstrate how data science is applied in real-world situations.

45.6 Why Data Science Is a Great Career

We should encourage anyone interested in data science to pursue it because:

- It is a great career.
- It will definitely continue to be in demand in the future.
- It is one of the highly sought-after knowledge professions.
- It is important to businesses around the world.

Data scientists help companies:

- Grow effectively.
- Learn how to do things more efficiently.
- Learn how to do things smarter.

There will always be a need for people with these skills, and data scientists are those people.

45.7 Don't Be Discouraged by Mathematics

Many successful and popular data scientists were not great mathematics students.

Arithmetic and mathematics in school are not necessarily everyone's strongest subjects.

However, when mathematics is connected to a real problem rather than hypothetical numbers or problem statements, it becomes much easier to understand and apply.

Knowing the people who will benefit from the mathematics we use makes the learning process more meaningful and rewarding.

46 The Report Structure

46.1 Course Information

Course Textbook: *Getting Started with Data Science*

Publisher: IBM Press; 1st Edition (December 13, 2015) (Print)

Author: Murtaza Haider

Prescribed Reading: Chapter 3, Pages 60–62

46.2 Planning the Report Structure

Before starting the analysis, we should think about the structure of the report. The structure depends on the length of the document.

- A **brief report of five or fewer pages** is more direct and presents a summary of the key findings.
- A **detailed report of more than 100 pages** incrementally builds the argument and contains:
 - Details about other relevant works
 - Research methodology
 - Data sources
 - Intermediate findings
 - Main results

The author reviewed reports by leading consulting firms, including **Deloitte** and **McKinsey**, and observed that report length largely depends on its purpose.

- **Brief reports** are generally written as commentaries on current trends and developments that attract public or media attention.
- **Detailed and comprehensive reports** provide:
 - A critical review of the subject matter
 - Extensive data analysis
 - Commentary
 - Newly collected data or interviews with industry experts to answer research questions

46.3 Essential Front Matter

Even if we expect the report to be **five or fewer pages**, it is recommended that the report follows a prescribed structure that includes:

- Cover Page
- Table of Contents
- Executive Summary
- Detailed Contents
- Acknowledgments
- References
- Appendices (if needed)

46.3.1 Cover Page

The author frequently finds that reports are submitted without a cover page. This issue is not limited to undergraduate students—it is also common among doctoral candidates and even reports published by reputed business firms.

At a minimum, the cover page should include:

- Title of the report
- Names of the authors
- Author affiliations
- Contact information
- Name of the institutional publisher (if any)
- Date of publication

The publication date is especially important because reports without the year and month cannot be properly cited. From a business perspective, including contact information also makes it easier for readers to reach the authors.

46.3.2 Table of Contents (ToC)

A **Table of Contents (ToC)** is compared to a map for a journey never taken before. It gives readers an overview of the document before they begin reading.

A ToC should include:

- Main headings

- List of tables
- List of figures

We should never hesitate to include a ToC, especially when the document (excluding the cover page, table of contents, and references) is **five or more pages** long.

46.3.3 Executive Summary or Abstract

Even for a short document, it is recommended to include an **Abstract** or an **Executive Summary**.

- The executive summary should explain the main arguments in **three paragraphs or less**.
- For reports running **a few hundred pages**, the executive summary may be longer.

46.4 Core Sections of an Analytical Report

46.4.1 Introduction

An introductory section helps readers who are new to the topic understand the problem before moving into more detailed analysis.

46.4.2 Literature Review

A review of relevant research should generally follow the introduction.

The length of the literature review depends on how contested the subject matter is.

- If most researchers agree on the topic, the literature review can be brief and cite only the most influential authors.
- If the topic contains numerous differing viewpoints and caveats, we should cite the relevant research to provide sufficient context before beginning the analysis.

The literature review is also the appropriate place to:

- Highlight gaps in existing knowledge.
- Introduce the research questions.
- Present the hypothesis.

46.4.3 Methodology

The methodology section explains:

- Research methods
- Data sources used in the analysis

If new data has been collected, we should explain the data collection process in sufficient detail.

The literature review should also support our choice of:

- Variables
- Data
- Methods

and explain how these choices help answer the research questions.

46.4.4 Results

The results section presents the empirical findings.

The analysis typically begins with:

- Descriptive statistics (see **Chapter 4, "Serving Tables"**)
- Illustrative graphics
 - Plots (see **Chapter 5, "Graphic Details"**)
 - Maps (see **Chapter 10, "Spatial Data Analytics"**)

The analysis then progresses toward formally testing the hypothesis (see **Chapter 6, "Hypothetically Speaking"**).

Depending on the type of analysis, we may use:

- Regression models (see **Chapter 7, "Why Tall Parents Don't Have Even Taller Children"**)
- Categorical analysis
 - **Chapter 8, "To Be or Not to Be"**
 - **Chapter 9, "Categorically Speaking About Categorical Data"**
- Time-series analysis (see **Chapter 11, "Doing Serious Time with Time Series"**)
- Data mining techniques (see **Chapter 12, "Data Mining for Gold"**)

Many business reports present results in a more understandable manner by minimizing statistical details and relying on illustrative graphics to summarize findings.

46.5 From Results to Discussion and Conclusion

46.5.1 Discussion

The discussion section builds the main arguments using the results presented earlier.

This section relies on narrative to help numbers communicate the central thesis.

During the discussion, we should:

- Refer back to the research question.
- Address the knowledge gaps identified earlier.
- Explain how the findings contribute the missing piece to the overall understanding.

The author also notes that not every analytics project produces definitive evidence. Often, the results provide only a partial answer, accompanied by numerous caveats(safety measures).

46.5.2 Conclusion

The conclusion section generalizes the specific findings and presents them in a way that emphasizes their overall importance.

In this section, we should:

- Summarize the key findings.
- Ensure readers focus on the overall contribution rather than only the caveats.
- Identify future developments in research and practical applications that may result from the work.

46.6 Final Elements

The remaining components of the report include:

- References
- Acknowledgment section (acknowledging those who supported the work)
- Appendices (if needed)

46.7 Have We Done Our Job as Writers?

As data scientists, we are expected to perform thorough analysis using appropriate data and suitable analytical tools.

As writers, we are responsible for communicating our findings effectively to readers.

The journal **Transport Policy**, a leading research publication in transportation planning, provides a checklist for authors before submitting manuscripts. The author believes this checklist is equally valuable for aspiring data scientists.

Before finalizing a report, we should ask ourselves the following questions:

- Have we told readers, at the outset, what they might gain by reading our paper?
- Have we made the aim of our work clear?
- Have we explained the significance of our contribution?
- Have we set our work in the appropriate context by providing sufficient background, including a complete set of relevant references?
- Have we addressed the question of practicality and usefulness?
- Have we identified future developments that might result from our work?
- Have we structured our paper in a clear and logical fashion?

47 Careers and Recruiting in Data Science – Lesson Summary

47.1 Overview

In this topic, we investigate what companies seek in a competent, experienced data scientist and learn how to position ourselves to get hired as a data scientist.

Although data scientists come from diverse backgrounds, they share qualities and skills that consistently set them apart from other data-related roles.

47.2 What Companies Look for in a Data Scientist

Companies recruiting data scientists may seem to want data scientists to have it all. They may seek a person who has all the desired skills, ranging from:

- Domain-specific knowledge
- Proficiency in analyzing structured data
- Proficiency in analyzing unstructured data
- Skills in presenting findings
- Storytelling skills

Realistically, this person is a rare find.

Instead, companies need to develop teams of people who work together and collectively possess these skills rather than seeking a single individual who has all of these qualities. They may also need to identify individuals with potential and help them develop the skills they need.

47.3 Passion and Curiosity

What really matters is finding someone who is passionate about the kind of business the company does.

Companies should look for someone who:

- Is excited about working with data in their particular industry.
- Is curious and can ask interesting, meaningful questions about the types of data they intend to collect.

For example, a person who could be a great data scientist in the retail environment may not be excited about working in IT-related firms or working with gigabytes of health-related web logs.

Companies will find that excitement results in high productivity.

That excitement also relates back to curiosity, which we discussed in another lesson as an important trait for a data scientist. Someone excited about their field is likely to be curious and find the right questions to ask.

Constantly wondering why things are the way they are helps us stay motivated and engaged.

Similarly, self-learning and tinkering (experimenting and exploring technologies through hands-on practice) are also helpful traits.

47.4 Essential Technical Skills

As data scientists, we should love to play with data and create data visualizations.

We need to:

- Think analytically and computationally.
- Have a strong background in mathematics, statistics, and probability to reach valid conclusions with data.
- Develop computer programming skills.

The tools we use may vary depending on the type of data we work with. However, common programming languages include:

- Python
- R (an open-source programming language developed for statistical analysis)

Because data scientists work with large quantities of stored data, we must also understand:

- Data storage systems
- Data retrieval systems
- Structured data
- Unstructured data

With artificial intelligence, we also need to know common machine learning algorithms to gain insights from data.

47.5 Communication and Storytelling

One or more members of the data science team at an organization will write a report as the final step of the analysis. Therefore, the following skills are also important:

- Communication skills
- Instructional skills
- Presentation skills
- Storytelling skills

The report synthesizes (combines and summarizes) large amounts of data to create a narrative that engages and surprises the reader.

A clearly organized and logical report should communicate:

- What the reader gains by reading the report.
- Clearly defined goals.

- The significance of our contribution.
- Appropriate context by providing sufficient background.
- Why the work is practical and useful.
- Plausible future developments that might result from the work.

47.6 Dr. Hader's Storytelling Analogy

As **Dr. Hader** explained, imagine the reader is driving on a mountain. There is a sharp turn ahead, and they cannot see what lies beyond the turn. They go around the curve, and suddenly they see a tremendous valley in front of them. They experience a great sense of awe.

Similarly, when we present a significant finding and communicate it effectively, this is what people feel because they were not expecting it.

47.7 Key Takeaways

Companies should focus on building data science teams rather than searching for a single individual with every desired skill.

An effective data science team should include people who:

- Are curious.
- Understand the subject matter well.
- Love working with data.
- Have expertise in statistics.
- Have expertise in mathematics.
- Have expertise in machine learning.
- Have expertise in computer programming.

Lastly, and possibly most importantly, the team should include a skilled storyteller who can present the team's findings creatively and engagingly.

48 Glossary: Careers and Recruiting in Data Science

This glossary contains many of the important terms covered in this course. These terms are essential for us to recognize when working in the data science industry, participating in user groups, and taking part in other certificate programs.

Term	Definition	Topic Where the Term is Introduced
Analytical skills	The ability to analyze information systematically, logically, and in an organized manner.	<i>Recruiting for Data Science</i>
Chief Information Officer (CIO)	A business executive responsible for an organization's information technology systems and technology-related initiatives.	<i>How Can Someone Become a Data Scientist</i>

Computational thinking	Breaking problems into smaller parts and using algorithms, logic, and abstraction to develop solutions. It is often used, but not limited to, computer science.	<i>How Can Someone Become a Data Scientist</i>
Data clusters	A group of similar, related data points that are distinct from other clusters.	<i>How Can Someone Become a Data Scientist</i>
Executive summary	Usually occurring at the beginning of a research paper, this section summarizes the important parts of the paper, including its key findings.	<i>The Report Structure</i>
High-Performing Computing (HPC) cluster	A computing technology that uses a system of networked computers designed to solve complex and computationally intensive problems in traditional environments.	<i>How Can Someone Become a Data Scientist</i>
Mathematical computing	The use of computers to calculate, simulate, and model mathematical problems.	<i>Importance of Mathematics and Statistics for Data Science</i>
Matrices	Plural of matrix , matrices are a rectangular (tabular) array of numbers often used in mathematics, statistics, and computer science.	<i>Recruiting for Data Science</i>
Stata	A software package used for statistical analysis.	<i>Recruiting for Data Science</i>
Statistical distributions	A way of describing the likelihood of different outcomes based on a dataset. The bell curve is a common statistical distribution.	<i>How Can Someone Become a Data Scientist</i>
Structured Query Language (SQL)	A language used for managing data in a relational database.	<i>Importance of Mathematics and Statistics for Data Science</i>
TCP/IP network	A network that uses the TCP/IP protocol to communicate between connected devices on that network. The Internet uses TCP/IP.	<i>How Can Someone Become a Data Scientist</i>

49 Careers and Recruiting in Data Science

At this point, we know that:

- Data Science helps physicians provide the best treatment for their patients.
- Data Science helps meteorologists predict the extent of local weather events.
- Data Science can even help predict natural disasters like earthquakes and tornadoes.
- Companies can start their data science journey by capturing data. Once they have data, they can begin analyzing it.
- Everyone who uses the Internet generates mass amounts of data daily.
- Amazon and Netflix use recommendation engines, and UPS uses data from customers, drivers, and vehicles to use the drivers' time and fuel efficiently.

- The purpose of the final deliverable of a Data Science project is to communicate new information and insights from the data analysis to key decision-makers.
- The report should present a thorough analysis of the data and communicate the project findings.
- Companies should look for someone excited about working with the data in their particular industry. They should seek out someone curious who can ask interesting, meaningful questions about the types of data they intend to collect. They should hire people who love working with data, are fluent in statistics, and are competent in applying machine learning algorithms.

49.1 Characteristics of an Effective Data Science Report

A clearly organized and logical report should communicate the following to the reader:

- What they gain by reading the report.
- Clearly defined goals.
- The significance of our contribution.
- Appropriate context by giving sufficient background.
- Why this work is practical and useful.
- Conjecture plausible future developments that might result from our work.

50 Case Study: Final Assignment

This case study explores the data scientist's career path and key attributes, highlighting the skills, education, and experiences required to excel in this dynamic field. We follow the story of **Lila**, a fictional individual who aspires to become a successful data scientist.

50.1 Education and Skill Acquisition

With an economics undergraduate degree and a substantial data analysis background, Lila finds data science and its potential to drive meaningful change captivating. Inspired by her experiences, she makes a determined decision to transition her career and step into the role of a data scientist.

Lila realizes that to embark on her data science journey, she needs to enhance her skills and knowledge. She enrolls in the **IBM Data Science Professional Certificate** online program, which covers key topics such as:

- Statistics
- Machine Learning
- Data Analysis
- Programming languages, including **Python** and **SQL**

She diligently completes the coursework and practices her coding skills on real datasets.

50.2 Building a Strong Foundation

As she progresses in her studies, Lila gains a deep understanding of data science fundamentals, including:

- Data manipulation
- Data visualization

She learns to use Python libraries such as:

- **NumPy**
- **Pandas**
- **Matplotlib**

This strong foundation equips her with essential skills for data analysis.

50.3 Visualization for Storytelling

Lila understands that she must communicate her findings effectively, so she learns which types of data visualizations are most informative.

She learns to create charts and graphs that visually represent data, including:

- Sales trends
- Customer segmentation
- Product popularity

These visualizations allow stakeholders to grasp the significance of the data and support effective storytelling and decision-making.

50.4 Hands-On Experience

Lila understands that practical experience is invaluable in data science.

To strengthen her practical skills, she:

- Participates in **Kaggle** competitions.
- Works on personal data projects.
- Solves real-world data problems.
- Develops problem-solving skills.
- Creates a **GitHub** account.
- Uploads her projects to build her professional profile.

50.5 Data Wrangling and Preprocessing

Lila learns that data scientists spend a significant portion of their time on data cleaning and preprocessing.

She works on various datasets and learns data preprocessing using the **NumPy** and **Pandas** Python libraries. She becomes skilled in:

- Handling missing data
- Outlier detection
- Feature engineering to improve model performance

50.6 Communication and Storytelling

Recognizing that data scientists must communicate their findings effectively, Lila hones her data storytelling skills.

While pursuing the **IBM Data Science Professional Certificate**, she learns visualization tools such as:

- **Matplotlib**
- **Plotly**

She learns how to create compelling visualizations and present insights in a clear and understandable manner.

50.7 Networking and Collaboration

Lila actively participates in data science communities and attends meetups and conferences.

She:

- Collaborates on open-source projects.
- Connects with fellow data scientists.
- Gains exposure to various industries by attending the **IBM TechXchange Conference**.

50.8 Domain Expertise

Understanding that domain knowledge is crucial, Lila chooses a niche that aligns with her interests.

She explores several domains, including:

- E-commerce
- Healthcare
- Finance
- Several other fields where data science skills can be effectively applied

Because of her master's in economics, she chooses **e-commerce** as her core domain to pursue a data science career.

50.9 Landing the First Job

After months of preparation, Lila begins applying for data scientist positions.

She:

- Tailors her resume to highlight relevant skills and projects.
- Builds an online portfolio that showcases her capabilities.
- Demonstrates her commitment to the field through her work.

50.10 Lila's Approach to Working on Her First Task as a Data Scientist

As a newly hired junior data scientist at a retail company, Lila uses data insights to improve customer service.

Her first assignment involves analyzing customer data to identify patterns and anomalies that could impact customer service. She uses data analysis to enhance the overall customer experience.

50.11 Dataset Selection and Sourcing

In the initial phase of her data science journey, Lila faces the challenge of selecting a suitable dataset and procuring it from different sources.

Apart from the historical data available for the organization for the **past four years**, she:

- Explores various repositories, websites, and databases to find suitable datasets.
- Collects data from diverse sources.
- Decides how to harmonize and integrate disparate datasets into a cohesive whole.
- Consults product professionals, data engineers, and domain specialists for guidance on merging datasets effectively.

50.12 Data Understanding and Cleaning

Lila begins by importing the dataset into her data analysis environment using **Python** and **SQL**.

She:

- Loads the dataset.
- Examines the first few rows to understand its structure and contents.

During data cleaning, she:

- Checks for missing values.
- Identifies duplicate records.
- Detects outliers.

She addresses missing data by:

- Imputing missing values.
- Removing rows or columns containing missing values.

Outliers are identified and treated appropriately based on their impact on the analysis.

50.13 Exploratory Data Analysis (EDA)

As Lila performs **Exploratory Data Analysis (EDA)**, she carefully decides which summary statistics, visualizations, and distribution analyses will best reveal insights into customer behavior and sales trends.

She conducts EDA by:

- Generating summary statistics.
- Creating visualizations such as histograms and scatter plots.
- Exploring the distribution of variables.

EDA helps her understand:

- Customer behavior
- Popular products
- Sales trends

Each analytical choice influences the story the data tells.

50.14 Feature Engineering

Lila recognizes the value of feature engineering in improving her analysis.

She evaluates whether creating new features, such as calculating **total purchase amounts**, will enhance the usefulness of the dataset for her project.

50.15 Statistical Analysis and Machine Learning

Lila evaluates whether statistical tests or machine learning algorithms are necessary for her analysis.

She:

- Uses regression analysis to understand relationships between variables.
- Explores machine learning models for:
 - Demand forecasting
 - Customer segmentation
- Performs statistical tests to uncover patterns in the data.
- Uses regression analysis to understand the impact of **unit price on sales**.

50.16 Presentation and Reporting

At the culmination of her analysis, Lila focuses on effectively presenting her findings.

She:

- Documents her analysis using a **Jupyter Notebook**.
- Compiles a comprehensive report.
- Prepares a presentation.
- Highlights actionable insights.
- Provides recommendations for the e-commerce platform's stakeholders.

50.17 Continuous Learning

After completing her first project, Lila continues to:

- Refine her skills.
- Explore more complex datasets.
- Tackle increasingly challenging data science tasks.

50.18 Machine Learning Skills

Although Lila completes an introductory Machine Learning course as part of the **IBM Data Science Professional Certificate**, the field continues to intrigue her.

To further develop her expertise, she decides to pursue the **IBM Machine Learning Professional Certificate**.

She:

- Identifies **Machine Learning Repository** datasets in the course.
- Experiments with various machine learning algorithms.
- Studies algorithms such as:
 - Linear Regression
 - Decision Trees
 - Deep Learning models
- Continues developing expertise in selecting and fine-tuning algorithms based on specific data problems.